JOINT UNDERTAKING
Bio Based Industries

ANNEX 1 (part B)
Bio-based Industries Research and Innovation action
NUMBER – 745791 – REFUCOAT

History of changes

Version	Date	Changes
1.0	23.01.2017	Maicerias Españolas, S.A. short name is changed to "DACSA" (Table 1.2 List of Beneficiaries)
1.0	23.01.2017	FRAUNHOFER GESELLSCHAFT ZUR FOERDERUNG DER ANGEWANDTEN FORSCHUNG E.V.' short name "Fraunhofer" is changed in all tables (before was FhG-ICT)
1.0	23.01.2017	3.1.3 Timing of the different work packages and their components (Gantt chart): Partners participation changes: T1.2 Partner 16 added, T2.2 partner 11 eliminated, T4.1 partner 9 eliminated. Missing task 4.2 is added Missing task 5.7 and task 5.8 are added Associated deliverables and timing updated according Annex 1 (part A)
1.0	23.01.2017	New subtitle "3.2.1 Management structure and procedures" is added. "3.2.3. Innovation management strategy" is changed to "3.2.2 Innovation management strategy".
1.0	23.01.2017	Table 3.4.a Summary of staff efforts: changes produced in WP1: Miplast (2PM), WP2: AIMPLAS (9 PM), IRIS (0), WP4: AIMPLAS (3), IRIS (31), WP5: MIPLAST (26), WP6: MIPLAST (2)
1.0	25.01.2017	The shortcoming identified in the ESR regarding the Intellectual Property and Exploitation Board is corrected: "VITO" is changed to "Biopolis" (page 44)
1.0	27.01.2017	3.1.3 Timing of the different work packages and their components (Gantt chart): Deliverables 7.9, 7.10, 7.11 are added
1.0	27.01.2017	3.1.3 Timing of the different work packages and their components (Gantt chart): WP9 is added
2.0	31/03/2017	Figures 1.1 a, 2.1 a) removal of FARMAMAK, figure 3.3a) Inclusion of AIMPLAS as coordinator, New partner: CSIC
2.0	31/03/2017	Redefine governing boards to be consistence with Figure 3.2a). Steering Committee and Governing Board will be named as Executive Board
2.0	12/04/2017	Tables 3.4 a) and b) update with the new partners
3.0	26/06/2017	Update tables, figures and text removing UNILEVER and adding GRUPO APEX.

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1. Excellence

1.1. Objectives

Global high barrier packaging film consumption is at approximately 1.76 million tonnes in 2014 with a value of €12 billion. This is forecast to grow at a CAGR of 5.0% between 2014 and 2019 according to a recent market report[1]. According to the report, there is a growing demand for packaging materials that give even greater protection in the food, beverage and pharmaceutical industries. As plastics have become more and more common, concerns have arisen about their ability to allow the exchange of gases and vapours that can compromise the quality and safety of packaged products. Therefore, a variety of barrier technologies have been commercialised that preserve and protect, optimise shelf life, reduce the need for preservatives, provide transparency and gloss, etc. Without this barrier packaging, perishable goods would be susceptible to a wide range of deterioration processes. However, **barrier packaging is restricted by factors such as lack of biodegradation, recycling complexity and cost**. Mechanical recycling also becomes a problem with multilayer structures containing more than **one type of plastic or combination with other materials such as aluminium**. Environmental pressure groups have also raised concerns about increases in the amount of food packaging, with many companies responding by reducing their packaging.

Bio-based polymers have been widely studied in recent years. Replacing the oil-based packaging materials with bio-based ones might give a competitive advantage due to more sustainable and greener image, maintain properties. Special attention should be paid **to barrier properties which are extremely important especially for bio-based food packaging materials**.

Moisture resistance with hydrophilic polymers is in many cases inadequate, thus, excessive water vapour transmission through packaging dilutes the quality of foods, resulting in shorter shelf-lives, increased costs and eventually more waste. Among the full biodegradable bioplastics, polyhydroxyalkanoate (PHA) is a cost competitive polymer with good water barrier properties and polyglycolic acid (PGA) has excellent water barrier properties being one of the most promising new commercially available barrier polymers. However, their current high cost (>20 €/kg) reduces their use as a single film for food packaging applications.

RefuCoat project aims to develop **hybrid bio-based high oxygen/water barrier and active coatings** to be used in a **monolayer bio-based packages** (films and trays**)** as alternative to current **metallised** and **modified atmosphere (MAP) packages**. Currently, both types of packages are based on the use of non-renewable materials in multilayer structures that lead to complex and expensive recycling steps.

Figure 1.1.a shows the shows the scheme of the project including the role and relation between the partners.

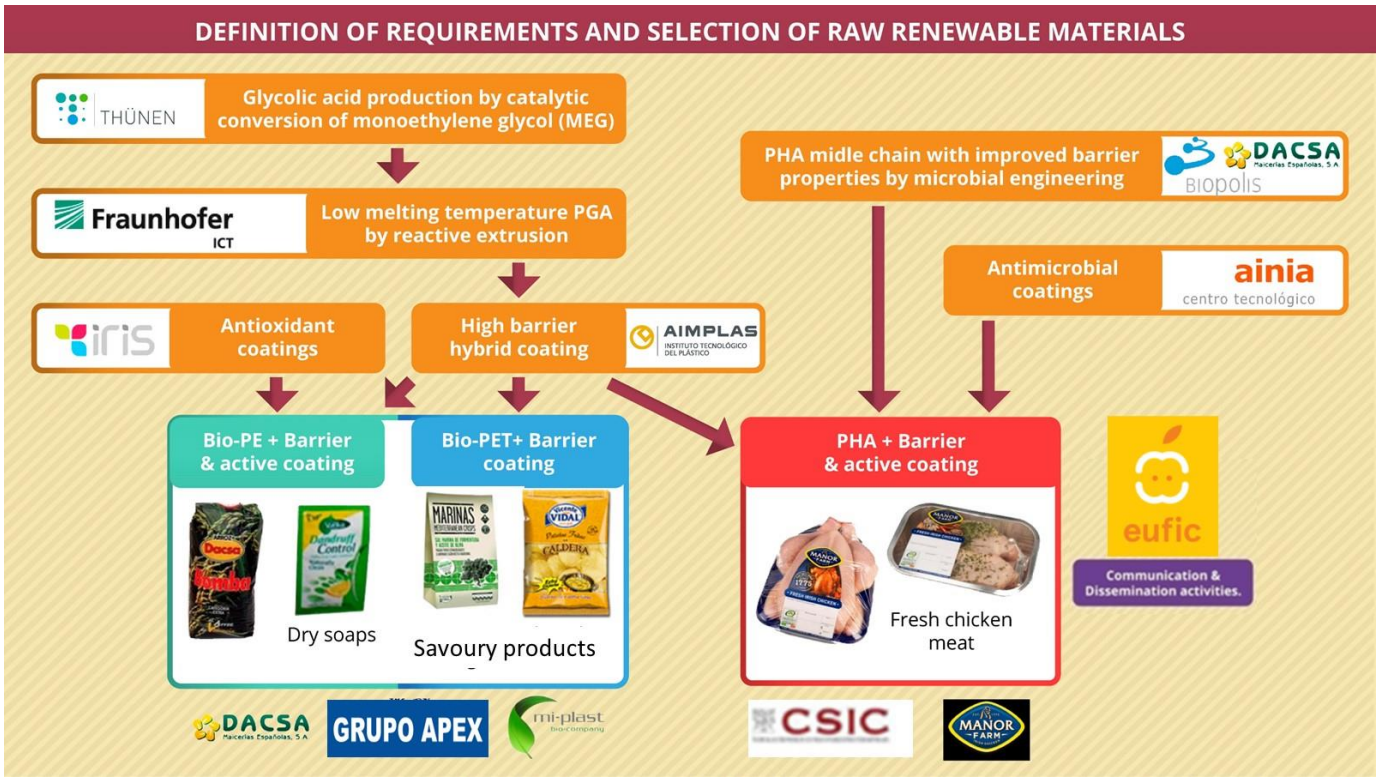


Figure 1.1.a Main proposed innovations in RefuCoat's project and involved partners.

[1] www.smitherspira.com/news/2014/april/high-barrier-packaging-film-consumption-forecast.

1.1.1. Overall Objective.

Objective 1: Microbiological modification of middle chain polyhydroxyalkanoates (PHA) to achieve a polymer with high water barrier. Target WVTR < 5. *(Linked to WP1 & 2).*
Current 50 µm film of PHA has an average WVTR of 30 g/m^2/day, the inclusion of longer chains and hydrophobic functional groups in the PHA backbone increases water barrier properties of the polymer. At the same time, it is important to improve processability of the polymer. Therefore, a battery of different PHA molecules will be microbially synthesized and their thermo, mechanical and barrier properties will be analyzed. The production process of the PHA that meets the processing and target barrier requirements will be scaled-up in order to provide material for the processing and coating tests. ***Performance indicators:*** by M18, a report on the thermal and barrier properties of the mcl-PHA; by M30, a report on the improvements of processability; and by M30 a report on selected mcl-PHA scale-up.
Objective 2: Production and purification of glycolic acid (GA) for reactive extrusion polymerization from mono ethylene glycol (MEG) using gold based catalysts. Target yield > 90% & purity >99% *(Linked to WP2).*
Development of active, highly selective and long-term stable catalysts, preferably supported gold-based catalysts, for the conversion of MEG to GA. In order to achieve high space-time yields the reaction will be carried out in concentrated MEG solutions. To establish a green process water will be used as solvent and oxygen or hydrogen peroxide, respectively, as green oxidizing agents which ensure high atom efficiency. Suitable GA recovery and purification protocols will be established. ***Performance indicators:*** by M24, two reports about chemocatalytic GA production process and GA isolation and purification protocols, respectively, and by M30 a report about GA production scale-up.
Objective 3: Development of polyglycolic acid (PGA) low melting temperature based adhesives to be used in the hybrid coating formulations *(Linked to WP3).*
Fraunhofer will develop and benchmark most efficient, economic and ecologic viable processes and process combinations for the polymerisation of glycolic acid. This will be realized by investigating different promising processing technologies: • (Semi) continuous reactor based reaction polymerisation will be evaluated for maximum control of the resulting material properties and easy state of art further material utilisation • Reactive extrusion polymerisation will be evaluated for highly efficient, solvent free continuous polymerisation • Finally, the combination of both will be evaluated, realizing initial pre-polymerisation completed by reactive extrusion, enabling the opportunity for further material modification in continuous, solvent free processes. ***Performance indicators:*** by M18 a report on benchmarked polymerisation technologies and ranking thereof, by month 30 an in-line polymerisation demonstration line and a descriptive report thereof.
Objective 4: Development of biodegradable Organic–Inorganic Hybrid Coatings based on the combination of modified silica oxide and PGA as a real alternative to metallised coatings. Target OTR < 2 cc/m^2/day and WVTR < 2 cc/m^2/day *(Linked to WP3).*
A new class of transparent organic–inorganic composite materials can be prepared with the sol–gel methodology, through the incorporation of oligomeric or polymeric molecules of PGA into a solution formed by a precursor of the inorganic phase. All components used for the coatings will be food contact approved. Coating will be suitable to form a thin coating layer (less than 10 g/m^2 or 8 microns) using standard or slight modified plastic coating machines. Coating will be used also as a thermal seal layer and a minimum seal strength of 2 N/mm ***Performance indicators:*** by M30, a report will be prepared on sol-gel reactions for PGA hybrid coating formulation.
Objective 5: Incorporation of an active substance (antimicrobial and antioxidant) to the hybrid coating to enlarge the shelf life of selected foods. Shelf life target increase of selected food products up to 15% (*Linked to WP4).*

Active coatings suitable to be applied on multilayer films will be developed to extend the shelf life of the packed product. The possibility of joining such function with that of the previously described barrier hybrid coating in a single layer will be studied to facilitate the production process and further recycling operations. The active substance will require combining different active substances: bacteriophages or their endolysins against specific pathogens and substances with a broad spectrum of microorganisms targeted against spoilage ones. The application process on the substrates from the previous WP will be developed and the resulting properties of coated films validated. ***Performance indicators:*** by M26, a report on the active hybrid barrier packaging development will be ready demonstrating its capacity to extend shelf life of snacks by at least 15% vs. the passive solution using the hybrid coating only and equal to petrochemical counterparts therefore enhancing the capability of the developed Refucoat packaging solutions to substitute currently used materials. Recommendations for further upscaling will be provided. On the other hand, for poultry products, the benefits on shelf-life and food safety will be tested and evaluated. By M30, a report with the analytical results showing the differences between using or no using the active coating will be prepared.

Objective 6: To produce new packages based on Bio-PET + hybrid and active coating with high water and oxygen barriers (OTR< 1 cc/m^2/day and WVTR<1 g/m^2/day), *(Linked to WP5).*

Commercial bio-based polyester (for example Globio™ Bio-PET-Toyota Tsusho) will be extruded in a monolayer blown film extrusion line (AIMPLAS or MI-PLAST facilities) and coated with the suitable hybrid coatings produced in WP3&4. Flexible packages will be produced to contain savoury food products. The full package will be validated by GRUPO APEX in comparison with current metallised-PET following their internal procedures. The target shelf life time differences between both packages are to be less than 10%. ***Performance indicators:*** by M28, suitable packages based on Bio-PET and developed coatings will be available for processing at pilot plant scales. By M35, at least two packaging demonstrators (savoury products) will be available for their validation, as well as in terms of legal compliance.

Objective 7: To produce new packages based on Bio-PE + hybrid and active coating with high water and medium oxygen barriers (OTR< 5 cc/m^2/day and WVTR<1 g/m^2/day), *(Linked to WP5).*

Commercial bio-based polyethylene (for example Green PE of Braskem) will be extruded in a monolayer blown film extrusion line (MI-PLAST facilities) and coated with the suitable hybrid coatings produced in WP3&4. Flexible packages will be produced to contain food products sensitive to moisture. The full package will be validated by GRUPO APEX and DACSA in comparison with current metallised polyethylene following their internal procedures. The target shelf life time difference between both packages is to be less than 10%. To improve the project impact, the expansion of the technology to non-food packaging applications will be analysed. ***Performance indicators:*** by M28, suitable packages based on Bio-PE and develop coatings will be available for processing at pilot plant scales. By M35, at least two packaging demonstrators (cereal products) will be available for their validation, as well as in terms of legal compliance.

Objective 8: To produce new packages based on PHA middle chain + hybrid and active coating with high water and medium oxygen barriers (OTR< 5 cc/m2/day and WVTR<1 g/m2/day), *(Linked to WP5).*

PHA produced in WP2, the suitable coatings of WP3 and 4 and the packaging technologies proposed in WP5, trays and lips for chicken meat, will be produced in the facilities of MI-PLAST and AIMPLAS. The new trays will be full validated by MANOR (including migration tests and shelf life) in comparison with current ones based on PET and PE. The target shelf life time differences between both packages are to be less than 10%. ***Performance indicators:*** by M36, suitable packages based on middle chain PHA and developed coatings will be available for processing at pilot plant scales. By M42, at least one packaging demonstrator (chicken meat: tray and lid) will be available for their validation, as well as in terms of legal compliance.

Objective 9: To demonstrate the safety & regulatory compliance, as well as environmental & economic sustainability of the developed processes and products *(linked to WP6).*

LCA will be performed on the processes developed for the selected food packaging in the demonstration of the reduced environmental impact vs. current traditional materials. Standard and legislative aspects that apply to the project results will be continuously monitored. A Safety & Regulatory assessment will be performed during the project. Aspects in relation to waste management and potential health risks will be addressed. Finally, the techno-economic viability will be evaluated with inputs from all previous WPs. ***Performance indicators:*** by M12, a report analysing the main regulatory frameworks in place in the EU regions considered in the project will be produced. By M30, a report for the safe operation of the pilot plant will be produced, as well as recommendations for a DEMO plant. By M36, the LCA and techno-economic assessments will fuel into recommendations for future final improvements and they will be taken into account to assess commercial viability.
Objective 10: *To maximize the innovation impacts of the project for contributing to the uptake of the project results for companies' business and employment growth (linked to WP7).*
The project knowledge and results will be managed and protected. Business and exploitation plans will be delivered for all Key Exploitable Results (KERs) (in addition to the specific ones previously described in the project) to map the post-project phase. A comprehensive dissemination and communication campaign will be designed and implemented. Regular knowledge transfer activities will take place among partners. Workshops with researchers and participation in educative actions of next generation stakeholders will be organized. ***Performance Indicators:*** by M12, a robust dissemination and communication plan to support the recognition of RefuCoat as a reference in the maximization of value generated to create new opportunities in the packaging and food processing industries will be in place. By M24 and 36, a suite of training activities (>3) will be organized. By M36, final plans will be in place for the transition to the post-project phase to advance the TRLs, source of further funding to full DEMO scale processes will be analysed focusing on applications where a return on investment is feasible.

1.1.2. The drivers and opportunities for the development of "recyclable and bio-based food package"

The project relies primarily on a strong technological opportunity offered by the innovative character of technologies and processes for food packaging applications by developing a new range of bio-based barrier and active packages with a competitive cost and fully recyclable. In addition to this, the section below highlights the key socio-economic and regulatory drivers which further justify the implementation of the project.

- Towards new sustainable bioplastics: RefuCoat intends to demonstrate the potential and economic opportunity offered by the use bio-based polymers, including PHA from food by-products and other commercial biopolymers. With biodegradable materials RefuCoat will contribute to EC's Landfill Directive in order to reduce saturation in landfills.
- Reducing the package footprint of future industrial processes: by the use of: i) petrol based polymers and the removal of materials that reduce the recyclability such as aluminium or multilayer film structures, ii) increase the food shelf life by incorporation of safe and novel active additives in the package.

The "Ecological Footprint", one of the most important reports assessing key indicators related to the overall status of the planet, indicates that all of the 28 European Union (EU) countries surveyed are living beyond the means of "a single planet" and also rely heavily on the natural resources of other countries. Climate change, green economy and sustainable management of resources are among other topics of great concern at a European and global level, as indicated by the Memorandum of Understanding between the EU and the United Nations Environment Programme (UNEP) and the organization of the COP21 conference in France in the beginning of December 2015. A drastic change of our modern way of using different natural resources has to be carried out and accepted. As part of this general trend, the development of new technologies for packaging is urgent and essential, taking into account the global challenges with regards to pollution, global warming, and fossil oil shortage. In RefuCoat the transition to sustainable value chains will be initiated following the principles of these core values, i.e. the entire footprint of the value chains, ranging from feedstock to process to materials, including aspects related to transportation & logistics, side streams valorisation and recycling will be taken into account and act as underlying drivers behind technological choices.

- Innovation in materials through disruptive processing technologies. RefuCoat include the follows: i) new middle chain PHA with improved moisture barrier using microbiological modification routes, ii) Bio-based hybrid coating with high gas barrier properties at low thickness, iii) active packages containing essential oils and bacteriophages or their endolysins against specific pathogens and substances with a broad spectrum of microorganisms targeted against spoilage ones.
- Economic and societal drivers for a transition to the bio-economy: RefuCoat aims to replace current

metallised and multilayer barrier packages for meat packaging. Due to the innovations proposed, RefuCoat expects to achieve a package with improved properties and costs similar to current ones, but using bio-based safer and fully recyclable materials

In addition, meaningful contributions to sustainable energy and chemical supplies provide social and economic benefits. The transition towards a more sustainable, bio-based economy generates multiple benefits also from a societal point of view, linked to health, local industry & economy. From a general point of view, the development of sustainable and environmentally friendly processes and products is expected to contribute to improve the environment and the overall footprint. Local industry and jobs creation should follow, as recognized by the European industries and as perfectly stated by the European Commission. These different aspects will be tackled within RefuCoat project.

1.2 Relation to the BBI JU annual Work Plan

The RefuCoat project addresses the topic: BBI.2016.R5. First on a wider point of view, the project is totally aligned with to the BBI road map. The BBI aims at accelerating the building of bio-based value chains, starting from sustainable feedstock production and mobilisation towards the implementation and use of bio-based materials and products such as packaging (as defined in Value Chain 1, 2, 3 and 5). As such, we consider as main innovations in the new packages the use of bio-based products and their recyclability, in comparison with current multilayer packages made by non-renewable polymers. We furthermore validate the derived bio-based products in most of the strategic area of the Strategic Innovation and Research Agenda (SIRA). The detailed relations of the project to the specific challenges and scope addressed by the topic are justified in table 1.2 below.

Table 1.2 How RefuCoat relates to the scope and challenges of the BBI 2016 R05 topic.

Challenges of BBI 2016 R05	***How RefuCoat addresses these challenges***	***WP***
Develop **bio-based materials with new functionalities** for food packaging to fulfil the following objectives (i) Prove performances at least equal to existing, comparable packaging materials in terms of mechanical properties & temperature resistance or liquid/oxygen/UV/ CO_2-barriers or antimicrobial activity, (ii) To improve end-of-life options: provide biodegradable or compostable alternatives; or provide mono-material (easily recyclable) alternatives while matching the same properties (e.g. barrier) of existing multi-material ones.	RefuCoat aims to develop a full recyclable monolayer bio-based barrier and active packages for food applications including cereals, meat, snacks and savoury products. The raw material to produce the coating is polyglycolic acid (PGA). It is a full biodegradable material with thermoplastic behaviour and good compatibility with other bio-polyesters and easy to compatibility with polyolefins. The new packages to develop will have the same mechanical, thermal and barrier properties than petrol-based counterparts. RefuCoat aims to develop the two types of packages: one full biodegradable and compostable (including the coating material) and the second one bio-based and non-biodegradable but easily recyclable, due to the thermoplastic behaviour of the coating material.	2, 3 & 4
Proposals should address a small number of well-focused applications, identifying the benchmark products to prove the better characteristics of the developed materials.	The project is focused on two packages; i) the replacement of metallised packages by the use of hybrid barrier coatings for savoury products and cereals and ii) the production of middle chain resins to develop a fully recyclable & biodegradable package for chicken meat products.	1 to 5
Proposals should include the direct link with end-market actors to provide clear requirements of the developed products and materials.	GRUPO APEX, DACSA and MANOR are large end-user companies. They are in charge of clear requirements definitions (WP1) and package validation (WP5) in comparison with current ones..	5,6
Proposals should comply with existing legislation for food contact.	All selected materials and additives fulfil the current legislation related to plastics for food-contact applications. For example, Regulation EC 10/2011 on plastics intended to come into contact with food and Regulation EC 450/2009 on active and intelligent materials and articles intended to come into contact with food.	1,6
For specific applications, biodegradable packaging should be the preferred route. Design for recycling at end-of-life must be part of the early development stages for non-biodegradable packaging material.	For chicken meat products RefuCoat plans to develop a fully biodegradable package. Whereas for cereals and savoury products, we plan to develop fully recyclable but non-biodegradable packages. For selection of the suitable package structure considerations such as cost, ease to recycle and required properties will be taken into account.	6

The projects should cover any Technology Readiness Level (TRL) from 3 to 5.	RefuCoat targets TRL3-5 as explained in table 1.3.a. The project consortium includes 3 SMEs and 4 large industries with a significant role in the project.	1to 5
The new bio-based material should be at a competitive price to enable subsequent downstream steps for the production of films and flexible food packaging to suit market applications.	To calculate the cost, the full process will be taken into account from raw materials production to the management costs after the use. Hybrid coatings will employ resins such as PGA and cheap fillers such as SiO_2, to minimize costs. Metallised packages are also expensive to produce, due to that few companies have vacuum metallization facilities and that this is an energetic costly process. The cost of bio-PET or Bio-PE are similar to non-renewable plastics. The extra cost needed to manage the metallised package vs RefuCoat's packages will compensate the higher costs of the resin and the coating.	5,6
Proposals should also include an environmental and socio-economic assessment, for example with an LCA, with particular focus on the end-of-life phase.	WP6 is dedicated to the evaluation of the environmental, societal and safety aspects (in terms of alignment with existing regulations). In addition, LCA and socio-economic assessment will be delivered for the products and processes from the project. When applicable, eco-toxicity tests will be performed along with the biodegradation trials of the materials for packaging.	6

1.3 Concept and approach

1.3.1 Overall concept

Metallised films (or metalised films figure 1.3.a) are polymer films coated with a thin layer of metal, usually aluminium. They offer the glossy metallic appearance of an aluminium foil at a reduced weight and cost. Metallised films are widely used for decorative purposes and food packaging, and also for specialty applications including insulation and electronics. Metallised films are essential when the product requires extra protection to ensure a longer shelf life, aroma and flavour retention and for the luxury industry to enhance the visibility of the packaging on the shelf. Metallised film output in Europe was 124 kt in 2012, with the market forecast to rise 2.0-2.5% a year in 2014-18.

Figure 1.3.a. Example of a metallised film

Metallisation is performed using a physical vapour deposition process. The metal is heated and evaporated under vacuum. This condenses on the cold polymer film, which is unwound near the metal vapour source. This coating is much thinner than a metal foil and they are lower than one micron. While oriented polypropylene (OPP) & polyethylene terephthalate (PET) are the most common films used for metallisation, others such as polyamide, polyethylene (PE) or polylactic acid (PLA) are also used.

As figure 1.3.b shows the high barrier properties of metallised films with OTR and WVTR barriers lower than 1, i.e., up to fifty times for PE, up to ten times for PET

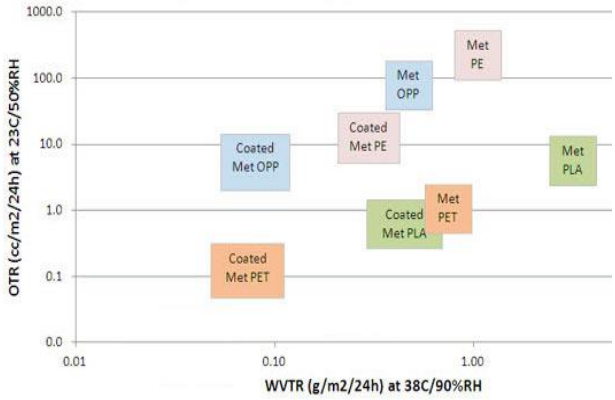


Figure 1.3.b Barrier properties of metallised films.

In Europe, demand for metallised film – replacing aluminium foil (that has a current market of 550 kt/y) – is showing strong growth levels, with transfer-metalizing the favoured technology. Food products packed in metallised or coated films include sausages, pasta, cheese, instant drink powder, jam, milk powder, health drinks, tea, coffee, herbs, spices, fruit juice (pouches), snacks, biscuits and other bakery products, etc. Although metallised PET film market continues to grow, there are no established sustainable solutions to recover and recycle post-consumer and post-industrial metallised films waste or any application of this material.

While plastic recycling industry has been established for several decades, metallised films are discarded as waste and end up in the landfill (figure 1.3.a). There are no known commercially existent metallised films recycling facilities in the world due to numerous technical reasons that make recycling of metallised films difficult and the possibility that they could contaminate current recycling streams leading to recycled plastics of low quality. Main large food companies such as GRUPO APEX are looking for real alternative solutions to metallized packages.

An alternative coating to metallisation *with high gas barrier, savoury retention, applicable to the same polymers (PE and PET), with high cadence of production using standard coating machines, similar cost and recyclable* ***is a real industrial need. Hybrid organic-inorganic coating*** *is the best solution beyond of the state of the art solution,* ***it is based on the combination of high oxygen barrier bio-polymers and SiO_2 particles.*** *Previous laboratory tests have demonstrated high barrier properties using coating layers of few microns.*

Figure 1.3.c Out-of-specifications post-industrial metallised film

Middle polyhydroxyalkanoate (PHA) as a biodegradable barrier polymer. One of the major benefits of flexible packaging, especially for food applications, is that it can be developed and used for high barrier, shelf life extending or niche applications. This can be done using co-extruded and laminated film structures, or by taking advantage of advances in modified atmosphere packaging (MAP). These film structures are made by combination of polyolefins and oxygen barrier polymers (EVOH, PVDC or PEN). Barrier polymers have a huge market in the packaging sector (more than 400 kt)[2]. Current barrier plastic materials industry is an oligopoly, that is, a market with a very limited number of producers, due to packaging manufacturers having high market entry barriers and high cost of raw materials. Bioplastics, however, generally have lower barrier to gas and moisture than oil-based plastics.

This is not a problem, when bio packaging is used with fresh products such as fruits, vegetables and bread but it can be a problem for foods that require a longer shelf life such as fresh meat or fish. As table 1.3.a shows, PHAs are a fully biodegradable biopolymer with the lower moisture permeability, a clear advantage in order to protect the hybrid coating from moisture.

Table 1.3.a WVTR of biodegradable polymer films (25 µm thickness & 25ºC).

Biopolymer	WVRT ($g/m^2/day$)
Starch	200 – 900
PBS	450
PBAT	326
PCL	172
PLA	45-52
PHA	25-35
PGA	***8***

RefuCoat aims to produce fully biodegradable packages for **fresh food products with MAP properties** by using **middle chain PHAs** modified with **lipids groups** in order to increase the PHAs hydrophobicity and combine it with the **hybrid coating with high gas barrier properties** containing low gas barrier polymer**, i.e., polyglycolic acid (PGA)**

One of the mechanisms suggested to improve microbiological safety and to avoid potential spoilage in minimally processed foods is the incorporation of antimicrobials into the package materials. In recent years, developments have been underway in the area of “natural” and “controlled-release” antimicrobials in packaging materials, known as active packaging.

Active packaging absorbs or releases substances in order to improve the quality of packaged food or to extend its shelf life. RefuCoat project will be based in the incorporation of some natural antifungal, antibacterial and/or antioxidant in the package and thanks to the capacity of releasing these additives the preservation of selected food (chicken fresh meat, snacks and cereals) and the shelf life will be increased. Consequently, there would be no need to introduce synthetic additives in processed food formulations and this would result in more natural and consumer-friendly foodstuff. Active packages are already being successfully applied to extend shelf life while maintaining nutritional quality and ensuring microbiological safety.

The safety of food-contact materials requires evaluation as chemicals can migrate from these materials into food. The materials should be manufactured in compliance with European Union (EU) regulations[3], including good

[2] www.prnewswire.com/news-releases/global-and-china-high-barrier-material-pvdc-evoh-pen-industry-report-2014-2017-275191651.html

[3] Active materials and articles may be placed on the European market if they comply with the restrictions set out in Regulation (EC) 1935/2004, articles, 3, 4 and 15, and the European Regulation (EC) 450/2009.

manufacturing practices, so that any potential transfer to foods does not raise safety concerns, change the composition of the food in an unacceptable way or have adverse effects on quality (for instance, taste and/or odour). The use of bacteriophages to control pathogens is a promising technology that has been proved effective in several applications. The addition of these agents or their enzymes (endolysins) to act as preservatives in active packaging can be very advantageous, allowing to combine the benefits of both technologies.

The raw poultry meat is still a vehicle of bacterial contamination at domestic level, as it contains persistent pathogens such as *Salmonella* and *Campylobacter*, the latter being responsible for the most frequent foodborne illness in the EU with an estimate of 9M cases/year and a cost of EUR 2.4 billion a year. The development of active packaging systems that act specifically against these bacteria as the last stage of the food chain can have a big impact in the safety of these products.

1.3.2 Positioning of the Project.

The activities developed in **RefuCoat** project will be implemented at TRL3 with a TRL target of 5. Most of the proposed innovations are based on technologies studied at laboratory level, as it is shown in Table 1.3.b:

Table 1.3.b. TRL justification of RefuCoat developments.

Technology proposed	TRL	Justify current and final TRL
Modified mcl-PHA.	3→5	Laboratory protocols developed to achieve TRL3 for modification of PHAs. TRL5 will be developed by optimizing process and pilot scale-up.
Catalytic oxidation of MEG to GA.	3 →4/5	Very few lab studies report on the catalytic oxidation of MEG to GA. No information is available on the long-term stability of the catalysts nor on the recovery of GA. TRL 5 will be achieved by optimizing the catalyst, reaction conditions and GA purification protocols in larger lab/small pilot scale.
Reactor production of PGA.	4 → 5	Batch-wise production of PGA has been described often. Materials are commercially available for medical applications. The project will increase the TRL focusing on tailored properties for coating applications.
Reactive extrusion production of PGA.	3 → 5	Reactive extrusion polymerisation has already been shown with the ability to greatly influence molecular weight and viscosity. In the project, the reactive extrusion of this particular material will be further evolved solving the challenges in polycondensation and resulting in high quality materials.
Polymerisation & further modification in single reactive extrusion step	3 →4/5	Reactive extrusion with further material modification has already been shown. Yet the here addressed application asks for special viscosity and shelf-life requirements. TRL of the combined processing will be increased including reactor upscale and build-up of demonstration line.
Hybrid coating	3 →4/5	Hybrid coating showed good barrier properties results in films, based on sol-gel well know reaction. Some preliminary papers show PVOH and bioORMOCER® as a hybrid coatings using chitosan or PCL. TRL will be increased with hybrid coatings in continuous and full controllable process.
Active packaging (AP).	3 → 5	AP technology generally has TRL above 3, based on past experience at partners and literature. Nonetheless, adaptation needed to the specific food and format selected will be addressed to a relevant demonstrator scale in WP4-5.
Bacteriphages (endolysins).	3→4/5	Antimicrobial technology is starting at lab-scale with published references in USA.and Brazil. RefuCoat will validate the use of bacteriophages (endolysins) carried by formulated coatings in an industrially relevant environment.

1.3.3 Positioning of RefuCoat within the European Context.

We were unable to identify public European projects with the same research objectives as those proposed in **RefuCoat**. However, there are a number of national and European projects with strong synergy with **RefuCoat**, and details are summarized in Table 1.3.c.

Table 1.3.c. Partners' projects related to **RefuCoat**.

Description of experience/project	Scope	End Year	Website for information (RefuCoat's Partner)
BUGWORKERS: New tailor-made PHB-based nanocomposites for high performance applications.	FP7-NMP	2014	www.bugworkersproject.eu/ (AIMPLAS).
BRIGIT: New biopolymers produced from lignocellulosic residues for flame retardancy highly demanding applications.	FP7-KBBE	2016	www.brigit-project.eu (AIMPLAS).
BIOREFINE-2G: Production of dicarboxylic acids and bio-based polymers**.**	FP7-KBBE	2017	www.biorefine2g.eu/ (AIMPLAS,).
OLI-PHA A novel and efficient method for the production of PHA polymer-based packaging from olive oil waste water.	FP7-NMP	2015	www.olipha.eu (IRIS)
WHEYLAYER2: Barrier polymers for sustainable packaging.	FP7-SME	2014	www.wheylayer.eu (IRIS)
POAP: Production of organic acids for polyester synthesis.	ERA-IB	2016	www.era-ib.net/poap (BIOPOLIS / THUNEN)
FAPA: Fumaric acid for polymer applications.	ERA-IB	2019	www.era-ib.net/6thjointcall/FAPA (THUNEN / BIOPOLIS)
INNOREX: Realising the continuous, metal-free polymerisation of PLA using alternative energies for an innovative reactive extrusion.	FP7-NMP	2016	www.innorex.eu (Fraunhofer)
StarTex: Developping melt spinnable, starch based materials starting from native starch, including reactive extrusion modification and bio-polymer blending for thermoplastic behavior, increased melt strength and enhanced water resistance.	German project	2018	www.ita.rwth-aachen.de/3-f-und-d/kurzberichte/2015/2015_12_09_Projektsteckbrief%20Star-Tex%20Manvi.pdf (Fraunhofer)
PLA4FOOD: Active multilayer packaging based on improved performance PLA formulations for 4th range food products.	FP7 – R4SME	2013	www.aimplas.es/proyectos/pla4food/ (AIMPLAS)
PH-BOTTLE: New sustainable, active and competitive PHB material based in fruit by-products getting advanced solutions for packaging applications.	FP7-NMP	2015	www.phbottle.eu (AINIA, AIMPLAS)
SUCCIPACK: Active, intelligent and sustainable food packaging using polybutylene succinate.	FP7 KBBE	2011	www.succipack.eu (AINIA)
WHEYPACK: Reduction of CO_2 emissions by PHB obtained from whey: demonstration in dairy products packaging	LIFE	2016	www.wheypack.eu (AINIA, AIMPLAS)

A thorough freedom to operate search has been done on existing patents and technical articles. EU and world-wide data bases have been consulted and there are no patents invalidating the proposed RefuCoat project objectives. Besides product innovations, the innovation potential for the partners regarding the improvement of the municipal solid waste and fish waste fractionation and purification, micro biotechnological processes, downstream, catalysis, green chemical and final production processes is substantial.

1.3.4 Approach and methodology.

RefuCoat is a **3 year project**, structured into 9 work packages. **RefuCoat** will take into account all development steps to obtain valuable products from low value by-products to fit real industrial needs, including a work plan with a well-balanced structure that combines research activities for the development of required biopolymers (WP2 and WP3), production of food packages applications (WP4 and WP5) with piloting activities in WP6. Environmental and economic studies are placed in WP6. RefuCoat is a well-balanced consortium covering all the value chain from waste valorisation to end-users.

In WP1, DACSA will lead the selection of most representative case studies where currently metallised packages are used to define their main requirements. Commercial biopolymers (Bio-PE and Bio-PET) will be selected to replace metallised packages. In WP2 CSIC will produce low permeability medium chain PHA from DACSA by-products

and novel microbial catalysts to incorporate high hydrophobic groups in the PHAs backbone. Thünen will produce GA at a competitive cost from bio-based MEG using own developed gold catalysts. In WP3 Fraunhofer will produce low melting point PGA using the monomers produced by Thunen in WP2. AIMPLAS will be in charge of production of hybrid coating by chemical bonding of PGA and SiO_2. In WP4, IRIS and AINIA will be in charge of antifungal and antimicrobial or antioxidant coating development according the needs of the products selected in WP1. In WP5 AIMPLAS with the support of industrial packer manufacturer (MIPLAST) will produce initial packages to pack the selected foods using the materials developed in previous WPs (2 to 4). WP5 industrial partners will produce a representative number of packages, fully characterized and validated for food applications. The comparative shelf life of the food products packed will be done by GRUPO APEX, MANOR and DACSA with the support of AINIA. LCA analyses, safety, standardization and economic assessment to validate the main objectives of the RefuCoat project are addressed in WP6 led by DACSA. Innovation management, business plan and Communication, Dissemination & Exploitation activities will be done in WP7 led by EUFIC. BIOPOLIS partner will be in charge of market analysis and elaborate the preliminary business plan for all the project partners and initial contacts (Stakeholders) with third-parties interested in the project results. WP8 will be dedicated to management and risk assessment and it will be led by AIMPLAS. WP9. Ethics requirements will have the objective to ensure compliance with the "ethics requirements" set out in this work package, and it will also be led by AIMPLAS.

1.3.5 Sex and Gender analysis.

The technologies, processes and materials developed in RefuCoat are gender-independent. All RefuCoat Partners have internal gender equality jobs creation and promotion policies. The consortium will foster the presence of women in the technical and management teams and decision board implemented to govern RefuCoat project. The coordinator of RefuCoat is Ms. Itziar Carracedo. Project aims to contribute to these objectives for enhancing Gender Equality by involving numerous female researchers (see section 4) in the project and by promoting women researchers to play a leading role in work packages. Actually, more half of the WPs are led by women: WP5 (Ms. Lorena Rodriguez), WP6 (Dr. Mari Carmen Vidal), and WP7 (Dr. Nina McGrath) are women.

The Treaty on European Union, the European policy establishes equal opportunities between women and men. In particular articles 2 and 3 establish equality between women and men as a specific task of the Community and also to promote equality. Monitoring the progress toward gender equality in science has become a well-established activity of the European Union research policy. In accordance with these principles, the approach of RefuCoat is fully in line with Directive 2002/73/EC of the European Parliament and of the Council of 23 September 2002 on the implementation of the principle of equal treatment for men and women as regards access to employment, vocational training and promotion, and working conditions.

1.4. Ambition.

1.4.1 State of the art and progresses in the field of microbiological modifications of mcl-PHA to enhance their barrier properties.

Polyhydroxyalkanoates (PHA) are polyesters made of (*R)*-3-hydroxyfatty acids. The monomer composition of PHA has considerable effects on its physical properties. The PHA structure can effectively be controlled by adjusting the carbon substrate to achieve the monomer content, by engineering metabolic pathways in the host or by feeding the culture with carbon substrates containing functional side chains that can be chemically modified in a second step[4]. PHAs offer great potential for packaging due to their good thermochemical and barrier properties.[5]

To improve flexibility for potential packaging applications, several copolymers can be synthesized, *e.g.* PHB-co-PHV [poly (3-hydroxybutyrate)-co-poly (3-hydroxyvalerate)] leads to a decrease of the glass transition and melting temperatures, and broadens the processing windows since there is improved melt stability and lower processing temperatures. PHA is sensitive to high processing temperature. Therefore, lower melting temperatures help to prevent the degradation of the chains in processing.

Middle chain PHAs (mcl-PHAs) are considered to be promising candidates for special bioplastic applications due to properties derived from their longer side-chains and altered crystalline structure, such as elasticity, hydrophobicity, low oxygen permeability, water resistance and biodegradability[6]. Furthermore, unconventional mcl-PHAs bearing different functional moieties in their side chains can be produced through different biotechnological strategies. These

[4] Visakh P.M. (2015) Polyhydroxyalkanoates (PHAs), their blends, composites and nanocomposites: state of the art, new challenges and opportunities. *RSC Green Chemistry*, chap. 1., The Royal Society of Chemistry.

[5] E. Bugnicourt et al. (2014). Polyhydroxyalkanoate (PHA): review of synthesis, characteristics, processing and potential applications in packaging. *eXPRESS Polym. Letters* 8: 791-808

[6] Tortajada M. et al. (2013). Second-generation functioanlized médium-chain-length polyhydroxyalkanoates: the Gateway to high-value bioplastic applications. *Int. Microbiol.* 16: 1-15.

reactive groups enable tuning of the physical and chemical properties of the polymer, and they are also potential targets for post-biosynthetic modifications. For example, the higher the molar fraction of unsaturated constituents in the monomers, the lower the resulting melting and glass transition temperature. Double bonds are easily attacked in chemical reactions, allowing polymer properties to be diversified even more. Chemical epoxydation of the pendant vinyl groups has also been applied to decrease melting temperature an increase glass transition temperature. Currently, the improvement of PHAs properties for packaging is obtained by blending them with other polymers or plasticizers[2]; or by means of nanotechnology[7]. In this project modifications will be achieved microbially and in some cases with combinations of chemical reactions. In order to increase market feasibility, low cost renewable feedstock such as starch rich residues from DACSA production process will be employed.

Beyond of the state of the art, **RefuCoat** will focus on the production of a range of mcl-PHA molecules with improved barrier properties, at the same time attending to high processability performance.

1.4.2 State of the art and progresses in the field of production of glycolic acid (GA) by catalyst technology.

Glycolic acid currently is produced in industrial scale up to 40000 t/a mainly by carbonylation of formaldehyde or hydrolysis of chloroacetic acid[8]. Minor production pathways involve the hydrolysis of the corresponding nitrile or the electrochemical reduction of oxalic acid[9]. All these production pathways use fossil raw materials. Microbial production from renewables such as glucose or xylose using different GMOs have been studied intensively in the recent years but due the low yields (< 0.4 g/g) and titers (< 60 g/L), biotechnological processes are not competitive to replace the current petrol based technologies[10-11]. GA may also be produced from glycerol by chemocatalytic oxidation with oxygen using supported gold-based catalysts. In this reaction GA is one of several by-products which is formed in only moderate selectivity below 50 % whereas glyceric acid is the main product[12].

Gold-based catalysts may also be used for the selective oxidation of mono ethylene glycol (MEG) to GA. Surprisingly, only very few studies are available for this reaction[13] and all of them target primarily the catalyst characterization. However, selectivity of > 90 % described in these studies is very promising. Only one study for the production of methyl glycolate, a derivative of GA, is of industrial relevance[14-15]. So far, no study targeted the development of an efficient process for GA production under industrial relevant conditions, which will be done for the first time in this project according to our prior experience in related selective oxidation reactions[16-17].

Bio-based mono ethylene glycol (Bio-MEG) is produced mainly from bio-ethanol via bio-ethylene or in minor amounts from sorbitol via hydrogenolysis. Production of Bio-MEG has shifted from the United States and Central and Eastern Europe to the Middle East, China, India and Canada, with a production capacity of more than 500 kt/year. MEG has a huge number of industrial applications such as polyesters, antifreeze, paints, etc. The bio-MEG from bio-ethanol is used mainly for Coca Cola's Plant Bottle. The production costs are low (below 1.5 €/kg) due to the large production facilities[18].

Beyond of the state of the art, RefuCoat will focus on the development of an efficient, green and robust chemocatalytic process using gold-based catalysts for the production of glycolic acid from bio-sourced mono ethylene glycol using industrially relevant conditions in larger lab/small pilot scale.

[7] Khosravi et al. (2015). Applications of PHA in food packaging: improvements by nanotechnology. *Chem. Biochem. Eng. Q.* 29: 275-285.
[8] Miltenberger K. (2012) Hydroxycarboxylic acids, aliphatic. In: Ullmann's Encyclopedia of Industrial Chemistry, Wiley-VCH, Weinheim
[9] Panova A et al. (2007) Chemoenzymatic synthesis of glycolic acid. *Adv. Synth. Catal*. 349: 1462-1474.
[10] Becker J. et al. (2015) Top value platform chemicals: bio-based production of organic acids. *Curr. Opin. Biotechnol*. 36: 168-175.
[11] Dichert W. et al. (2012) Improved glycolic acid fermentatative production with a modified microorganism. WO2012/025780.
[12] Bianchi CL et al. (2005) Selective oxidation of glycerol with oxygen using mono and bimetallic catalyst... Catal. Today 102-103: 203-212.
[13] Griffin MB et al. (2013) The selective oxidation of ethylene glycol and 1,2-propanediol on Au, Pd and Au-Pd …. *J. Catal.* 307: 111-120.
[14] Hayashi T, Inagaki T. (2004) Catalyst for carboxylic ester production and process for producing carboxylic esters. EP 1393800.
[15] Hayashi T et al. (2006) Selective oxydation of alcohol over supported gold catalysts: methyl glycolate formation from ethylene glycol and methanol. *Catal Today* 117: 210-213.
[16] Heidkamp K. et al.(2013) Ceria supported gold-platinum catalysts for the oxidation of alkyl ethoxylates. *Catal. Sci. Technol*. 3: 2984-92.
[17] Prüße U et al.(2011) Gold-catalyzed selective oxidation of glucose in concentrated solutions at elevated .. *Appl. Catal. A: Gen.* 406: 89-93.
[18] E4tech, RE-CORD and WUR (2015) "From the Sugar Platform to biofuels and biochemicals". Final report for the European Commission, contract No. ENER/C2/423-2012/SI2.673791

1.4.3 State of the art and progresses in the field of production of polyglycolic acid by reactive extrusion.

Polyglycolic acid (PGA) resin has very low permeability to O_2 and CO_2 –one hundredth that of PET (as figure 1.4.b shows). Additionally, PGA barrier properties are largely insensitive to humidity conditions, making it ideally suited for a variety of beverage and perishable food packaging applications. PGA shows similar biodegradation to cellulose and it degrades into CO_2 and water in compost within one month. Resultant compost has been proven safe. PGA is fully compatible with widely practiced industrial PET or other polyesters recycling processes, ensuring the quality of recycled polyesters.

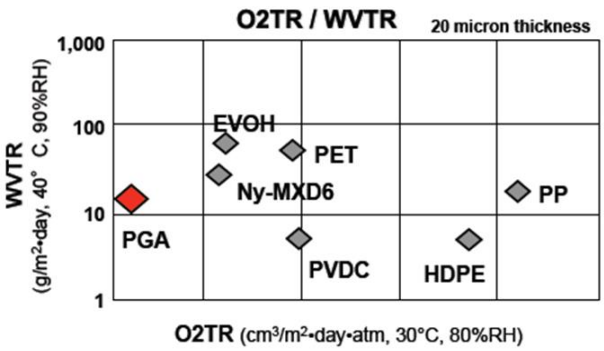


Figure 1.4.b. Barrier properties of PLA films.

Polyglycolic acid (PGA) production has been described in several patents.[19] Due to production hurdles and its use in mostly medical application, produced amounts remained low and high of price.[20] In 2014, Kureha Co. reported a process to mass-produce high-molecular-weight PGA, including the production of large yields of the intermediate glycolide (GL) with high levels of purity with a following ring opening polymerisation in bulk. The simplest method for synthesizing PGA is through condensation of glycolic acid using a dehydrating reaction. However, as there is an equilibrium between GL generation and the chain extension for the hydroxyl termination of PGA, sufficiently high molecular weights cannot be achieved using this method. Conventional manufacturing method for GL entails melting a glycolic acid oligomer (GAO) by heating and then recovering, through evaporation, GL as a depolymerization product from the surface of the melted GAO. However, this method is unpractical because: (1) melted GAO is unstable and highly viscous. Because heat transfer efficiency is difficult to improve, wall surface temperature must be increased, promoting adverse reactions, such as tar formation. (2) Evaporated GL deposits on the inside walls of distillation lines, promptly polymerizing, distillation lines easily become obstructed.[21]

Nevertheless, for the application described rather short chain/oligomeric substances are needed, and the project will overcome the above described problems by optimising economic and ecologic viable processes and process combinations for the polymerisation of glycolic acid with characteristics optimised for coating applications.

Beyond of the state of the art, **RefuCoat** will focus on the investigation of different promising production technologies for PGA materials: (i) (Semi) continuous reactor based polymerisation will be evaluated for *maximum control* of the resulting material properties and *easy state of art further material utilisation*, (ii) Reactive extrusion polymerisation will be evaluated for *highly efficient, solvent free continuous polymerisation* and (iii) Finally the combination of both will be evaluated, realizing initial pre-polymerisation completed by reactive extrusion, enabling the opportunity for *further material modification in continuous, solvent free processes*, coming up with *highly intensified processes minimizing the needed production steps* and such enabling new business opportunities with high economic and ecologic impact.

1.4.4 State of the art and progresses in the field of production of hybrid coatings.

Hybrid polymer coatings are a class of transparent organic-inorganic coating material that can obtain high barrier properties with respect to the permeation rates of oxygen, water vapour and volatile organic compounds that can be applied to bio-based polymers or oil based polymers by standard coating technologies.

Atomic layer deposition (ALD) and other detailed in section 1.4.6 are a different approach for the production of high barrier coatings. It is a chemical gas phase thin film deposition method bases on alternate saturate surface reactions, which a result in a unique self-limiting film growth mechanism with a number of advantageous features, such as excellent conformity and uniformity, and simple and accurate film thickness control[22]. ALD is not used commercially in the packaging market. Furthermore, when it is compare to sol-gel technique for coating deposition do not come out on top due to the need of a high cost and special equipment to take places and high temperatures process that goes against films degradation. For all considered, the need for hybrid coating goes through thinner coating that improve barrier properties and higher and conventional line speed production.

[19] Lowe, C. E.: "Preparation of high molecular weight polyhydroxyacetic ester", U.S. Pat 2 668 162, 1954

[20] K- M. Stridsberg et al. (2002) “Controlled Ring-Opening Polymerization: Polymers with designed Macromolecular Architecture”; Degradable Aliphatic Polyesters.157 Advances in Polymer Science, 41-65.

[21] Kazuyuki Yamane et al.; “Development of an industrial production technology for high-molecular-weight polyglycolic acid”; Polymer Journal (2014) 46, 769–775; doi:10.1038/pj.2014.69; published online 6 August 2014

[22] M. Ritala et al. (2002).Handbook of Thin films Materials, edited by H.S. Nalwa; Volume1: Deposition and Processing of thin films

Sol-gel approach has many advantages for the development of hybrid coatings due to the mild and low temperature conditions. Compatible with organic reactions, it allows associating organic and inorganic species into a single material/layer to obtain a hybrid coating with unexpected opportunities for new innovative surfaces. Sol-gel coatings are organically modified hybrid materials formed from a precursor colloidal solution (sol). The typical process involves the hydrolysis of alkoxysilanes to produce hydroxyl groups, followed by polycondensation among these groups and residual alkoxy groups to form a 3D polymeric network (Fig 1.4.a). Sol-gel coatings have been applied onto many substrates, such as paper, cellulose film, pullulan, PET, *etc*…

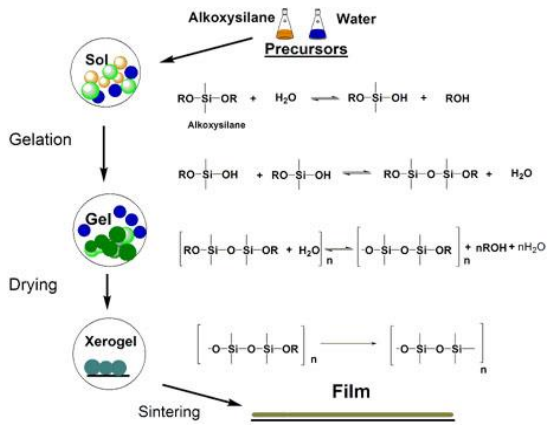


Fig1.4.a Sol-gel process.

Hybrid coating functionalized PLA and PP showed good results for barrier properties[23,24]. It has been shown that the oxygen, nitrogen and carbon dioxide transfer rates of barrier polymers such as PET and oriented PP can be reduced by up to two orders of magnitude, with a thin coating (1–2 µm) of PVOH/Si-SiO_2. More notably, it has been observed that the material maintains this feature even after a prolonged contact with water, which is a strong solvent for PVOH. Direct moisture sorption measurements show that silica lowers water uptake of PVOH and inhibits swelling and plasticization of the polymer[25]. However, for a high barrier packaging solution hybrid polymer coating can be used as a laminating agent in combination with SiO_x vapor deposition polymer. Due to the good adhesion showed to SiOx coating, the structures show excellent adhesion and barrier properties even under high mechanical and thermal stress and storage in humidity. Macroscopic defects of the SiO_x vapor deposition polymer are compensated by the hybrid coating, furthermore the SiO_x and the hybrid polymer form covalent bonding when they meet Si-O-Si. This leads to an area of high inorganic network density, with additional barrier effect.

In conclusion, the combination of organic polymer and inorganic material in a single phase provides exceptional possibilities to tailor properties for different applications. There are a wide range of companies focusing in sol-gel technology for the of production functional coating based on sol-gel approach for different applications.

Fraunhofer ISC recently has developed a bioORMOCER® a that significantly improve the barrier performance of conventional bio-packaging systems using a biobased type of inorganic-organic hybrid polymers which is both biodegradable and compostable[26]. However, they are based on Chitosan and polycaprolactone (PCL), both polymers have lower barrier properties, PCL is not biobased and low flexibility than Polyglycolic Acid (PGA).

PGA resin has very low permeability to O_2 and CO_2 –one hundredth that of PET (as figure 1.4.b shows). Additionally, PGA barrier properties are largely insensitive to humidity conditions, making it ideally suited for beverage and perishable food packaging applications. PGA shows similar biodegradation to cellulose and it degrades into CO_2 and water in compost within one month. Resultant compost has been proven safe. PGA is fully compatible with widely practiced industrial PET or other polyesters recycling processes, ensuring the quality of recycled polyesters.

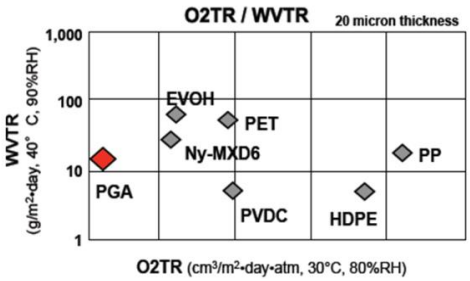


Figure 1.4.b. Barrier properties of PLA films.

Beyond of the state of the art, **RefuCoat** will focus on the development of new hybrid biobased type of inorganic-organic hybrid coating PGA/micro size Si-SiO_2. This hybrid coating will be applied using conventional coating technology, to obtain high barrier packaging structures based on Bio-PET, Bio-PE and PHA. Hybrid coating will be flexible to avoid microcracks during its manipulation, semi-transparent, thermo-conformable and the coating and its components must fulfil the EU regulations related to plastic in contact with food.

[23] Gree Bang et al. (2012); Biodegradable poly(lactic acid)-based hybrid coating materials for food packaging films with gas barrier properties. J. Industrial & Engineering Chemistry 18, 1063–1068

[24] K.H. Hass et al. (1999) Functionalized coating based on inorganic-organic polymers and their combination with vapor deposited inorganic thin films. Surface and Coating technology 11172-79

[25] M. Minelli et al. (2009), Barrier Properties of Organic-Inorganic Hybrid Coating Based on Polyvinyl with Improved wáter resistance. Published online in Wiley InterScience,. DOI 10.1002/pen.21440

[26] " Bioormocer", www.isc.Fraunhofer.de/focal-areas/barriereschichten/research-and-development/biodegradable-functional-coatings/?L=1

1.4.5 State of the art and progresses in the field of production of active coatings.

Active materials and articles refer to materials and articles that are intended to extend the shelf life or to maintain or improve the condition of packaged food; they are designed to deliberately incorporate components that would release or absorb substances into or from the packaged food or the environment surrounding the food[27]. Antioxidant (AO) and antimicrobial (AM) are among the most studied AP.

Furthermore, biopolymer-based packaging often fails to provide equivalent protection to food vs. their synthetic counterparts and can even be sensitive to microbiological spoilage themselves, so their enhancement via active substances can help making them more competitive. For example, IRIS developed AP with PHA substrates among others[28] using polyphenols such as hydroxytyrosols contained in olive oil waste water leading to significant delays in cheese and meat lipoxidation and leading to an increased OTR by a factor 2.5.

AP can provide higher safety and quality products and are receiving increasing attention from the food industry due to consumer demand for minimally processed and more natural food products. Active substances considered for novel bioactive packaging include antimicrobials, vitamins, phytochemicals, prebiotics, marine oils, and immobilized enzymes.[29] The use of natural antioxidants in such AP as opposed to the synthetic ones for food preservation like butylated hydroxytoluene (BHT), is also relevant. In view of the food products targeted in the project, essential oils from hears appear suited options both for snacks and chicken. Indeed, the antioxidant power of edible films was also reported to be significantly improved by adding oregano and rosemary extracts, rich in caffeic and chlorogenic acids among others[30]. A wide range of active compounds with antimicrobial properties are also reported, including silver-substituted zeolite, organic acids and their salts, bacteriocins such as nisin and pediocin, enzymes such as lysozyme, a chelator like ethylenediaminetetraacetic acid (EDTA), lactoferrin, and plant extracts have already been successfully incorporated in antimicrobial active packaging[31]. In addition, carvacrol has been suggested as one of the most promising antimicrobial agents, and it has been combined with different polymers such as PLA.

Specific consideration on active packaging for snacks

Dry foods such as dehydrated snacks, are very sensitive to oxidation. The addition of nonpolar antioxidants is effective, while polar antioxidants may lose their activity by passage into the aqueous phase, similarly to fat emulsions. Proteins and many other food components have protective action and act as synergists of other inhibitors, enhancing the effect of antioxidants. Snacks has long storage life at ambient temperatures.

Application dependent considerations

The efficiency of the active ingredients in food products depends very much on the water content. Because of the complexity of foods, it is necessary to test any combination of antioxidants/antimicrobial to stabilize the particular material and to optimize the mixture for a given packaging configuration (e.g. depending on its direct contact with the food or activity as volatile compound). In fact, also the mode of incorporation of the additive with the polymer matrix highly influences the release rate of the active substances. Compounding is the most described approach to process AP, but it may lead to thermally induced loss of activity as opposed to the superficial coating of the packaging with a solution of an active additive as targeted here. Nevertheless, the layer approach needs to use suitable binder or control layer[32] to allow, especially when used with wet food, progressive release at the required kinetics as opposed to its instantaneous overall release.

With a view to further improving performance of RefuCoat solutions, active packaging (AP) will be developed. Beyond of the state of the art, **RefuCoat** will deliver hybrid coating with barrier and active properties that will be tailored to the application on different substrates and validated for different food products (but have potential for many more polymers and food combinations). Besides, the use of natural additives such as essential oils as opposed to synthetic ones raising more and more safety concerns from consumers, will increment the innovation potential.

Specific consideration on active packaging for chicken

Chicken has a quick spoilage mechanism even at refrigerated storage due to mesophilic and psychrotropic bacterial

[27] EU REGULATION (EC) No 450/2009 of 29 May 2009 on active and intelligent materials and articles intended to come into contact with food, 30.5.2009

[28] FP7 project Oli-PHA (GA 280604-2).

[29] Lopez-Rubio A, et al. 2004. Overview of active polymer-based packaging technologies for food applications. Food Rev Int 20:357–87.

[30] Gómez-Estaca, J. et al. Food Chemistry 112(1): 18-25.

[31] Joerger RD. 2007. Antimicrobial films for food applications: a quantitative analysis of their effectiveness. Packag Technol Sci 20:231–73

[32] Ozdemir, M et al. (2004). "Active Food Packaging Technologies." Critical Reviews in Food Science & Nutrition 44(3): 185-193.

growth. As such, a lot of prior art describe the development of AP solution for poultry and AP is definitely in the trend of solutions that will ensure increase food safety, shelf life and packaging friendliness of fresh meat in general and more especially chicken[33].[34]. Food safety, in this case, tackles microorganisms as Salmonella, Campylobacter and E. coli. Not only poultry spoilages by microoganisms but also by oxidation reactions (colour and fats). For example, thyme oil (0.2 to 0.3%), rich in thymol, carvacrol and p-cymene, was applied on the surface of vacuum packed chicken liver leading to an improved preservation[35].

Currently, most of the antimicrobial active packaging materials are broad spectrum antimicrobials which do not target bacterial pathogenic species specifically. There is a significant need to develop novel antimicrobial packaging materials that have high specificity to target only pathogenic organisms while maintaining commensal bacteria. Thus, developng pathogen specific antimicrobial active packaging materials may improve the antimicrobial efficacy by reducing interactions with non-targeted microbes[36].

The use of bacteriophages to control pathogens is promising and is becoming a reality. Although the practice of primary bacteriophage therapy has been performed with a view to the treatment of bacterial infections in humans, the concept of removing undesirable bacterial populations using bacteriophages can be extended to foodstuffs. However, its incorporation in packaging is still unknown and it is necessary to better understand and evaluate its limitations: encapsulation, stabilization, release and effectiveness as antimicrobials[37]. The references tackling the use of bacteriophages and related substances with them as antimicrobials in active packaging are very scarce.

Related to bacteriophages are endolysins. They are the enzymes used by bacteriophages to degrade the cell wall structure of the bacterial host, resulting in cell lysis[38]. The use of endolysins is an alternative to the use of whole bacteriophages, with advantages on its survival.

Beyond of the state of the art, **RefuCoat** will target new antimicrobials focused on bacteriophages or their endolysins, that are specific against the pathogen microorganisms responsible for food safety (*Salmonella* and *Campylobacter*). Hardly, there are publications that use bacteriophages (endolysins) and they both are "*in vitro*" tests, with no food. **RefuCoat** will complement this literature lacks with "*in vivo*" tests, studying the effect of bacteriophages (endolysins) on poultry products.

[33] Z.F. Bhat et al. (2011), Recent Trends in Poultry Packaging: A Review. American Journal of Food Technology, 6: 531-540

[34] Z.F. Bhat et al. (2011), Recent Trends in Poultry Packaging: A Review. American Journal of Food Technology, 6: 531-540

[35] S.Papazoglou et al.. (2012). "Effect of Thyme Oil on the Preservation of Vacuum-Packaged Chicken Liver." J.Food Science 77, M473-80.

[36] E.Vonasek et al. (2014): Encapsulation of bacteriophages in whey protein films for extended storage and release, Food Hydrocol. 37, 7-13.

[37] D. M. Gouvea et al. (2015), Acetate cellulose film with bacteriophages for potential antimicrobial use in food pack, Food Sci. 63, 85-91.

[38] M.Schmelcher et. al, (2012): Bacteriophage endolysins as novel antimicrobials. Future Microbiol., 7(10): 1147–1171.

1.4.6 Main competitors.

Table 1.4.a shows a list of the main competitors for the different results that have been identified worldwide, identifying those features (at the different levels: technological, obtained results, state of development, prices, market niche, etc.) that RefuCoat results will have and comparing them with the ones that main competitors have.

Table 1.4.a Main competitors of packages and technologies developed in RefuCoat.

Result	Current competitor	Companies	Advantages	Main limitations of the commercial solution.	RefuCoat Advantages
Transparent high gas barrier coating.	Ceramis® silicon oxide coated films.	Amcor, Oike (JP), Celplas & Toyobo.	Low material costs, No yellow tint, High productivity, Good barrier properties, low thickness	High investment machine, only available in large few industries, expensive, fragile easy to form microcracks and no formable.	Low investment, technol. available in a large & SMEs companies, flexible coating (no cracks and thermoformed).
	Re-Active Evaporation of Aluminium.	Amcor (Camvac)	Low material costs, No yellow tint, High productivity and low investment	Process difficult to control, Layer sensitive to elongation, medium barrier properties & no formable.	
	Plasma SiO_2 CVD.	Tetra Pak & Dai Nippon	Low material costs and No yellow tint.	High investment, low productivity and no formable.	
GA	Parchem, DuPont, CrossChem LP, Phibro Animal Health Corporation and Chemsolv Inc.		Good quality and high purity	Use mainly in the medical and cosmetic sector	Low cost, high quality, Renewable source
PGA		Kaneka	High MW, good properties, market availability.	High cost, petrol origin	
PHAs	Meredian Holdins Group, Bio-ON, Metabolix		Market availability	Only short chain PHA available High cost, pure sugars as carbon source	Tailor-made mcl-PHA. Low cost, renewable carbon sources.
Anti-microbial	A wide range of commercially available AP solutions	Cryovac's	Antimicrobial specificity against pathogen microorganisms	None of them appear to be based on natural biobased substances having a joint anti-oxidant and antimicrobial activity and a special focus on the food stuff selected in the project. None the less meat, and poultry, are the object of more solutions at pre-competitive stage and is expected to indeed help savings currently high wastage in this food class[39] than dehydrated products but not always for similar purposes as in the project.	Improve in Food Safety
Anti-oxidant (AO)	Some films with AO & O_2 scavenging properties: based on Iron or light activated scavenger	Ciba, Cryovac Sealed Air Co	Market availability		Bio-sourced, natural and with more health / nutritional benefits for the consumer from selected essential oils, which also play an antimicrobial effect

[39] www.itene.com/en/press/press-releases/i/3934/60/active-and-intelligent-packaging-a-top-priority-in-fight-to-save-food-waste

2. Impact

2.1 Expected impacts

The RefuCoat project is fully in line with the European BBI Call "H2020-BBI-R05. 2016- Advanced biomaterials for smart food packaging" and with the wider objectives, guidelines and policies of several leading initiatives (e.g. the European Bioeconomy Alliance; EBA, RIS3, the bio-based cluster, and the European bioeconomy in 2030). The ultimate foreseen impacts of the proposed technologies are:

Table 2.1.a Contribution to the expected impacts defined in the *BBI R05:2016* topic text.

Impact defined in the topic BBI2016.R5
Improved mechanical and/or functional properties of the developed packaging products in a specific field as compared with state-of-the-art.
The main outcome of the project will be bio-based food packaging solutions with increased oxygen/water barrier, as alternative to current metallised and multilayer modified atmosphere packages. Both types of packages are made with non-renewable sources and they cannot be recyclable to obtain other high quality packages. For this, a new class of transparent organic-inorganic composite materials will be prepared with the sol-gel methodology, through the incorporation of oligomeric or polymeric molecules of PGA into a solution formed by a precursor of the organic phase. Al components used will be suitable for food contact applications.
In case the project focuses on biodegradability or composability: contribution to fulfilling the European objective of reducing the recyclable content in landfilled waste down to the targets as per the EC's Landfill Directive, by showing reduction in landfilling as end-of-life option for packaging material as compared to baseline. Biodegradability other than under industrial conditions should also be considered.
Some of the bio-based materials developed in the RefuCoat project will be biodegradable, i.e. medium chain PHA grades and organic-inorganic coatings based on PGA. For poultry products, the projected package is fully biodegradable; whereas for cereals, snacks, and savoury products, the packages will be fully recyclable, but non-biodegradable. PHA and PGA have high biodegradation rate than other biodegradable materials[40] such as PLA, PBAT, PCL… in non-industrial conditions such as sea water or home compost
Reduction of costs, on a life cycle analysis basis, as compared to on-market materials.
WP6 devoted to the evaluation of environmental, socioeconomic, societal and safety aspects for the processes and products developed in RefuCoat. To calculate the cost, the full process will be taken into account, from production until the cost of management after the use. The economic analysis of the processes developed throughout the project aims to determine the total costs through different tools and methodologies such as Life Cycle Cost Analysis (LCA) as well as evaluation of the economic feasibility of bringing the RefuCoat products/process to the market by means of techno-economic assessment (TEA). RefuCoat aims to use medium cost materials and high output process in order to have a final package easy to recycle or full biodegradable, reducing significantly the management cost of current metallised materials at post-industrial level (scrap in the package production) and post-consume. Based on the current cost of metalized packages and their difficulties to be managed at the end of their life, we expected a competitive cost package, with reduction of some ct€ by m2, based on to have a PGA at cost below 10€/kg as section 2.1.2 detailed. However, we are in a TRL 3, and to have an accurate cost is very complex. In the case of PHA, the expected cost of mcl-PHA will be times higher than PET. BIOPOLIS has a broad expertise in this field and we expected a significant cost reduction of the cost by the use of food by-products (from DACSA) and the use of high yield strain capable to join lipid groups to the PHA backbone.
In case the project focuses on the preservation of food products: Increase of at least 10% of the shelf life of the targeted products.
A target of the project is to increase at least 15% of the shelf life of the products with the incorporation of antimicrobial and antioxidant substances to the hybrid coating.
Opening new markets for new applications of biodegradable materials.

[40] PLA polylactic acid, PBAT polybutyrate adipate terephthalate, PCL polycaprolactone.

The proposal aims to look for real alternatives to metallised films, these are not recyclables or energy recovery is limited due to the metall. Metallized films have a higher cost than single monolayer packages where biodegradable polymers are used nowadays. Nowadays, there are not a real alternative to metallized films using bio-based materials. Some companies are producing PLA metallized or PLA with a SiOx coating (Ceramis®), however both are expensive and metallized PLA is not compostable or recyclable.

Contribution to the BBI JU Key Performance Indicators (KPI), specifically: ◦Creation of at least 1 new cross-sectorial interconnection in bio-based economy clusters (KPI 1): Set the basis for the creation of at least 1 new bio-based value chain based on the innovative products developed (KPI 2), new bio-based materials (KPI 5), and for new demonstrated consumer products based on bio-based chemicals and materials (KPI 6), or significantly improve the production process of existing ones.

KPI 1 When companies from different sectors start to cooperate on building new value chains, while these companies have not been active in the same value chain before / did not cooperate in business before (or did cooperate but in a totally different field).
KPI 2: New innovative products or existing products that have not been produced from the bio-based feedstock before. Nowadays, there is not in the market real alternatives to metallized films based on bio-based materials due to the limited gas barrier properties of many biopolyesters such as PLA or PBS. PGA and PHA are the biopolyesters with the best gas barrier properties. The combination of them, bio-PET and Bio-PE and inorganic particles will lead to a real alternative to metallized films. The processability and recyclability of the film will be ensured due to PGA is a thermoplastic material and compatible with polyesters and easy to modify to be compatible with polyolefins. GRUPO APEX and DACSA are looking for an alternative to their current metalized films. MIPLAST can direct supply the films to these companies or licencing the technology to other plastic coating companies. The equipment required to produce the coating (twin screw extruder) and the coated film (flexography, laminating or rotogravure technology). A business plan will be prepared in WP7.
KPI 5 New bio-based materials developed.
As table 2.1.d and section 2.1.4, the use of biobased products lead to better environmental sustainable packages, as the target packages of RefuCoat project. Based on the available literature of individual materials, the potential reduction in LCA terms such as Global Warming Potential (GWP in kg CO2-Eq.) and Energy Demand (MJ) per kilo of produced material will be between 21 to 133%.

Packages for	Current solution	Potential reduction using RefuCoat concept.			
		RefuCoat proposed solution	Rene-wable (%)	GWP (% less).	Energy Demand (% less).
Cereals products	Metallised polyolefin (0.5% Al /99.5% PE)	Bio-PE +3% of PGA Hybrid coating	100%	133%	64%
Savoury products	Metallised Polyester (0.5% Al /99.5% PET)	Bio-PET + 3% of PGA Hybrid coating	35%	54%	21%
Poultry products	PET/PE tray (94%/6%)	PHA + 3% of PGA Hybrid coating	100%	52%	28%

The market demand is full justified in section 2.1.4, a huge number of square meters of metallized films (3000-4000 million of m2) are produced each year in Europe, with a potential of use up to 15,000 million of m2, if aluminium foils is replaced by metallized ones.
In the case of meat, if we only consider poultry meat, the amount of packages used every year in Europe is 8 Million, based on the assumption that only 30% of the poultry meat is sold in plastic packages. Other meat (pork, beef, lamb...) and fish use similar packages, but change the type and amount of each gas used in the modified atmosphere.
KPI 6 New bio-based products and applications developed.
RefuCoat project aims to produce Glycolic acid from bioethanol and mcl-PHA from food by-products, as we commented both have a huge potential in the food packaging due to their good barrier properties alone or in combination with organic particles, thermoplastic behaviour and compatible with many biopolyesters (recyclability) and full compostable including in marine environments.

Hence, the project will aim to underpin key progress on these promising objectives together with associated knowledge of the underlying mechanisms to facilitate the transition towards a bio-based economy in the EU and

beyond. The expected impacts of the project with respect to the work programme are detailed in the following sections.

2.1.1 Unlocking the value of innovative processing technologies.

Developing new full recyclable solutions with improved barrier properties from renewable origin creates an important competitiveness opportunity for Europe as described in the European Commission strategy for 'innovating for Sustainable Growth: a bioeconomy for Europe"[41]. Currently, metallised and modified atmosphere packages are based on the use of non-renewable materials in multilayer structures that lead to complex and expensive recycling steps.

Global high barrier packaging film consumption was approximately 1.76 million tonnes in 2014, with a value of €12 billion. Moreover, the percentage of chemical production based on biotechnology is estimated to increase from less than 2% in 2005 to approximately a quarter of all chemical production[42] by 2025. In EU28 2013, the chemical and plastics represented a turnover of **42 billion €** and 200,000 employees[43]. Initiatives such as RefuCoat are an enabler for innovation in countless other sectors that can transform renewable feedstocks into sustainably raw materials as inputs for countless discreet manufacturing and process industries in Europe and internationally.

To increase the innovation capacity of the industry, our approach is to capitalise on a valuable opportunity for the biotech companies, chemicals, plastics and food industries to work together in order to substitute currently used metallised and modified atmosphere packages (MAP), with hybrid bio-based materials and active coatings to be used in monolayer and full recyclable packages.

The innovation process in only complete when new innovative knowledge is put to industrial use. The innovative potential of RefuCoat is enhanced by the **strong industrial composition of the RefuCoat consortium (7 partners out of 12; 3 of them SMEs and 4 large companies, convering the entire value chain**). Industrial partners are integrated in the project decision making structure to ensure that the involved industrial sectors drive project activities and benefit from the project results. To ensure success, industrial partners will lead all the technological WPs and enable the steering of project activities from the perspective of the whole supply and value chain. RefuCoat's success will be mainly based on the strong partnership of SMEs/industry and research organizations, all highly skilled in complementary fields (Figure 2.1.a).

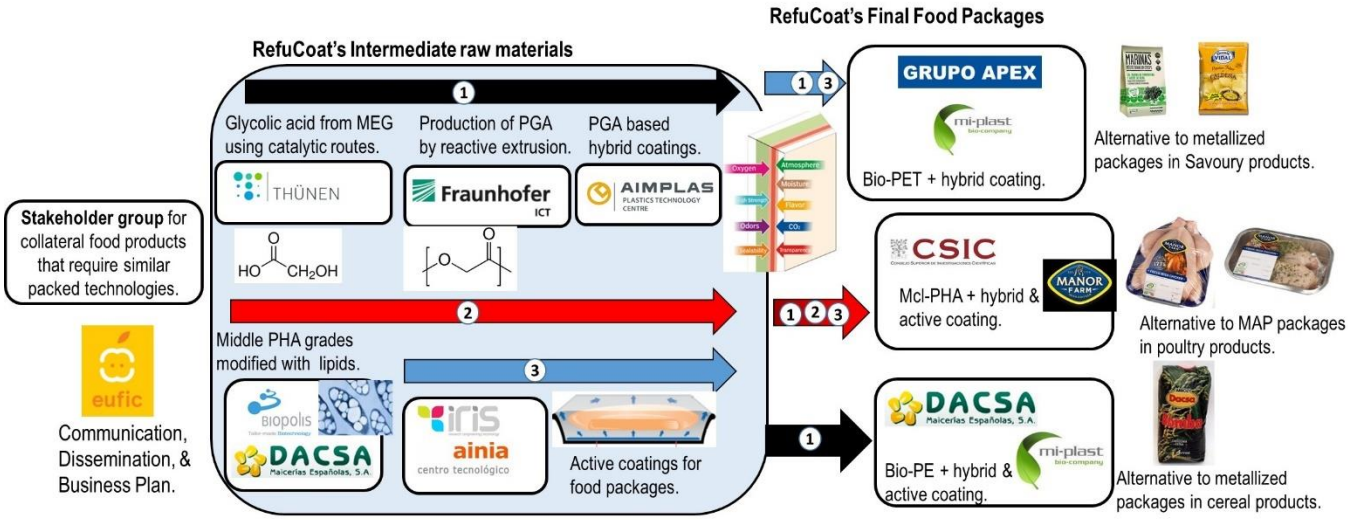


Figure 2.1.a. Structure and relation between partners in the consortium.

Three main routes are defined in RefuCoat project: In line①, THUNEN will be in charge of production of glycolic acid (GA) by conversion of monethylene glycol using novelty catalytic routes; Fraunhofer will produce low melting temperature PGA using the GA produced by THUNEN based on this product, whereas AIMPLAS and MI-PLAST will investigate the production and application of high barrier hybrid coatings. In line ② CSIC will be in charge of producing new mcl-PHA with improved water barrier properties with the support of DACSA as provider of renewable feedstocks; with this material together with high hybrid coating from line①, and active coatings develop by AINIA (antimicrobial) and IRIS (antioxidant) in line ③, AIMPLAS will develop barrier and active coatings. In The validation of mcl-PHA material will be performed by MANOR in fresh chicken meat, GRUPO APEX will

[41] http://ec.europa.eu/research/bioeconomy/pdf/official-strategy_en.pdf

[42] Smeets, E. et al. (2014) Evaluating the macroeconomic impacts of bio-based applications in the EU.

[43] http://biconsortium.eu/sites/biconsortium.eu/files/news-image/16-03-02-Bioeconomy-in-figures.pdf

validate the materials based on Bio-PET with coating with enhanced barrier and antioxidant properties and DACSA and MIPLAST will validate the materials based on Bio-PE coating with enhanced barrier and active properties.
Synergies among partners with expertise in complementary and cascaded processes, technologies and products will act as a booster for innovation placing better and more sustainable products on the market. The participation of large end-users with high market penetration guarantees high impact and the wide dissemination & exploitation of the project results at worldwide level.

2.1.2 Economic impact: strengthening the competitiveness and growth of companies.

The RefuCoat Project will take the promising technologies in development from the partners involved in this consortium formed by a research-industry complementation in order to provide an increase in the Technological Readiness Level toward a pre-commercial development. It will enable the European industries to better bridge the 'innovation gap' and the "valley of death" between technology development and commercialisation. Both very pertinently apply to the field of biorefineries which require significant investments, typically beyond the financial reach of individual private companies. Public intervention is required to foster industry leadership and to promote long term industry commitment in research and innovation related to bio-based industries.

Major drivers that can affect the demand of high barrier and functional packaging in future worldwide are increasing population, policies of different countries that support or promote use of renewable and recyclable materials, increasing investments by companies in search of a bio-based product or even the increasing efforts by countries reduce the use of fossil resources. Many companies are considering bio-based technologies for the production of chemicals with hopes of being able to use cheap sugars as a feedstock, thereby eliminating the need for fossil fuel-based feedstocks.

High economic impact will be achieved through strengthening the competitiveness of the Industrial Partners in the packaging sector and related ones (biopolymers, coatings, and antimicrobial/antioxidant) by: 1) promoting the use of RefuCoat materials, components and systems in high value-added applications while keeping costs competitive, 2) strengthening and differentiating the EU Industry, 3) increasing workers' skills. ***Detailed information of each market is presented in the follow points:***

High barrier food packaging market.

A key growth driver of high barrier packaging films is the need for extended shelf life of many products. Consumers are demanding those food products that are processed as little as possible and artificial additives and preservatives are not used for preserving the food quality. High barrier packaging films have unique properties like chemical properties, heat resistance, gas and aroma barrier, high tensile strength, puncture resistance, durability, special surface properties, and lighter packaging. Europe leads the global high barrier packaging films market in terms of both demand and production followed by Asia and South America[44].

Technologies and materials include multi-layer film, metallised films, biaxially oriented polyethylene terephthalate (BOPET), polyamide (BOPA) and polypropylene (BOPP or CPP), high oxygen barrier materials (ethylene vinyl alcohol [EVOH], polyacrylonitrile, polyvinyidene chloride [PVdC], polyvinyl alcohol [PVOH] and cyclic olefin copolymer), high barrier coatings, organic liquid coatings and inorganic vapour deposition high barrier coatings such as silicon oxide (SiOx), and aluminium oxide (AlOx).

Products such as bags and pouches, stand-up pouches, tray lidding film, forming webs, wrapping film, and blister pack base webs segment the global high barrier packaging films market. Applications market segment covers fresh food, retort convenience food, dried food, processed chilled food, snack foods and confectionery, cheese, baked goods, drinks, pharmaceutical and medical, personal care and cosmetics and other.

In the case of metallised films, its global market was 240 kt/y in 2011. It is rising at 5% CAGR to 2016 vs. 20 kt/yr of oxide-coated at 7% CAGR to 2016 vs. 550 kt/y of Al foil at 2.0% CAGR to 2016

Savoury dry products and meat consumption.

According to GRUPO APEX, savoury snacks are enjoyed by consumers of all ages across Europe being consumed at various eating occasions. In 2014 the retail value of the savoury snacks market in Europe amounted to around €14 billion with nearly 1,850,000 tonnes of savoury snack products being purchased. Consumption of savoury snacks across Europe varies from country to country but on average around 3.6kg are purchased per year per capita[45].

Other products where metallised films are employed are sausages, ice-creams, pasta, cheese, instant drink powder, milk powder, health drinks, biscuits and other bakery products, etc. and non-food applications such as personal care. Based on the current use of metallized films (124 kt in 2014) and the increase of use in the next years as aluminium foil alternative, the consumption of metallized films can be calculated in more than 5,000 m^2/year.

[44] www.decisiondatabases.com/ip/239-high-barrier-packaging-films-market-report#sthash.fVOy5hdn.dpuf.
[45] www.esasnacks.eu/europe-statistics.php.

Figure 2.1.b Examples of food and non-food products packed in metallised films.

The figures for the consumption of poultry meat in the EU also illustrate that the preference for poultry meat differs in the EU[46]. While on average a European citizen consumed annually almost 27 kilos of poultry meat in 2014 (i.e., a production of 13.5 Mt). Globally, poultry meat consumption is predicted to overcome pork consumption in 2020 and increase by 27 % through 2023. More than 8 million of packages are used every year to pack poultry meat, assuming that the 30% of poultry is packed, the packed size to pack 500 g of meat and the pack has a weight of 8-9 g. It is packed in tray and lids using modified atmosphere protection (MAP) technology. Due to the need of high barrier properties, several types of materials are used, but normally, they need the combination of several immiscible materials (for example: PET+LDPE+EVOH or EPS/LDPE/EVOH) that difficult their recycling and finally, they are land field or used for energy recovery. The required plastic materials for pack poultry meat is calculated in 64 kt/y.

Cost of barrier polymers[47] such as EVOH or PVDC is 6-8€/kg, and the cost of metallized increase the cost of the un-metallized film in more than 20%. Therefore in view of the expected competitive cost of the PGA hybrid coatings and considering processing and additives costs, it is realistically feasible for RefuCoat to be cost competitive with current supply.

Based on polyglycolic cost and the chemical routes involved, the cost of a hybrid coating is estimated in 17 €/kg and 20 €/kg if a hybrid and active coating is considered. For savory products, we expected to obtain similar cost than current metallized packages, due to the cost of coating will be 5ct€/m^2 based on a 2.4 µm thickness coating (i.e., 3 g/m^2). In the case of the food will require a hybrid and active coating, the coating cost is estimated in 8ct€/m^2 for a thickness of 3.2 µm (i.e. 4 g/m2). However, if we remove 7 µm of EVOH, plus 10 µm of a tie layer with a full cost of 6 ct€/m^2, this extra cost is considered attractive to the market given the accompanying improvement in shelf life and product quality, recyclability and compostability.

Glycolic and Polyglycolic acid market

Approximately 10% of the EU's €491 billion in chemical sales in 2010 was estimated to be generated by bio-based chemicals. The value of bio-based chemicals is expected to rise to €150 billion globally by 2030, as some 1800 new bio-refineries are expected to be commissioned before 2022.

Global monoethylene glycol (MEG) market, which stood at 25 Million tonnes and reported turnover of $ 27.21 billion in 2014, is anticipated to reach $ 40.85 billion in 2023, expanding at a CAGR of 6.1 percent between 2015 and 2023. However, bio-based mono ethylene glycol (bio-MEG) is produced mainly from bio-ethanol via bio-ethylene or in minor amounts from sorbitol via hydrogenolysis. The main producers are from far-east countries: India Glycols Ltd (MEG from Bioethanol with a plant capacity: 170 kt/a); Greencol Taiwan (MEG from Bioethanol with a plant capacity: 100 kt/a) and Changchun Dacheng Industrial Group (MEG and polyols from sorbitol with a plant capacity: 200 kt/a).

Bio based PET is a biodegradable product made from MEG (monoethylene glycol) and PTA (purified terephthalic acid) materials. Around 30% of bio based PETs are made of MEG, a sugarcane ethanol product, and the rest is made of PTA, a chemical-based crude oil product. They emit around 30-70% less carbon dioxide compared to traditional plastics during landfills. The bio-MEG from bio-ethanol is used for Coca Cola's Plant Bottle, the first-ever recyclable PET plastic bottle made partially from plants[48].

[46] www.avec-poultry.eu/system/files/archive/new-structure/avec/Annual_Report/2015/Annual%20Report%202015.pdf.
[47] http://omnexus.specialchem.com/tech-library/article/materials-developments-for-barrier-food-packaging-applications
[48] http://www.coca-colacompany.com/plantbottle-technology

The global glycolic acid market[49] is basically driven by the market for personal care products. Glycolic acid is used extensively in personal care products like moisturizers, face peels, creams, lotions, and others. Glycolic acid is basically a crystalline, odourless, and a colourless solid and is by far the smallest Alpha Hydroxy Acid (AHA). It is naturally found in vegetables and fruits like unripe grapes, sugarcane, sugar beets, pineapple, and others. On a commercial scale, glycolic acid is synthesized artificially via several chemical reactions, e.g., it is made from the reaction between chloroacetic acid and sodium hydroxide. The global glycolic acid market is expected to reach USD 415.0 million by 2024.

Polyglycolic acid or PGA also known as polyglycolide is a polymer formed by polycondensation of glycolic acid. They are mainly used in cosmetic formulations, household and industrial cleaning products and in medical sutures. DuPont Corporation (U.S) is a key participant in the global glycolic and polyglycolic acid market.

Bioplastic market

Currently, bioplastics represent about one per cent of the about 300 million tonnes of plastic produced annually. But as demand is rising and with more sophisticated materials, applications, and products emerging, the market is already growing by about 20 to 100 per cent per year. From the 61% are bio-based/non-biodegradable and 39% are bio-based and biodegradable as shown in figure 2.1.c. Bio-based, non-biodegradable plastics, such as bio-based PE and bio-based PET, are the main drivers of this growth. More than 60 percent of the bioplastics production capacity worldwide in 2014 was bio-based, durable plastics. This share will increase to over 80 percent in 2019. Production capacities of biodegradable plastics, such as PLA, PHA, and starch blends, are also growing steadily, nearly doubling from 0.7 million tonnes in 2014 to well over 1.2 million tonnes in 2019. PHA production will double by 2019 compared to 2014, due to a ramp-up of old and new capacities in Asia and the USA.

Bioplastics are used in an increasing number of markets, from packaging, catering products, consumer electronics, automotive, agriculture/horticulture and toys to textiles and a number of other segments.

The data compiled by European Bioplastics [50] shows that packaging remains the single largest field of application for bioplastics with almost 70 percent (1.2 million tonnes) of the total bioplastics market. Due to the excellent fit of bioplastics in the packaging market, this number is expected to increase to more than 80 percent (6.5 million tonnes) in 2019. The data also confirms a decisive increase in the uptake of bioplastics materials in many other sectors, including textiles, automotive applications, and consumer goods.

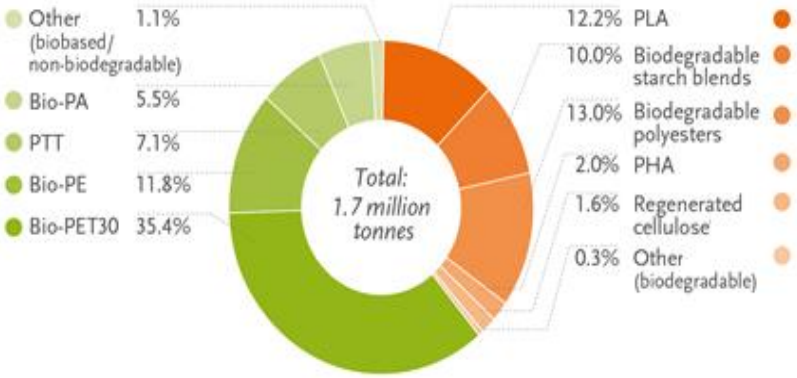


Figure 2.1.c Global production per material type of bioplastics 2014. (Source: www.european-bioplastics.org/bioplastics/materials/)

The data compiled by European Bioplastics[51] shows that packaging remains the single largest field of application for bioplastics with almost 70 percent (1.2 million tonnes) of the total bioplastics market. Due to the excellent fit of bioplastics in the packaging market, this number is expected to increase to more than 80 percent (6.5 million tonnes) in 2019. The data also confirms a decisive increase in the uptake of bioplastics materials in many other sectors, including textiles, automotive applications, and consumer goods.

Active packaging market: antioxidant and antimicrobial.

Functional additives and **barrier coatings market** for plastic packaging is projected to reach €3.2bn by 2018, at a 5.7% annual growth, according to a recent report from Smithers Pira[52]. In addition, the **active & intelligent packaging market** was valued at €1.9billion in 2015 and is demonstrating a strong growth trend with, for example the US demand forecast to increase 8.0 percent annually to €3 billion in 2017 and Europe being the second biggest market behind US.

[49] www.transparencymarketresearch.com/pressrelease/glycolic-acid-market.htm
[50] http://www.european-bioplastics.org/market/
[51] http://www.european-bioplastics.org/market/
[52] www.smitherspira.com/news/2014/april/high-barrier-packaging-film-consumption-forecast.

In summary, the potential impact of the products to develop in RefuCoat project and the current and expected cost are summarized in the table 2.1.b. Expected costs are evaluated based on the expertise of the RefuCoat's partners and the methodology proposed to systematically evaluate the techno-economic prospects of future production routes of bio-based bulk chemicals by Burk et al.[53]

Table 2.1.b Potential economic impact of RefuCoat target products.

Target products.	Current// Expected price (€/kg)	Expected Worldwide consumption (kt/y).
Glycolic acid	5-10 // <4	120
Polyglycolic acid	17-15 // <10	9
Hybrid Coatings	5.5 ct€/m2 metalized// 5ct€/m2 (for estimated coating cost 17€/kg)	7 kt/y to cover 170 kt/y of metallized films.
Antioxidant & Antimicrobial	6ct€/m2 EVOh+tie layer//8 ct€/m2	2.5kt/y to cover 8,000 million packages.
mcl-PHA	No data for mcl-PHA, for scl-PHA 3-6.5 // <4	68 (in 2019)
PHA Sheets+ Hybrid coating	2.5 (PET)// 6.5 (PHA+ coating)	64

2.1.3 Business model proposed by each partner for future commercialization of RefuCoat developments.

Regarding the **segments in the value chain directly represented in the project**, and thus project partners, the expected impact is described below, which will result in an increase of the potential market and increased high added value product portfolio of RefuCoat's partners:

- **Renewable chemicals by chemical routes (BIOPOLIS, AIMPLAS, Fraunhofer and THUNEN):** will select and optimize strains and catalysts to maximize the yield of the reaction. The combination of novel catalysts and strains to obtain valuable products from bio-based products is the **main and innovative product to develop in the RefuCoat project**. The typical business model for these partners would be to patent the chemical routes and catalysts, and use the validations in RefuCoat to either license out the strains/catalysts and their applications or enter into joint ventures with companies which are already active in this area for the production of chemicals. The RefuCoat's partners are very active in the international industrial biotechnology market as service providers and are investing in their own IP portfolio in order to progress in the value chain and retain more value of their activities.
- **Barrier coatings for food and packaging (AIMPLAS, AINIA, IRIS, CSIC and MI-PLAST):** novel, low cost affordable formulations of active and barrier coatings will be developed; the formulations will be optionally enhanced with active ingredients that will be food (contact) compliant, toxic-substance free, flexible, increase food conservation and/or packaging recycling, cost effective, etc. RefuCoat partners plans to either patent the formulations if novelty allows it or to keep them as trade secrets that could be licensed to GRUPO APEX, MANOR and DACSA, and to other coatings manufacturers so they can provide royalties while being commercialised in the broad packaging and food markets. MI-PLAST will have an exclusive use of the coated film for their current market places. Licensing to other companies will be proposed.

Business models and plans will be developed to support the commercialisation of each key project result and partner. The Business Model and Plan developed in WP7 will integrate IPR, innovation protection and exploitation strategies, and the accompanying techno-economic analysis and LCA developed in WP6. This will pave the way for the successful exploitation of the project results. The expected impact, based on a conservative scenario of the value chains, is described in qualitative and quantitative terms in table 2.1.c. This results in an estimated benefit (an average 15% of turnover) of 68 M€ over the period 2025-2030 (5 years after project end). It is important to highlight that the environmental benefits have not been included in these calculations.

[53] M. J. Burk et al. Biotechnology for Chemical Production: Challenges and Opportunities, Trends in Biotechnology. 34, 187–190, 2016.

Table 2.1.c. Preliminary evaluation of RefuCoat economic impact after 5 years project finish (when TRL8 is achieved).

Potential applications	Expected price (€/kg)	World Market size (kt/y) /Market share[54]	Expected Turnover[55] (M€/y)	Direct Employ-ment[56]	IP management	Required investment
Glycolic acid	4,0	120 /5% (6kt/y)	24,0	68,6	License to toll manufacturer	15 - 20 M€
Polyglycolic acid	10,0	9/2.5%	4,5	12,9	Licencing to chemical & compounding companies	No Investment required. Industrial facilities available in conventional chemical, compounding and coating companies.
Hybrid Coatings	17,0	7 / 5%	6,0	17,0	Licensing to packaging manufacturers and coatings manufacturers.	
Active & hybrid coating	21,0	2.5 /5%	2.6	8,8		
mcl-PHA	4,0	68 / 5%	12,8	36,6		
PHA coated sheets	6,5	64/4%	17,7	70,7	License to toll manufacturer	3-5 M€

The RefuCoat project will have a positive impact on the competitiveness of SMEs and INDUSTRY by reducing technical bottlenecks making the whole sector more attractive to investments. In particular the project will strengthen the competitiveness of the European packaging industry and by reducing technical bottlenecks in this area making the whole sector more attractive to investments contributing to new, sustainable economic activity and the creation of new employment.

2.1.4 Wider environmental and social impacts of the project.

On 13 February 2012, the European Commission adopted a strategy for «Innovating for Sustainable Growth: A Bioeconomy for Europe» [57]. This strategy proposes a comprehensive approach to address the ecological, environmental, energy, food supply and natural resource challenges that Europe and the world is currently facing. In the RefuCoat project chemical and biotechnological approaches are employed in order to obtain the new products of interest from renewable biological resources such as mlc-PHA & PGA

Environmental impact will be reduced because new bio-based products will be marketed using environmentally friendly technologies from the raw materials to the final production, showing a positive CO_2 balance relative to existing fossil-fuel derived products or other bio-based sources. Table 2.1.d shows the estimated differences between current (fossil based) and RefuCoat approaches in terms of Global Warming Potential (kg CO_2-Eq.) and Energy Demand (MJ) per kilo of produced material. Due to the use of renewable resources, preferential use of microbial synthesis, highly effective catalysts, low reaction temperatures, green chemistry routes, we expect a significant reduction in both parameters.

Table 2.1.d RefuCoat expected impact on renewable source content and energy demand.

Current solution	Oil based materials[58].		Potential reduction using RefuCoat concept.					
	GWP (kg CO_2-Eq.)	Energy Demand (MJ/kg).	RefuCoat proposed solution	Rene-wable (%)	GWP (kg CO2-Eq/ % less).		Energy Demand (MJ/% less).	
Metallised polyolefin	2,02	70,66	Bio-PE +3% of PGA	100%	-0,67	133	25,7	64

[54] Expected conservative market share (%) based on current worldwide consumption and production capacity of RefuCoat partners. In brackets t/y of RefuCoat's products that will be produced for target market share.
[55] Only considers the turnover the amount of material that RefuCoat industrial partners can produce and commercialize (DACSA, MIPLAST) or licensing (BIOPOLIS & IRIS) .
[56] We consider a turnover of 350 k€ for new job created for the chemical and biological industry and 250 k€ for plastic & coating industry.
[57] //biobs.jrc.ec.europa.eu/sites/default/files/generated/files/policy/EU%20Bioeconomy%20Communication%20Strategy%20Brochure.pdf.
[58] Diferent sources: Ecoinvent database; EVOH www.sustainableminds.com/showroom/kuraray/; Tsiropoulos, I., et al (2015). Life cycle impact assessment of bio-based plastics from sugarcane ethanol. J. Cleaner Production, 90, 114-127; Narodoslawsky, Met al. H. (2015). LCA of PHA Production–Identifying the Ecological Potential of Bio-plastic, Chemical & Biochem. Eng. Quarterly, 29, 299-305.

(0.5% Al /99.5% PE)			Hybrid coating			%	8	%
Metallised Polyester (0.5% Al /99.5% PET)	2,58	68,77	Bio-PET + 3% of PGA Hybrid coating	35%	1,19	54%	54,57	21 %
PET/PE tray (94%/6%)	2,50	68,65	PHA + 3% of PGA Hybrid coating	100%	1,19	52%	49,72	28 %

Using biopolymers (from removable origin such as bio-PE or bio-PET, or from removable origin and biodegradable such as PHA) as basis for the new materials, the environmental advantages have been proved. In a metanalysis[59] from 2012, 30 life cycle assessment studies on different PHA types were compared and it concluded that this kind of products have clear environmental advantages in comparison with their fossil counterparts. Specifically, the substitution of fossil derived plastics by PHA brings positive impacts in the categories of climate change, in terms of emissions, and exhaustion of fossil resources. The savings on fossil resources is calculated for PHA between 40 to 90 MJ per kg of plastic, when compared to PC, and between 20 to 40 MJ per kg of plastic, comparing with PP, HDPE, LDPE, PET and PS. Regarding emissions, a containment of 5.8 kgCO_2eq per kg of plastic was calculated for PHA. This saving is higher than the one calculated for PET and PS, 2.5 and 4.2 kg CO_2eq per kg of plastic respectively. Polyglycolic acid has also significant environmental advantages in comparison with high barrier polymers such as EVOH or PVDC[60].

Additionally, the enlargement of food products shelf life by the use of active substances will significantly reduce the food waste production by improving the time in the distribution line.

Furthermore, the adoption of safe-by-design approaches will support the European competitiveness in this particular research and innovation field. Regulators can additionally use the developed predictive tools that underlie the concept of safe-by-design and thus improve the regulatory preparedness by timely interactions with innovators developing novel applications of polymers in the packaging field. These interactions focus on sharing views and expertise on how best to deal with new safety aspects of innovative products for workers, consumers and the environment. A LCA of the key exploitable products will be undertaken in WP6 led by DACSA, with the support of commercial software and the real data obtained throughout the whole value chain

Employment impact:

RefuCoat aims at a significant contribution in the large number of jobs that green chemistry and packaging sector will create in the coming years. We expect at least **220 qualified jobs directly** related to the sale of novelty packages five years after the end of the project (when TRL8 will be achieved), based on a very conservative European market share of 1-3% as table 2.1.c detailed. According the PlasticsEurope (the Association of Plastics Manufacturers in Europe)[61], the sector generated an even greater number of indirect and induced jobs – up to three times higher than through direct employment, i.e., **RefuCoat can create up to 660 of indirect jobs.**

Packaging sector is affected by a **drain of employments to the Far East competitors.** Even the European market of plastic converters, supporting over 1.45 Million direct jobs, suffered an employment decrease by about 9% in the last six years[56]. The new technology developed will provide to all of them competitive advantages that will contribute to stop their loss of employment and to keep European standards of living.

For all these new jobs, education and training of workers will be necessary to adapt their skills to the new biodegradable package production proposed in RefuCoat. This training will be through the RTDs to SMEs and through its wider roll out via licenses and dissemination activities for the third party licensees all over Europe and particularly in the less technologically advanced regions of the EU.

2.1.5 Additional investments

Industrial partners foresee future investments in order to adapt the current facilities to the requirements for the novel products (during the project or afterwards for adapting manufacturing lines). In relation to microbial culture and biopolymer production, BIOPOLIS expects to enlarge and optimize current facilities. Other SMEs (MIPLAST) will continuously improve facilities and capabilities for the developed applications. In addition, in the case of large industries (GRUPO APEX, MANOR, DACSA), it is expected that the new products will be oriented to the creation of new business lines, with also possible new investments at long term. Particularly DACSA foresees the endorsement of new manufacturing facilities for obtaining larger amounts of raw materials.

2.1.6 Policy framework

RefuCoat will also contribute to the fulfilment of the following European norms and the legislation in force:

[59] Metanalysis of 30 LCAs. Essel R, Carus M, bioplastics, 2012, 7, 46-50

[60] www.fda.gov/downloads/Food/FoodIngredientsPackaging/EnvironmentalDecisions/UCM143273.pdf

[61] www.plasticseurope.org/Document/plastics---the-facts-2015.aspx

- Regulation (EC) 1935/2004 of the European Parliament and of the Council of 27 October 2004 on materials and articles intended to come into contact with food.
- Regulation (EC) 10/2011, and its amendments, on plastic materials and articles intended to come into contact with foodstuffs.
- Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008, which lays down measures to protect the environment and human health by preventing or reducing the adverse impacts of the generation and management of waste and by reducing overall impacts of resource use and improving the efficiency of such use.
- Regulation (EC) 2073/2005 on microbiological criteria for foodstuffs.

RefuCoat will also consider the application of the **directives for packaging and packaging waste** to the final products, according to the European Parliament and Council Directive 94/62/EC of 20 December 1994 and the amendments introduced to it by Directive 2004/12/EC of 11 February 2004.

Moreover, REFUCOAT contributes to the Council resolution of 24 February 1997 on a Community strategy for waste management promoting waste recovery with a view to reduce the quantity of waste for disposal and saving natural resources, in particular by reusing, recycling, composting and recovering energy from waste.

Likewise, RefuCoat will contribute to the following EU horizontal policies:

- Chapter XVI of TCE industry. Article 157 states "The Community and member states ensure the existence of the necessary conditions for competition within the industrial community".
- The Treaty of Amsterdam: Title VIII (articles 125 to 130 of TCE) for a coordinated strategy against unemployment on a European level.
- FP7 policy objectives: strengthening the innovation capacity of EU SMEs and their contribution to the development of new technology, bridging the gap between research and innovation by outsourcing research.
- Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal.

Global concern, sustainable solution: The use of fossil fuels can be separated into two classes: energy use and non-energy use. The former is by far the most important class where fossil fuels are used for producing electricity, heat and in the transport sector. However, **about 8-9%** of all fossil fuels in the EU are consumed as feed-stocks (**non-energy use**) in various industrial processes, **of which roughly half are used to manufacture polymers**. The fossil carbon is hence transferred from fossil deposits to the techno-sphere and may finally enter the atmosphere and will contribute to the greenhouse effect and global warming.

Despite all this, according to SPI: The Plastics Industry Trade Association [62] (2015), the global polymer industry is expected to grow with a CAGR of 3.9 percent over 2015-2020 and reach more than 400 million of tones in 2020, from which only the 1.5% will be produced from renewable raw materials. Therefore, initiatives such as **RefuCoat** that will demonstrate the use and modifications of biopolymers, at least 75 % from renewable resources are of key importance to increase the use of renewable materials as base materials for the new sustainable EU Industry. Summing up, the main advantages of bioplastics are:[63]

- Their production uses 65% less energy than it takes to produce petroleum-based plastics.
- Bioplastics generate 68% fewer greenhouse gases than fossil-fuel-based plastics.
- Manufacturing petroleum-based plastics uses approximately 200,000 barrels of oil per day.
- As they degrade, bioplastics will remain non-toxic and will not leach dangerous chemicals into the soil and therefore, bioplastics are more energy-efficient, better for the environment (safer and recyclable) and make EU economy less dependent on foreign oil.

RefuCoatwill contribute to improve the **quality of life, health and safety** based on several standpoints:

1) By **improving the properties of biopolymers** from renewable resources, fully recyclable and biodegradable without residue able to substitute a wide range of polymers coming from fossil sources. In the particular case of replacement of metallised films and MAP multilayer structures, the availability of new alternative tailor-made coatings grades will contribute to fulfil the Waste Framework Directive (WFD) 2006/12/EC and in particular the EU Parliament and Council Directive 94/62/EC of 20 December 1994 on packaging and packaging waste.

2) **Contributing to the EU Landfill Directive 1999/31/EC** which aims at reducing the amounts of biodegradable waste going to landfills. Member states are required to reduce these amounts by 65% by 2015. However, 12.4 million tons (50% of the total) of post-consumer plastic was wasted in landfill in 2007, so the introduction of a material 95%

[62] http://www.plasticsindustry.org/files/Global%20Trends%202015.pdf
[63] The Bioplastic Debate [on line].www.foodservicewarehouse.com

from renewable resources that can be recycled or composted to produce a fertiliser for non-food crops, and which is biodegradable without residue, will contribute to improve waste management indicators.

2.1.7. Potential barriers to impact realization

The following barriers are considered specifically relevant to the work programme:

✓ *Non-competitive price*. The technical risks inherent to an R&D project may lead to: 1) commercial low productivities 2) high downstream process costs 3) high process energy costs. A risk and contingency plan has been outlined and risk monitoring strategies (section 3.1.1) have been implemented.

✓ REACH regulation. Current legislation is concerned by the potential health, safety and environmental risks in relation to bio-based products and food packaging materials. The RefuCoat project will review the legislation in: i) Regulations REACH and CPL (classification, labelling and packaging of substances and mixtures) and Legislative Decree EU 10/2011, ii) Directive 52/97 defining hazardous substance*:* assessment of the risk to the health and safety of workers, iii) Environmental impact assessment, strategic impact assessment, air and waste sectors, Legislative Decree 152/2006 and Decision 2000/532/EC to ensure compliance of the developed products.

✓ *Conservatism*: Imperfect flow of existing information about green chemistry and biotechnology safety. Lack of information about service life, durability and advantages of the new products. RefuCoat will provide well documented results, dissemination materials and product datasheets including characterization and environmental data, etc, that will form a sound base to support high level development of regulations and green procurement policies

✓ *Governance support*. This kind of initiatives and the dissemination of the new products and technologies will contribute to the creation of enabling policies settings introducing a new raw material policy, providing a clear political commitment and continuous coordination between relevant bio-based economy sectors and policies, addressing bio-based economy needs in updates to major EU and national legislation (CAP, Waste Framework Directive, EU support to biofuels and bio-energy, research and innovation programmes, EU Cohesion Policy and development programmes).

✓ *Lack of commercial incentives to enter the biomaterials for smart packaging market*: initial small packet size, brand resistance to bio-based products; European policy and support for bio-based products and bioeconomy is unclear. To tackle these barriers different market products will be proposed, with huge potential demand and medium cost

✓ *Competition from alternative routes for the production of biomaterials*: The RefuCoat approach will be to perform the processes by using the routes with the least environmental impact taking into account low processes costs.

The main hurdles related to the commercial exploitation of project technologies/Results and their impact/probability are summarized in Table 2.1.e.

Table 2.1.e. Risks description, impact & probability.

Description of risk	***Impact***	***Probability***
Shortage of resources	*Medium*	*Low*
No customer interest in trials	*High*	*Low*
Customer acceptance	*High*	*Medium*
Processes cannot be made economically feasible.	*High*	*Low*
Regulatory issues	*High*	*Low*
IP issues	*High*	*Low*
Logistics issues	*Medium*	*Low*
Not being able to supply for industrial demand	*Medium*	*Low*

2.2 Measures to maximise impact

In the framework of BBI Joint Undertaking, dissemination and exploitation activities, and communication plan are key elements to enhance private and public return on investment, respectively. The overall strategy of RefuCoat includes a substantial number of impact amplification actions with the over-arching aim to distribute excellence and knowledge within and outside the project. A specific Work Package (WP7) has been devoted to dissemination, exploitation and communication plan definition, implementation and monitoring

2.2.1 Dissemination, communication and exploitation of results

In order to achieve considerable visibility of the RefuCoat project a media network of press and media contacts will be established during the first 6 months of the project. Besides, dissemination activities will be monitored to assess their effectiveness (though impact measurement). Several dissemination channels will be used:

- Both internal and external events: especially those conference/forums related to biotechnologies and bio based products but also those that enhance visibility of the BBI initiative towards other stakeholders including the general public.

- Website: the content will be driven to ensure political and public awareness of the RefuCoat project, show other ongoing projects and to gain acceptance from various audiences at European and national level.
- Partnering platforms: partners will be encouraged to pro-actively publish their suggestions, doubts, comments about the project and exchange their knowledge.
- Newsletters: specially lead to show the potential market opportunity of the final products. Created in electronic format to be more environmentally-friendly.
- Publications: articles in specialized and general magazines, press releases, leaflets and factsheets with the key achievements.
- Events: partners will organize a forum on the specific topic of smart food packaging. Besides, a consortium will be encouraged to promote BBI within the EU institutional area attending to events inside the European Parliament, Info Days and/or Open days.
- One or two conferences related to bio-based and smart food packaging products will be organized.

Communication between partners, EC, and target stakeholders (through a Stakeholder board) is vital to ensure the appropriate dissemination of RefuCoat results and make the project public in an effective way. In order to achieve that, a "Communication Strategic Plan" with the activities ("what") to be performed during the project ("when") and partners and stakeholders to be engaged ("who") will be developed.
The Dissemination Plan will endeavour broad impact among specialized scientific and industrial audiences following the EC policies on this matter. As a general rule, the main project results will be first protected by patents, for subsequent exploitation by partners and/or third parties, and then presented at different scientific/technical conferences, meetings and trade fairs, and published in specialized and peer-reviewed scientific/technical journals.
RefuCoat partners include companies with substantial patent portfolios, as well as research and academic groups with a large experience in scientific/technical publication (see section 4: Members of the Consortium), that will be applied to the dissemination of the project results. In addition, the partners will disseminate the results to business stakeholders in their respective customer bases, to further promote the exploitation of the results. Such contacts will be conducted upon prior agreement with all partners, to ensure the preferential rights of the companies within the consortium.
A significant number of **publications in high-impact journals and books of different** areas is expected. Attendance of partners to **international conferences**, **congresses**, **workshops**, and **meetings** will be also encouraged. Complimentary knowledge areas, such as biotechnology, chemical catalysis, polymer engineering and processing, bioplastics, biorefineries, and biomaterials will be covered owing to the multidisciplinary nature of the project. The commitments of the different partners are shown in Table 2.2.a.

Table 2.2.a. Publications and participation in international events

Partners	Conferences & Seminars	Publications
BIOPOLIS + DACSA	European Congress on Biotechnology, BIO World Congress on Industrial Biotechnology, Microbiology Society Annual Conference…	**Biotechnology and Microbiology:** Microbial Biotechnology, Applied & Environmental Microbiology, Applied Microbiology & Biotechnology, Biocatalysis & Biotransformation, Enzyme & Microbial Technology, Microbial Cell Factories, Trends in Biotechnology
THUNEN + Fraunhofer	Chemical Biology EMBO Conference, International IUPAC Congress on Green Chemistry, International Conference on Combinatorial Chemistry and Applications, International Symposium on Green Chemistry	**Chemistry:** Chemistry & Biology, ChemCatChem, ChemSusChem, Journal of Catalysis, Combinatorial Chemistry & High Throughput Screening, Green Chemistry, Journal of Agricultural & Food Chemistry, Journal of the American Chemical Society, Journal of Chemical Theory & Computation, Journal of Computational Chemistry and Tetrahedron
BIOPOLIS + DACSA	International Conference on Chemical, Biological and Environmental Sciences, International Conference on Biological, Ecologic Environmental Sciences and Engineering…	**Bioengineering** and **Environmental Technologies:** Advanced Synthesis & Catalysis, Bioprocesses and Biosystems Engineering, Bioresource Technology, Biotechnology and Bioengineering, Environmental Science & Technology, Journal of Chemical Technology & Biotechnology and Journal of Polymer Science
AIMPLAS +	World Biomaterials Congress,	**Biomaterials and polymer science**: Journal of Applied

Partners	Conferences & Seminars	Publications
CSIC+ MI-PLAST	European Bioplastics Conference, International Symposium Frontiers in Polymer Science, International Seminar of bioplastics and composites, Functional and sustainable coating seminar, IAPRI's Symposium	Polymer science, Journal of coatings technology and research, Surface and Coatings Technology, European Coatings Journal, Journal of Polymers and the Environment.
IRIS + AINIA + GRUPO APEX + MANOR FARM + EUFIC	Biopol, Biopolpack, World Congress on Food Science and Technology, AIPIA conference	**Food Science & Technology,** Packaging technology, bioplastics magazine, etc.

At least one of the publications will be a **comprehensive review on the technical background and market significance of the project**, innovative bio-based, sustainable packaging applications, to which every partner will be invited to contribute.

In addition, **contacts with associations, thematic networks and discussion groups** will be encouraged, in order to use their dissemination tools (website, sectorial newsletters and magazines, invited workshops…) as channels to further promote the project. RefuCoat partners have access by professional membership to different sectorial platforms, as detailed below. This will generate additional resonance effects within industrial associations to accelerate the transfer of RefuCoat concept to the European Industry and research institutions (Table 2.2.b).

Table 2.2.b. International associations with direct link with RefuCoat's partners.

- *Food and Beverages* FOOD DRINK EUROPE - International association of food and drink producers FOOD4LIFE SPAIN – Spanish Technological platform focused on the food value chain FIAB – Spanish Federation of Food and Beverages (BIOPOLIS)
- *Plastics and Packaging* European Bioplastics (AIMPLAS) ASOBIOCOM - Spanish Association Compostable Biodegradable Plastics (AIMPLAS). FPE -Flexible Packaging Europe (AIMPLAS). AIPIA - Industrial association for active and intelligent packaging (AIMPIAS, AINIA) EuPC - European Plastics Converters (AIMPLAS) EPRO-European Association of Plastics Recycling and Recovery Organisations (AIMPLAS) ECP4- The European Composites, Plastics and Polymer Processing Platform (AIMPLAS).
- *Biotechnology and Bio-Based Industries* ASEBIO - Spanish Association of Biocompanies (BIOPOLIS) SUSCHEM - Spanish Association of Sustainable Chemistry (BIOPOLIS) Farmaindustria – Spanish Association of Pharma Industry (BIOPOLIS) EUROPABIO – European Association of Biotechnology (BIOPOLIS) BIC – Bio-based Industries Consortium (BIOPOLIS, MIPLAST) CEFIC- The European Chemical Industry Council SPIRE-Sustainable Process Industry (IRIS)

In relation to **training and knowledge transfer**, the training of specialized researchers and technicians will be pursued, by organizing **training courses for researchers**, as at least two satellite events of the annual meetings (M12, M24, M36) whose topics will be selected one year ahead. Whenever possible, the workshops will be organized in agreement to major conferences/exhibitions to maximize their impact. The events will be specifically tailored for SME researchers and technical staff. These courses will include keynote lectures from partners as well as for specially invited speakers, such as potential technology users of RefuCoat achievements. In addition, we will establish **specific exchange and mobility programs** for interchanging personnel between academic and industrial partners, attending to their complementarity and common grounds, in order to stimulate the transfer of knowledge, to increase cooperation between partners and to favour the flow of ideas across the partners' facilities.

Finally, a **technical guide** to explore the methods and technologies developed during RefuCoat project will be published by the consortium, aiming at early adopters, such as SME researchers, technical staff from academia and industrial potential end-users. The guide will include information valid to translate the results to parallel applications (analytical methods, processing techniques, equipment set-up, prototype development), however committing to necessary privative information for exploitation purposes.

Communication with stakeholders and to the general public: the project partners recognize that, in order to accelerate research and promote validation of the RefuCoat concepts, there is a need to widen communication activities to the wider community and specifically inform policy makers, active industrial groups and societies and the general public on progress and potential impact of the research. Publication of articles and participation to interviews in the non-specialist press using all media available (regional TV programs, radio programs) will widen the visibility of RefuCoat and its achievements. Focus of this communication will be on demonstrating impact of chemical technologies on the important societal challenges such as climate change and resource depletion as well as changing a general negative perception of broad public towards chemistry which is often associated with air and water pollution, industrial disasters and similar negative stereotypes. Each partner will maximize the visibility of RefuCoat in its already established annual or biannual outreaching events focusing on different generations and societal groups.

Table 2.2.c below summarises the various communication and dissemination channels used during the project.

- **Table 2.2.c**. Publications and participation in international events

Commercial Dissemination	
Patents	The companies need to ensure their competitiveness and innovation through patents applications. According to their specific application/market sector, the companies will deposit application patents.
Market oriented magazines	Articles will be published in dedicated industrial magazines to enhance the perceptibility of the industrial development of bio-based chemicals and materials in different sectors of activity.
Events	Participation to events, fairs … can enhance the visibility of the project, especially towards commercial exploitation.
Website	Besides the link to the dedicated RefuCoat website to be present on each partner's website, additional information and developments can be provided on dedicated websites such as: http://www.omnexus.com/; http://www.specialchem.com/ for plastic & elastomers, adhesives & sealants, coatings & inks; http://www.specialchem4bio.com/ FISCH: www.fi-sch.be/ ; CLIB2021: www.clib2021.de/en
Social media	A group related to RefuCoat product line will be created on LinkedIn to give additional visibility towards this project and its importance.
Scientific Dissemination	
Scientific publications	From a scientific point of view, publications will have to be reported by the RTOs/academics in recognized scientific journals such as reflected above.
conferences	Besides attending more market-oriented conferences and events, the developments realized within the project have to be presented at a higher scientific level where research and industrials meet as indicated above.
newsletter	Another way of reaching a broader public in the scientific community is through the publication of newsletter on a regular basis. Here also, a newsletter on a 6-months basis could be elaborated and provided to a broad network of partners' contacts. This newsletter can also be made available on the RefuCoat website, LinkedIn, partners' website, conferences, events & workshops.
Public Dissemination	
Press	The impact of the press contacts and editorial contributions can influence the perception of the broader public, and unlock the awareness towards bio-based economy. Different press canals will be used: National journals; Regional press; Dedicated press linked to the green economy but for a non-specialized public.
Website	All partners' website Dedicated RefuCoat website enabling clear understanding for a neophyte public
Social media	The aim is to make RefuCoat's role and the contributions it makes in the area of sustainable development known to a wide public in an understandable form. This type of dissemination aims at sharing these values with society in order to improve the general

	public's perception of the impact and social relevance of innovation.
State Dissemination & civil society	
Newsletter	Providing newsletter on a regular basis to our stakeholders, as described above, is an additional way of reaching different communities
website	Link to RefuCoat website
Networking	The partners in this consortium have access to different supporting networks: www.biorizon.eu; www.biobaseddelta.nl; www.fi-sch.be/
Reporting EU	Enterprise & Industry Magazine on EU website (http://ec.europa.eu/enterprise/magazine/index_en.htm)
REACH	Chemical Safety Assessment under REACH

2.2.2 Business exploitation Plan

The general aim of this task is to provide the grounds to successfully exploit the outcomes of the project. Being RefuCoat a private-driven project, the exploitation plan will comply to the underlying commercial structure of a business plan. All the data produced in the project (with a focus on the results generated by WP7) will be analyzed in order to make a general product/market correlation. Research will be conducted also to gather information that will allow to potentially understand i) the positioning/pricing of the product and its unique selling properties; ii) how the market would react to the new product; iii) the necessity of additional studies/approvals that may be needed before a launch of the product; iv) design of different business models under which the product may be produced with respect to socio-economic impact; v) consumer interest in the product.

Successful commercialization of new food technologies: The success of food packaging innovation depends on consumers accepting novel products and technologies. Effective communication among key actors involved in the process of developing food innovations (from food technologists to consumer scientists) is needed to increase consumer acceptance and lower failure rates and investment losses. A set of tools for achieving effective internal and external communication during new product development is available (Connect4Action website, Toolbox section).

The main duty of the Innovation and Exploitation Board will be to review the results and developments within the project to decide about their exploitation policy in the form of patents, licenses and transfer of know-how.

The potential products and technologies planned within RefuCoat are summarized in Table 2.2.b. The partners involved and preliminary IPR strategy and Key Performance Indicators are shown.

Table 2.2.b Summary of the key exploitable results.

WP	Key Exploitable Results (KER)	Partner Interested In Exploitation	IPR strategy
2	Functionalized PHA with enhanced water barrier by microbiological routes.	BIOPOLIS, DACSA,	To protect by patent the most relevant outcomes.
2	Process for production of GA from Bio-MEG	THUNEN	To protect by patent the most relevant outcomes.
3	Polymerization of low molecular weight PGA suitable for coating formulations by reactive extrusion.	Fraunhofer, MI-PLAST.	Patent the full process or specific steps.
3	PGA- SiO_x hybrid coatings with excellent gas barrier.	AIMPLAS, MI-PLAST	To protect by patent for use in several bio-based substrates.
3	Production of PGA and the hybrid process in one processing step using a twin screw extruder.	Fraunhofer, MI-PLAST, AIMPLAS	Patent to protect the full process.
4	Antimicrobial active coating	AINIA	Patent to protect the full process.
4	Antioxidant active hybrid coating.	IRIS	Patent will be considered but by default, formulations will be kept trade secret
5	New packages based on modified PHA and hybrid and active coatings with high water and medium oxygen barriers	BIOPOLIS, MANOR, AINIA AIMPLAS	Industrial secret.
5	New packages based on bio-PET and hybrid coatings with high water and oxygen barriers for savoury products.	GRUPO APEX, MI-PLAST, AIMPLAS	Industrial secret.

5	New packages based on bio-PE and hybrid and active coatings with high water and medium oxygen barriers for cereals products.	DACSA, IRIS, MI-PLAST, AIMPLAS.	Industrial secret.

Commercialization of the most promising products of the project will be conducted by the industrial partners, via market launch of new product applications. Indirect exploitation will also be attempted, should the specific results not be of the interest of the industrial partners for internal use within the Consortium. In this case, these generic developments will be transferred to sectors outside the Consortium.

The **potential end-users or clients** of RefuCoat results have been identified as:

- Food and non-food companies that require new packaging solutions with improved barrier properties
- Food companies that produce substantial amounts of by-products interested in their valorisation
- Packaging companies that currently design and produce packaging solutions
- Plastic film and sheet companies that formulate and process materials for packaging applications
- Engineering companies that commercialize technology and equipment for packaging applications
- Chemical companies that synthesize and distribute biopolymers
- Chemical companies that synthesize and distribute intermediate bio-based chemicals.
- Engineering companies that develop projects for turn-key manufacturing plants

In this way, RefuCoat customer base is formed by (i) internal clients, that will be industrial, large companies for direct exploitation of the results, and SMEs, for indirect exploitation of the results, by licensing agreements to third parties, and (ii) external clients, in the event internal clients are not interested or cooperation agreements can be reached, for different geographical or market targets. External clients are within close reach of the partners, as current customers, partners or shareholders of the participating SMEs and RTOs. Each partner has experience and dedicated resources (commercial department, exploitation board, technological transfer offices) to promote its own research outcomes, products and services. The Exploitation Plan of the project will organize such resources for the synergistic promotion of RefuCoat.

The contents of the Exploitation Plan are further described as follows:

- Market study: to gather information on the potential applications in food, packaging and related sectors, defining per specific segment: competitors, price, market volume, growth rate, strategic partners and current alliances, geographical trends…

- Technical and legal feasibility: QC and specific market regulations compliance, Freedom to Operate update, IPR strategy update, additional technical activities required (product upgrade),

- Economic study: Financial projections will be specified to include: i) marketing feasibility (estimated market size, market demand, and market gap), ii) technical feasibility (fix assets value, man power wages and salaries, materials and utilities), iii) financial feasibility (upfront investment and future operating cost). Net present value (NVP), Cost-benefit calculations, and estimated balance will be provided. Sensitivity analysis will be applied to identify further steps which could strengthen or threaten the marketing of the technology.

- Marketing and Commercial Plan: to ensure commercial compatibility and readiness of the identified parameters from the technical, economical and legal feasibility studies and to define and position the final product (*e.g.* design, layout, specifications, price, models…) within the European ambit and especially in the partners countries as potential key investors.

- Business plan: A new business model will be prepared according to the information gathered and will be firstly applied to the consortium level. SWOT analysis and risk assessment will be conducted. The required investment and planned outcomes will be translated into a projected profit and loss account and balance sheet.

The Exploitation Plan will be developed for the whole project but also for each of the participants. Every partner in the consortium will benefit in accordance to the nature of the entity. For RTO centres, the consolidation of the research activities will derive in new research lines to offer to different companies. The impact arisen from the project will translate into the consolidation of R&D business lines related to biomaterial design and applications. The extra knowledge about active packaging with improved properties and bio-based processes will allow them to support industry in Europe by feasibility projects and developments. In the case of AIMPLAS, the technological centre will further develop R&D lines devoted to innovative packaging solutions and production of bio-based hybrid coatings. THUNEN will expand its knowledge on applied catalysis for polymer precursor synthesis. AINIA will further

develop R&D lines related to active coatings and strengthen partnership with food industries. Fraunhofer will expand and exploit capabilities in biopolymer synthesis and processing techniques development.

The commercialization of RefuCoat products will be most probably accomplished by large companies involved in the Project. Industrial end users (GRUPO APEX, DACSA and MANOR) will include the developed products in their manufacturing lines, for different food applications. Apart from end users, exploitation of capabilities along the whole production chain will be an opportunity for business development of SMEs and RTOs. The markets addressed are global, so target customers addressed are distributed worldwide, becoming a significant volume of sales for every of the sectors involved in RefuCoat, thanks to the high replication potential. Both will benefit from indirect exploitation schemes, as technology providers.

SWOT analysis describes the barriers, obstacles and overall framework conditions of RefuCoat project.

Table 2.2.c SWOT analysis of RefuCoat project.

	Drivers	Barriers & risks
Internal origin	**STRENGTHS** - Integration and synergistic collaboration among partners with complementary profiles within the consortium. - End-users and key stakeholders directly involved in product development. - Consideration of complimentary applications for further impact. - Prototypes validated within the consortium. - Previous experience in product market launch within the consortium.	**WEAKNESSES** - Technical risks inherent to product development (unmet specifications, excessive consumable costs, excessive energy consumption costs in comparison to current benchmarks). - Lack of information regarding mid-term product service life and durability.
External origin	**OPPORTUNITIES** - No comparable technical approach has been marketed so far. - Competitive pricing based upon preliminary market analysis. - Possibility to extend product positioning to different industry applications. - Closeness of partners to key opinion leaders for maximized impact and dissemination. - Current socio-political situation pushing Europe for improved competitiveness and leadership in innovative food packaging solutions.	**THREATS** - Low demand leading to high unitary costs, resulting in unmet break-even point. - Legal and regulatory barriers delaying time to market for certification issues. - Intercepted market launch by competitors.

RefuCoat aims to the development of cost-effective solutions to obtain novel and sustainable packages with improved barrier properties. Therefore, it is essential to evaluate and demonstrate the actual economical and environmental benefits of such products. For this reason, we have considered a number of activities directed to estimating product cost distribution, through manufacturing process simulation at industrial scale. The technical details required for the implementation of the process at industrial scale will be defined. The evaluation of the scaled protocols using economic indicators will allow comparison to current market benchmarks. Environmental evaluation of the process will also support the reduced impacts of these new added value chains. In terms of exploitation, the participation of all value chain agents will support the development of competitive and performing final products.

RefuCoat project will be mainly developed by industrial partners. The consortium is composed of 7 companies supported by 5 RTD centres. Among the 7 companies, 4 of them are large industrial companies from the different sectors where the products will be applied for commercial exploitation: GRUPO APEX, DACSA and MANOR. The participation of large companies as end users is of main importance in order to focus the needs from final customers. With wide experience in their markets, they will be able to guide the other technological partners, not only in the technical specifications, but also with a cost-effective vision to be taken into account for further exploitation success. These companies will develop final products which will be tested and validated at the end of the project, obtaining an optimal solution for new products with improved performance at competitive prices.

2.2.3 Intellectual Property, knowledge protection and regulatory issues

IP monitoring and IP rights attribution strategy

The following key activities are proposed by the RefuCoat consortium to ensure the proper knowledge management and adequate and effective IP protection and exploitation by the right partner:

- Knowledge and IP management follow the legal requirements defined in official documents published by the European Commission, such as the document: "Participation Rules and Contracts".
- The Executive Management Committee (EMC) will manage knowledge generated from RefuCoat and associated IPR. The EMC will seek professional advice from each relevant technology transfer organizations active in all partners when these are needed.
- The "Pre-existing know-how" will be identified by individual partners during negotiations for the Consortium Agreement.
- As the project outcomes and deliverables will be prepared, the consortium will establish details of the knowledge to be generated by the project. The strategy for using and disseminating knowledge within and beyond the consortium during the lifetime of the project and afterwards will be reviewed at each project meeting.
- After the second year of the project, the consortium, via the EMC, will establish a list of the Intellectual Properties to be protected, work out a timetable for filing the patent applications and establishing other protections (Utility Models, Trade Marks, Software utilities, Copyright, etc.)
- The knowledge and IP will be documented in detail, including the definition of knowledge, ownership of the knowledge, IPs and associated patents/copyrights/trade-marks, plan for exploitation/dissemination of knowledge and technology transfer, etc.
- Working with the partners concerned, the EMC will with the support of technology transfer associates deal with the applications for accessing the joint knowledge/IP generated by RefuCoat and will monitor the fulfilment of the Consortium Agreement on the "Access Rights", "Use of the Knowledge", "Confidentiality", "Publications, Press Releases", etc.
- Disputes on the access to the knowledge, use of the knowledge, IP protection and technology transfer will be resolved using the mechanisms that will be defined in the Consortium Agreement, for which the process will be coordinated by the project coordinator.

Consortium Agreement and background IP

Confidentiality will be reinforced through the Consortium Agreement that will be signed prior to the contract with the European Commission. According to the Consortium Agreement, knowledge generated shall be the property of the partner(s) carrying out the work leading to that knowledge. When several partners have jointly carried out work generating knowledge and where their respective share of work cannot be ascertained, they shall have joint ownership of that knowledge. In this particular case, partners shall jointly apply to obtain and/or maintain the relevant intellectual property rights and shall strive to set up amongst themselves appropriate agreements in order to do so. The share of each contributor to the knowledge development shall be defined proportionally to the resources implemented by each, whether human, financial or intellectual. The access rights to pre-existing know-how and knowledge, as defined in the general conditions of the EC contract, are thoroughly detailed in the Consortium Agreement. Such rights will be treated with or without financial compensation on a case-by-case basis, by considering each party's interest. A multi-party NDA is already in place between all partners and associate partners.
Open Access Publication: Peer-reviewed publication of the project's results will be pursued in full respect of international copyright laws. In line with the European Commission's policy, the RefuCoat consortium fully supports open access publication. All research will be published online in a form that is free-of-charge to the end user, and where possible research papers will be published in "gold" open access journals. Costs to cover publication charges in such journals have been included within the RefuCoat consortium budget (see Part A and section 3.4) and the consortium aims for at least 75% of publications to be provided in the "gold" open access mode.

Regulatory issues

The partners from RefuCoat consortium are committed to all applicable regulations and other legal requirements to the involved fields of research. A preliminary assessment of the Regulations that might impact the projects has been carried out as detailed in the Table below:

Table 2.2.f. Applicable regulations to RefuCoat knowledge area

Standard	Title/content
Regulation 14044	Environmental Management
UNEP specifications	Guidelines for Social Life Cycle Assessment of Products
ISO 13065:2015	Sustainability criteria for bioenergy
DIN Standards	Principles of Environmental Protection

In addition, standards are identified as a critical part for the commercialization potential of the results, particularly as standards for bio-based products, including bio-based plastic materials, are in development at the European Committee for Standardization (CEN). Contacts with the relevant standardization bodies have been planned (e.g. ISO), as relevant stakeholders of the project, that have been included in the communication plan as target audience. Several RefuCoat members, including AIMPLAS, BIOPOLIS and THUNEN, have strong connection to standardization bodies, and experience in developing specific guidance and references of industrial relevance. In such framework, a CEN workshop agreement will be proposed, to develop in a CEN workshop to welcome all experts interested. This document will probably include technical specifications or reports, but may also result in a broad reference standard for coated/active packages.

3. Implementation

3.1 Work plan — Work packages, deliverables and milestones

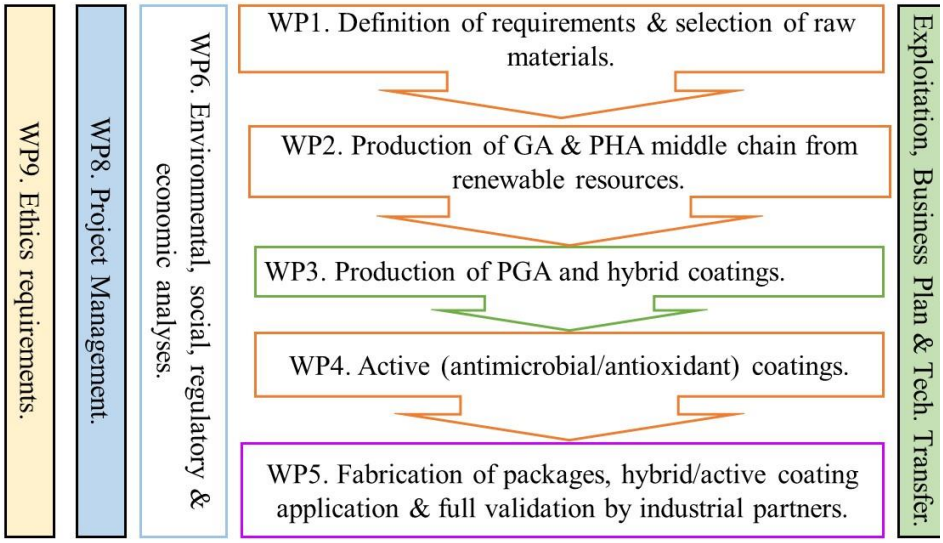
3.1.1 Overall structure of the work plan

This covers all the planned research activities from final products requirements and selection of raw materials (WP1), to production of GA and PHA from renewable resources (WP2), production of PGA and hybrid coatings and production of antifungal and antimicrobial coatings (WP3-WP4). WP5 aims to produce packages for their full validation. WP6 will study the environmental and economic viability for the products. WP7 activities are related to communication, dissemination, exploitation, business plan and technology transfer activities. WP8 will provide the full management of the project in order to make all the proposed developments in line with defined RefuCoat objectives.

WP	*Work package description.*	*WP leader*	*Participants*
WP1	Definition of requirements by the large end-users (MANOR, GRUPO APEX and DACSA). Selection of raw materials to produce the packages and the active and hybrid coatings.	DACSA	All
WP2	Production of glycolic acid from bio-based mono ethylene glycol using innovative gold catalyst. PHA middle chain from renewable resources (DACSA by-products) modified with lipids to be more hydrophobic and improved moisture barrier properties by the inclusion in the PHA chain suitable lipid groups.	CIB-CSIC	AIMPLAS, Fraunhofer, THUNEN, IRIS & DACSA
WP3	Production of PGA by reactive extrusion of the GA dimer. Development of a hybrid coatings by chemical combination of PGA and SiOx. Furthermore, to perform the full process (polymerization and SiOx bond) in one step by reactive extrusion will be evaluated and main parameters optimized.	Fraunhofer	AIMPLAS
WP4	Active coatings will be developed to extend the shelf life of the packed product. The active substance will require combining different active substances: bacteriophages or their endolysins against specific pathogens and substances with a broad spectrum of microorganisms targeted against spoilage ones. Active and barrier hybrid coating in a single layer will be studied to facilitate the production process and further recycling operations.	AINIA	IRIS

WP5	Fabrication of monolayer packages (PHA trays and lids for chicken meat packaging, bio-PET films for savoury products and Bio-PE for cereals packaging). Using conventional coating technologies, the active and hybrid coatings develop in WP3 and $ will be incorporated to the different packages according the requirements defined in WP1. The full packages will be characterized and validated by industrial partners in comparison with current commercial ones. The recyclability and/or compostability of packages will be evaluated using suitable EN/ISO standards.	AIMPLAS	BIOPOLIS, AINIA, MIPLAST, IRIS, & MANOR.
WP6	Assess the environmental and economic impact of the packages and technologies proposed in RefuCoat throughout each stage of development, to ensure compliance with legislation, to maximise market acceptance and facilitate increasing optimisation of these processes and products.	DACSA	All
WP7	Complete IPR strategy, dissemination and technology transfer activities to ensure the full exploitation of the innovations and scientific results will be carried out. The active involvement of companies in the project, representing the different sectors active in the value chain will ensure wide exploitation with realistic objectives and plans. Communication, Dissemination and Exploitation Plans will be developed, including the PEDR.	EUFIC	All
WP8	Project activities of coordination, efficient grant management (resources optimization), communication with the EC, risk management	AIMPLAS	All
WP9	Ethics requirements	AIMPLAS	All

3.1.2 Graphical presentation of the components showing how they inter-relate (Pert chart).



3.1.3 Timing of the different work packages and their components (Gantt chart).

Num.	Work Packages and Related Tasks	Partners Num.	Delivera-bles	Project Months																	
				2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36
WP1	**Definition of requirements & selection of raw materials.**	**All (9)**	**D1.1-D1.4**				H		H												
T1.1	Selection of food packages and their main requirements	1,3, (9)10,11	D1.1		♦																
T1.2	Selection of biopolymers and renewable intermediates	(1), 4,5,6, 9	D1.2			♦															
T1.3	Selection of by-products for middle chain PHAs	(11),8	D1.3				♦														
T1.4	Component definition for active coated materials	(3), 7, 10, 11	D1.4						♦												
WP2	**Production of GA & PHA middle chain from renewable resources.**	**1, 2,4,(5),6,8, 11**	**D2.1-D2.7**						H							H					
T2.1	Production of modified mcl-PHA	1, 2,(8), 11	D2.1																		
T2.2	Characterization of the modified mcl-PHA	1,(8)	D2.2										♦								
T2.3	Improve the processability of develop PHA middle chain	(1),6, 8	D2.3												♦						
T2.4	GA Production	(5)																			
T2.5	GA Isolation and Purification	4, (5)	D2.4													♦					
WP3	**Production of PGA and hybrid coatings.**	**1, (4)**	**D3.1-D3.3**									H						H H			
T3.1	Glycolic Polycondensation,	1, (4)																			
T3.2	PGA polymerisation	(4)	D3.1									♦									
T3.3	Process coupling – sol-gel reaction for a PGA hybrid coating	(1)	D3.2													♦					
T3.4	Process coupling – in situ modification	4,(1)	D3.3															♦			
WP4	**Antifungal and antimicrobial coatings.**	**3, (7),1**	**D4.1-D4.4**								H							H			
T4.1	Antioxidant Active coating	(7)	D4.1											♦							
T 4.2	Evaluation of activity against microorganisms: active substances and formulated coatings.	(3)	D4.2-4.3																		
T4.3	Antimicrob.	(3)	D4.2-4.3								♦			♦							
T4.4	Upscaling of the coating and lamination process	(1),3,7																			
T4.5	Prevalidation of pilot scale films with selected food	3,(7)	D4.4															♦			
WP5	**Fabrication of packages, coating application & full validation by industrial partners.**	**(1), 3, 6, 7, 9, 10, 11**	**D5.1-D5.5**												H				H H	H H	
T5.1	Production of monolayer films using Bio-PE and PHA	1, (6)	D5.1												♦						
T5.2	Production of monolayer sheets using PHA and bio-PET	(1), 10	D5.2																♦		
T5.3	Application of the hybrid and/or active coating over the developed films and sheets	(1),3,6,7	D5.3																♦		
T5.4	Manufacture of multilayer packages for snacks	1, (9),11	D5.4																		

Num.	Work Packages and Related Tasks	Partners Num.	Delivera-bles	Project Months																	
				2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36
T5.5	Characterization of the coated films and sheets	1,3,7,(9),10,11																		♦	
T5.6	Testing packaged foods	(3),7,9,10,11	D5.5																		♦
T 5.7	Analysis of recyclability.	All (1)	D5.6																		
T5.8	Biodegradation of the PHA package at lab scale.	(1)																			♦
WP6	**Environmental, regulatory, social & economic analyses.**	**All (11)**	**D6.1-D6.3**																H	H	H
T6.1	Economic Analysis	All (8)	D6.1, D6.4									♦									♦
T6.2	Product Life-Cycle-Assessment	All (1)	D6.2																		♦
T6.3	Regulatory and safety analysis	All (1)																			
T6.4	Study of consumer perception	All (3)	D6.3																		♦
WP7	**Communication, Dissemination, Exploitation, Business plan & Technology Transfer.**	**All (12)**	**D7.1-D7.9**		H	H						H									H H H
T7.1	Communication & Dissemination Plan	All (12)	D7.2, D7.3		♦							♦									♦
T7.2	Development of project identity and website	All (12)	D7.1		♦																
T7.3	Production of dissemination materials for a target audience	All (12)	D7.3, D7.8									♦								♦	♦
T7.4	Exploitation	All (2)	D7.4			♦															♦
T7.5	Business Models & Plans for KER	All (2)	D 7.6																		
T7.6	Intellectual Property Management	All (1)	D7.4, D7.5			♦															♦
T7.7	Technology Transfer	All (12)	D7.7									♦									♦
WP8	**Project Management.**	**All (1)**	**D8.1-D8.4**			H															
T8.1	Project activation and meetings	(1)	D8.1-D8.4	╬		♦			♦			╬			♦						♦╬
T8.2	Project monitoring and communication with the EC	(1)																			
T8.3	Consortium management, financial and legal issues	(1)																			
T8.4	Risk Management	(1),2																			
WP9	**Ethics requirements**	**(1)**	**D 9.1-D 9.2**																		
T 9,1	The ethical and societal impacts study	(1)	D 9.1			♦															
T 9.2	Ethics issues	(1)	D 9.2			♦															

3.1.4 Technical Risks.

The table 3.1.b summarizes the technical risks that have been anticipated during the preparation of this project, including contingency measures to prevent or mitigate them, in order to ensure that the outcomes of the project can be achieved. The previous experience, continuous collaboration and resources devoted to risk management will minimize potential problems when they arise.

Table 3.1.b. *Risks and mitigation proposals for different activities to be performed in the* RefuCoat *project.*

Description of the Risk. (Risk Level*, and Likelihood (%)).	**WP**	**Proposed risk-mitigation measures**
Characteristics of raw materials available unsuitable for the process envisaged (M, 10%)	1	Pretreatments and supplementation will be applied to modulate the composition of substrates to meet process demands. Preliminary analysis has been conducted and support substrate potential.
Broad variety of final products requiere different packaging characteristics (L, 20%)	1	Product specifications will be set and prioritized attending to the market potential of each application. Whenever possible consensus specifications will be agreed upon.
Low yield on mcl-PHA synthesis (H, 20%)	1, 2	Mutant strains with improved properties will be used for optimized PHA production. Co-feeding of supplementary substrates as inducers of mcl-PHA for enhanced yields.
Thermo-mechanical properties of mcl-PHA are inconvenient for processing.	2	Previous experience of partners enables additivation of raw mcl-PHA to meet processing requirements (including plasticizers, lubricants, and compounding).
Low reactivity of MEG due to the presence of inhibitors (H, 20%).	2	Alternative catalyst routes and processing parameters will be explored. Increase of residence time of reactive extrusion by combine more than one extruder in a tandem configuration. Remove impurities using the expertise of THUNEN.
Low yield of PGA synthesis (M, 30%)	3	
Thermo-mechanical resistance of PGA is not enough for further processing Reduction in biopolymer properties, bad smell. (H, 30 %).	3	Different compatible antioxidants, lubricants and plasticisers will be tested to reduce the action of compounding extruder over the material. Use of low shear screw elements and minimize residence time thanks to the modular configuration of multi screw extruders.
Low yield of PGA reduction. Bad inorganic network due to low number or hydroxyl groups. (M, 30%)	3	New strategies for improving the number of hydroxyl group in PGA (new reductant). Cost of new strategies will be considered.
There are too many active substances to be studied. (M, 50 %).	4	We will choose those with consolidate data from bibliography and 3-4 as supplement for research reasons. We will elaborate a reserve list.
The selected substances don't show the expected activity (L, 20 %)	4	Extend the tested substances to the reserve list.
Poor solution or dispersion of active substances in coating solution. (M, 50 %).	4	Change to coating formulations including emulsifiers and stabilizers
Poor wetting of film surface with the coating and/or inadequate adhesion coating to substrate: (L, 20 %).	4	Adjust of surface energy: corona treatment and/or use a new primer on substrate.
Active coated materials developed do not show activity. (L, 20 %).	4	Check the activity in formulation coatings, before applying on the substrate. Increase active substance concentration.Register and guarantee that the time between material preparation and the tests is minimum.
Shelf-life of tested food is not increased. (L, 20 %)	4	Check the "in vitro" activity in coated films Increase active substances concentration in coating formulation.

Hybrid coating mixed with active compounds does not allow the suitable release of the active compounds (M, 20%)	4	The active coating will then be applied as a subsequent layer and the controlled release will be tuned as described in subtask 5.1.3
Probe sizes not enough for standardized tests (L, 20 %)	5	Adapt the standards and register deviations from them.
Poor barrier properties of new films and sheets. (L, 70%)	5	Increase coating thickness or increase coating solid contents in order to obtain better barrier properties
Final film properties is not enough to fulfil the selected food package requirements (M, 20%).	5	Additional layers will be proposed and optimized using commercial biodegradable materials. At industrial level up to eleven layers can be used just for one films.
Non organoleptically neutral taste of the active coatings. (M, 20%)	4,5	The selection of the bioactive substances and binder will be made in tune with the foodstuff to be coated or packed respectively. Snacks is expected to accept well selected herb essential oil flavour.
Final film properties are not enough to fulfil the selected food package requirements (M, 20%).	5	Additional layers will be proposed and optimized using commercial biodegradable materials. At industrial level up to eleven layers can be used just for one films.
RefuCoat's products might be relatively expensive (M, 40%)	5	To revaluate and improve the production process to minimize energy consume or to look for alternative and cheaper energy source. To look for alternative catalysts that improve yield or reduce energy consumption.
LCA or Energy consume is higher than current process (H, 10%)	6	

*****L=low risk, M=medium risk, H=high risk

3.2 Management structure and procedures

3.2.1 Management structure and procedures

This BBI consortium will be effectively coordinated by Partner AIMPLAS. AIMPLAS has wide expertise in participation and coordination of international level, and most importantly, to support product development and technology transfer from lower Technology Readiness Levels to market launch, meeting the needs of end-users, generally large companies responsible for final product commercialization. The organizational structure has been developed to represent such transference process, key to the success of market-driven RefuCoat, and is shown in Figure 3.2a.

The main decision-making and arbitration body of the consortium will be the General Assembly (GA), consisting of up to one permanent representative per partner, that will be lead by AIMPLAS. GA will delegate day-to-day, short-term decision making into the Executive Board (EB), that will always require formal ratification from GA. GA will decide on the project objectives, consortium agreement, project work plan and deliverables, risk assessment and technical and financial modifications, nomination of EB members, acceptance of new partners or third parties in the consortium, action against non-committing partners, shifts in budget and allocation of resources and most importantly on approval of IPR actions and exploitation strategy.

The **Executive Board (EB):** one senior PI from each partner and will be chaired by AIMPLAS. It will meet at least once every six months, or more often should the need arise. The EB represents the highest level of decision-making within the project. It will make decisions regarding the management issues including technical exploitation, financial, planning and control matters e.g.: monitor the progress of the project against the set objectives, deliverables and milestones; and define, allocate and develop tasks in accordance with the Project Technical Annex. It will also make decisions regarding the overall project strategy (e.g. project re-organisation and contractual matters, exploitation matters when they arise (IPR)) and consult with the EC regarding contract items: If needed, the EB will have the ability to amend the scientific objectives of the project, identify potential new project partners, propose any amendment to the grant agreement (GA) or to the consortium agreement (CA) and if needed, convene the external experts on any specific Scientific, IPR, security and regulatory issues if any arise. The EB represents the interest of all partners. Its composition and the rules of proceedings will be established in the CA, complementary to the

Commission contracts and will take into account the individual aspects of the consortium and specific aspects of the project. All partners will sign the CA before the project starts.

An **Advisory Board (AB)** will be appointed, composed by up to four external members of worldwide reputation on the activities related to the project, such as final use of the product or product development strategies (specialty plastics, innovation management, IPR, marketing issues), to complement the views of industrial partners. The members of AC will be selected by the GA. The AB will meet once a year at the annual meetings. The role of AB will be to provide an independent assessment and external quality control on the progress of the project in order to advise GA and EB on priorities or strategies to be implemented by the consortium when required. The expert(s) will be allowed to follow the progress along the entire project after signing a confidential agreement.

The **Project Coordinator** (Ms. Itziar Carracedo, from AIMPLAS) will be the cornerstone of the consortium, being the only intermediate for communication and legal representative of the consortium before the EC, and reporting to the GA and the EB. PC will be in charge of overall technical, financial and administrative management of the project, submission of contractual reports, deliverables and financial reports to the EC and funding management, taking responsibility for receiving and distributing project funds, in accordance to EC guidelines and Consortium Agreement. The PC will foresee the signature and eventual amendment of the Consortium Agreement. The PC will be the channel for back and forth communication with the EC, for the efficient achievement of all duties of the consortium in accordance to the contract.

The PC will be supported by the **Administrative Manager (AM)** of the project, that will be responsible of the back-office activities related to general management of the project and contractual obligations towards EC, particularly, creation of project identity and templates, supervision and revision of periodic reports and deliverable and milestone schedule, day-to-day interaction with partners, foreseeing of risk-assessment and mitigation measures, preparation of meeting minutes, gathering and sharing of presentations by partners, completion of questionnaires required by the EC, organization and coordination of the periodic Project Meetings, particularly with the EC representative, including selection of meeting venue, agenda and logistical arrangements in cooperation with each hosting partner, collection of financial statements whenever required. Both PC and AM will be responsible for specifically controlling non-technical hurdles that may arise during execution of the project (see Section 3.2), supported by GB.

Work Package Leaders (WPL). WPL are already identified amongst the partners involved in each WP. All WPLs will meet at technical coordination meetings, arranged appropriately according to the needs of the project. Each WPL represents the Project Coordinator at WP level and acts as a sub-project coordinator, responsible for:

- Coordinating WP activities and ensuring collaboration between partners and among the WP,
- Reporting task activities to the coordinator in a quarterly Progress Reporting Form,
- Monitoring progress and ensuring that milestones and deliverables of the WP tasks are fulfilled according to the Work Plan,
- Facilitating communication and transfer of knowledge and materials between WPs,
- Scheduling, if needed, special meetings to determine suitable measures to be taken.

The Intellectual Property and Exploitation board (IPEB) will be appointed by EUFIC, at the onset of the project and will be composed of one senior representative of the technology transfer department of each partner and commercialization executives from industrial partners. This board will be in charge of assisting WP leaders, on request, in identifying knowledge that could be the subject matter of protection (foreground IP), reviewing the exploitation plans, and advise on any issues that could be related to dissemination of potentially IP sensitive data on the basis of unpublished articles and activity reports issued by activity leaders. An Exploitation Manager will lead this board, chosen at the outset of the project from amongst the partners.

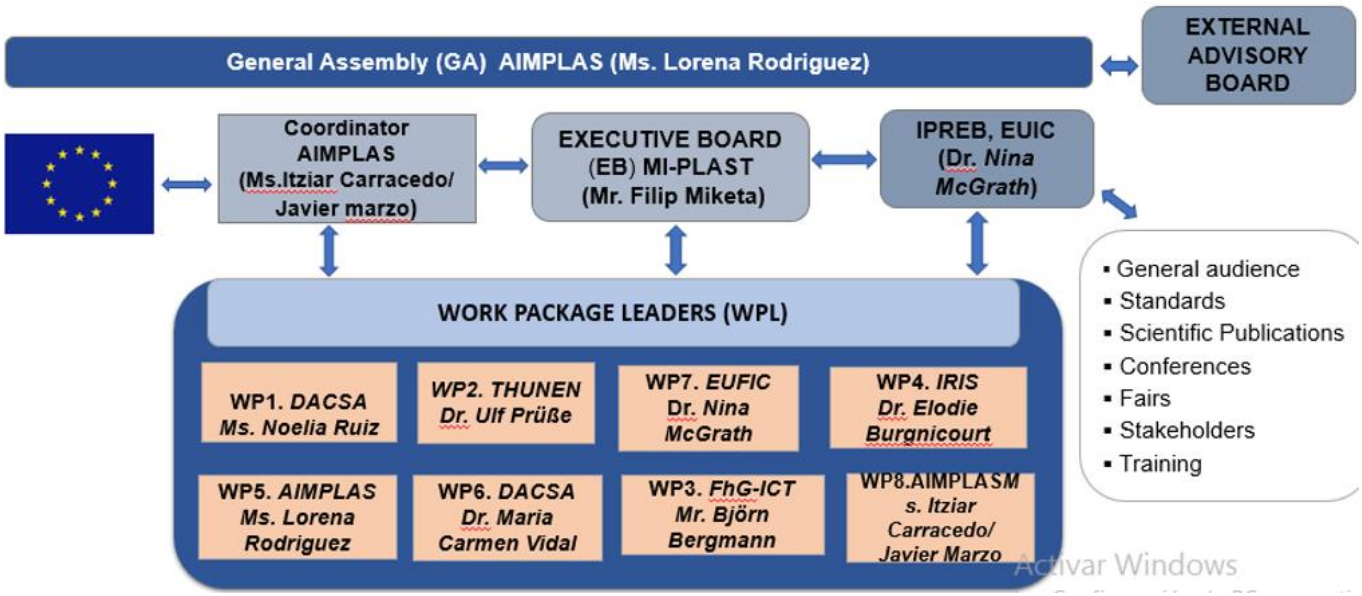


Figure 3.2a Organizational structure of the consortium

3.2.2. Innovation management strategy

Progress reporting

Each partner and WP Leader will formally report, twice a year, the progress of the work performed to the project coordinator by issuing a Progress Reporting Form. The reporting shall include information about the technical progress, the results obtained (e.g. deliverables) and the compliance with the work programme. The progress status will also be reported in terms of percentage of completion, estimated time for full completion, actual person-months spent and person-months still needed for full completion. The periodic reports (contractual), based on Technical/Quality/Cost/Delay information from all partners, will be issued according to the periods defined by the EC and presented to the EC for approval together with all financial audit certificates justifying the costs incurred for the reporting period. A final report and a Mid-Term Assessment Report (presented below) will fully complete the reporting activities of the RefuCoat project.

Protection of innovation and IP plan

An IP and exploitation plan will be prepared as part of WP8 with the cooperation of all partners at mid-term and updated at the end of the project in order to ensure adequate protection and exploitation of the developed database. A CA, which is under preparation and will be signed before the start of the project by all partners, will clearly state that knowledge shall be the property of the partner generating it. Where several contractors have jointly carried out work generating knowledge and where their respective share of the work cannot be ascertained, they shall have joint ownership of that knowledge and shall be entitled to use and license it without owing any financial compensation to each other. The partners shall agree amongst themselves the allocation and terms of exercising ownership of that knowledge

Mid-Term assessment

The Project Coordinator will organise a Mid-Term assessment meeting (before the end of the 30th month from the starting date of project) with all the partners and EC representatives. Prior to the meeting, the Mid-Term Assessment Report on the progress of the research and the partners' plans for future exploitation strategy will be submitted to the EC. The goal of the meeting will be to present the content and objectives of the second part of the contract as well as procedures to manage future exploitation of results. A decision will be made taking into account EC representatives advice regarding the objectives specified at this stage for the technical/scientific progress and the exploitation perspectives for the results.

Conflict management strategy (in addition and prior to Consortium Agreement)

The project's conflict management strategy is achieved through the following key goals:

- Discover and resolve issues before they become serious conflicts,
- Create a climate of trust where partners feel free to exchange any ideas,
- Encourage and engage partners to speak their minds and without hidden agendas.

The three key activities are organised as follows:

- Review the current project progress at periodical meetings to be able to detect any possible problems
- Create a list of activities (list of issues to be solved) where project issues are captured and their status

(open, under investigation, deferred, fixed etc.) is remembered,

- Monitor issues through an issue management process, consisting of: detection, recording, analysing, prioritising and allocating ownership of issues.

The following problem escalation path (to be solved on the lowest level, when possible) is defined as follows: 1) WP partner; 2) WP Leader; 3) Coordinator; 4) Governing Board.

Degree of excellence management

Throughout the duration of the project, the quality of the results achieved will be continuously monitored to determine whether the project is meeting the quality standards and to identify, when necessary, ways to eliminate causes of unsatisfactory results. Project results include deliverables as well as project management results such as cost and schedule performance. Quality monitoring will be performed by the "degree of excellence" management process and will be implemented within the first month of the project. A balanced scorecard will also be implemented. It will integrate key performance indicators to monitor the degree of excellence of the management process throughout the project's life.

Non technical risks management

Besides technical risks identified previously in Task 8.4 (please refer to Work Plan, WP8), management/operational risks have also been assessed. Such non-technical hurdles will be specifically controlled by PC and AM as part of their duties, supported by EB. AIMPLAS, have a track record of successfully implemented projects at EU level, as have many of RefuCoat partners. The management structure of the project has been devised to prevent problems and is enforced by the previous cooperation existing among partners, providing a safety network for the successful completion of the project, based in open and enduring cooperation.
Such non-technical risks are referred in Table 3.2b.

3.3 CONSORTIUM AS A WHOLE

3.3.4. Overall description: adequacy, complementarities, representatively of the value chain and synergies.

RefuCoat consortium has a well-balanced composition. The RefuCoat consortium brings together a wealth of expertise and resources within the areas, including partners that propose the best approaches for producing full recyclable food packaging with enhanced barrier properties and new functionalities. The presence of main actors in the whole value chain demonstrates the critical mass of complimentary resources that will enable the RefuCoat project to achieve its targeted societal, industrial, and scientific breakthroughs and commercial success. Each partner has a clearly defined role within the project and will contribute specific expertise that will enable the project success. **SME partners** (MIPLAST, IRIS and BIOPOLIS) and the **IND partner** (GRUPO APEX, MANOR and DACSA) will provide the fundamental knowledge, technologies and expertise in the abovementioned areas required to achieve the project objectives, as their activities within the project are fully in line with their own core business and research strategies.

Figure 3.3.a shows the expertise of each partner in the project and their distribution around Europe. **RTO Partners** (THUNEN, AIMPLAS, Fraunhofer, AINIA and CSIC) have the necessary research expertise and will provide support to the partners to achieve their objectives. Each partner has demonstrated a long-term commitment to the development of their own research, policy and commercial activities within the industrial targeted by the project.

All of them therefore offer excellent potential for a lasting integration of activities and the creation of a pan-EU culture for

Figure 3.3.a. *Sectors and expertise brought by the RefuCoat consortium*

the technologies, materials and processes developed.

The mobility of resources and personnel will be encouraged throughout the project, not only helping to form strong bonds between the consortium members. Those actions are going to be implemented in order to lead towards an integration of complimentary activities, services and future research projects to improve the composite multi stage processing at EU level. This is a complete storyboard of actors complemented by an advisory board to extend industrial and technical insight.

The table below (Table 3.3.a) gives a brief overview of each partner specific business and their expertise, and the main tasks in the project allocated to them based on that expertise.

Table 3.3.a RefuCoat Participant Expertise and Tasks Distribution.

Part num	Participant short name	Participant business / expertise-	Main role in the project
1	AIMPLAS	Research Technology Institute expert in polymer materials, nanotechnologies and processing.	Project management. Development of high barrier PGA hybrid coating. Preparation of packages for end-user's validation in collaboration with NIPLAS
2	BIOPOLIS	Leading biotechnology SME. Expertise on isolation, design, validation and production of microorganisms and cellular metabolites for agri-food, pharmaceutical and chemical industries.	. Economic analysis. Exploitation.
3	AINIA	Technological Center devoted to Research & Development in Agro-food industry integrating Biotechnology, Nano-technology; Food technology; Chemical technologies among others	Developing active coatings for spoilage and pathogens microorganisms, combining antimicrobials with a broad spectrum for the spoilage ones and bacteriophages (endolysins) against pathogens.
4	Fraunhofer	Leading applied science association in Germany, research thermoplastic and thermoset materials, strong background in integrated compounding processes, reaction setup and analytics thereof	Development of GA polycondensation, cyclising depolymerisation and glycolide purification, development of process options for PGA polymerization, development of a process for in situ PGA modification
5	THUNEN	Federal research institute with expertise in the biotechnological and chemo-catalytic conversion of renewables, the production of biobased polymers and sustainability assessment of bio-based systems.	Chemo-catalytic production of glycolic acid (GA)
6	MIPLAST	Croatian company focused on processing of LDPE and HDPE materials, their packaging and production, as well as their regeneration and recycling.	Production of PHA and Bio-PE films for package savoury products and cereals.
7	IRIS	Engineering company specialized in process monitoring and control, and process optimisation.	Antioxidant active coating for savoury and cereals products.
8	CSIC	The Biological Research Centre (CIB) is a public funded national research institute that belongs to the Spanish National Research Council (CSIC).	Production of PHA for new poultry packages. Sheet coating and thermoforming.
9	GRUPO APEX	Food company specialized in savory products.	Validation of packages for savoury products as alternative to metallized packages.

10	MANOR	Largest chicken processor in the Republic of Ireland, exporting to countries all over Europe among other areas.	Validation of packages for poultry products as alternative to multilayer MAP packages.
11	DACSA	Leading food company focused on cereal products, mainly those derived from corn and rice.	Validation of packages for cereals products as alternative to metallized packages. Production of modified mcl-PHA
12	EUFIC	Non-profit organisation that communicate science-based information about foodand health to different stakeholders	Dissemination and communication activities.

The participation and leadership of SMEs and IND in RefuCoat is clear as they represent the 50% of the Partnership and around 31% of the budget and 20% EC contribution**.** In section 3.2 (management structure and procedures), it can be seen that the relevant Project Governance Bodies (General Assembly, Executive board, Technical Management Committee, Innovation Management Board and IPR-Exploitation Board) **have a majority of SMEs**, having the decision-making power in all of them. Also 3 of 9 WPs are led by companies. All of them demonstrate a strong commitment to the project and understand the responsibilities that this brings with it and have assumed a fundamental role within the project work programme, as clearly summarized in Table 3.3.b.

Table 3.3.b *RefuCoat Partnership Distribution.*

Partner type	Nº	Percentage of partnership (%)	Total effort (p/m)	Percentage of effort (%)	Total Budget	EU/BBI REQUEST	EU/BBI %	In kind contribution	contribution (%)
SME	2	16,67%	77	15%	558.000,00 €	457.560,00 €	20%	100.440,00 €	11%
IND	4	33,33%	85,7	16%	487.200,00 €	0,00 €	0%	487.200,00 €	52%
RO	5	41,67%	343,4	66%	2.012.264,25 €	1.698.137,43 €	74%	314.126,82 €	34%
OTHER	1	8,33%	15	3%	176.875,00 €	145.037,50 €	6%	31.837,50 €	3%
TOTAL	12	100,00%	521,1	100%	3.234.339,25 €	2.300.734,93 €	100%	933.604,32 €	100%

To obtain the complementary range of expertise and the critical mass needed (both in human and financial terms), it is necessary to bring together, companies and ROs representing the entire supply chain and R&D areas, working in close cooperation to solve a common problem, which cannot be found within a single EU Region, requiring the integration of resources from across EU.

RefuCoat partnership is composed by 12 organizations from 4 EU countries, having the required pan EU approach essential for achieving the objectives of the project since it is required to address the EU social and Industry needs in the field with a wide EU perspective. Furthermore, to ensure that the long-term and widespread take-up and commercial success of the RefuCoat results, it is important that the project innovation and research activities are undertaken with a strong knowledge of the different industrial sectors addressed. The Consortium therefore ensures the participation of different EU regions, whose roles within the project are clearly defined and represent strategically important regional objectives, which will address the overall objectives of enhancing European industrial competitiveness, moving towards a circular economy and hence **providing a clear EU added value**.

3.3.5. Industrial/Commercial involvement to ensure exploitation of the results.

RefuCoat **project clearly has a number of benefits for the involved SME** as we explain on detail along the proposal. It enables the transfer and market replication of the new materials for packaging that contributes to the developed well-defined and high market products, supporting the take-up of the new advanced technologies. Besides, RefuCoat will create a new and more open market/supply chain for the materials, technologies and sales as previous figures shows.

The involvement of all the participants during and after the end of the project, in order to strengthen the industrial and commercial exploitation, is a key aspect of the consortium relationship, implemented through the innovation and IPR management strategies resulting in the Business Plan included in the Final PEDR.

During the project. Throughout the life of the project, all project partners will be responsible for updating the project 'Plan for the Exploitation and Dissemination of Results'. Dissemination will target 'take-up' actions within EU Local Authorities-Public Bodies, Stakeholders, SMEs and End-Users to promote the growth of the materials and technologies developed into new and existing market sectors; and to encourage SME participation within the RefuCoat supply chain. Publication and dissemination of project knowledge and results will be conducted on

agreement between the consortium partners, led by the EUFIC supported by BIOPOLIS ensuring that the SMEs and industry interests are not only taken into account but are the real drivers for the exploitation strategy. Managed by the **IPREB**, each project partner will be required to develop their own 'Exploitation Plan' that will demonstrate how the project results will be integrated into their own forward business plan and research actions. Exploitation activities will include: protection and management of innovations, IPR and project results, gathering of reliable market intelligence and forecasts, feasibility studies for the creation of spin-off activities, evaluation of the project results for commercialization and application within new markets, steps to implement the proper industrial manufacturing policies, and the development of effective support structures to reinforce supply chain. Such activities are integral part of WP7.

After the end of the project. As explained on section 2.1, all the partners will obtain economic profits from the commercialization of the innovative RefuCoat new materials and logistic implementation and will be able to introduce to the market cost competitive and sustainable products, increasing/opening markets and besides, to obtain profits by licensing the project results related to their knowledge field. However, as it will be detailed in the Business Plan and PEDR, the percentages of these profits will be different among the Partners depending on the exploitation agreement reached among co-owners which will reflect the percentage of contribution of each Partner during Project execution, the Background brought by them to achieve such result, and so on. Furthermore, these economic profits will encourage all the partners to contribute in the initial investment necessary to set up the exploitation process, once the project is finished. The SMEs participants are economically and managerially capable of performing the exploitation of the results, based on their turnover, their own investment in R&D and the level of economic contribution to the project. The other partners have also the right competences to guarantee the RefuCoat exploitation of results in their areas. All partners are highly convinced of the potential impact of the project results in their businesses, and neither the SMEs, nor the other partners have conflict of interests regarding the areas of commercialization/exploitation, from a geographical, technological or market sector point of view. More details are provided in section 2.1., where apart from the exploitation interest, the steps to achieve the foreseen impact are described.

Industrial participation is very strong in this project. The consortium includes seven industry partners, with multiple and complementary technological, research and business expertise, covering the full spectrum of the activities proposed in RefuCoat. They are dynamic enterprises with an active intention and outlook to expand their product/service portfolio into new markets. All project industrial partners planning to directly capitalize on the RefuCoat results as we explain in detail in impact section, either by expanding their existing products/services, or by licensing technology. The geographical dispersion of the industrial partners ensures that they will cover an important proportion of European markets.

3.3.6. Other countries.

In the RefuCoat consortium there is one participant from the Associated Countries, Turkey, that as the Article 7 of the H2020 Regulation says is fully entitled to participate in this call for proposal with exactly the same right as they are considered eligible partners.

3.4 RESOURCES TO BE COMMITTED

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4. MEMPERS OF THE CONSORTIUM

4.1. PARTICIPANTS

PARTNER 1 . COORDINATOR **Full name**: AIMPLAS-Asociación de Investigación de Materiales Plásticos y Conexas (AIMPLAS) **Web page**: www.aimplas.es	**Country:** Spain **Type:** RO	AIMPLAS PLASTICS TECHNOLOGY CENTRE

Brief description of the organization:
AIMPLAS, Technological Institute of Plastics located in Valencia, is a private, non-profit Association with more than 500 associated companies created in 1990. AIMPLAS is formed by +120 highly skilled professional, more than 65% with a Masters, Engineering or equivalent degree in Chemistry, polymer engineering, materials engineering or equivalent, including 15 PhD.
AIMPLAS' fields of work are related to technological research and development on thermoplastic and thermosetting plastic materials & products, its transformation processes and their recyclability and sustainability. AIMPLAS generates new knowledge and technologies that can be transferred to companies in order to help them to increase their effectiveness and competitiveness.
AIMPLAS has a broad expertise in the fields of recycling, compatibilization, reactive extrusion, biopolymers and renewable source materials, special assisted processing technologies and nanocomposites, gained during the last years in the frame of EU and national projects. Nowadays, AIMPLAS is involved in more than 25 European projects and has participated in 77 projects in FP5, FP6, FP7, LIFE+, CIP-EcoInnovation EU Programmes, among others, coordinating 27 of them. At National and Regional levels AIMPLAS participates in around 100 projects yearly.
AIMPLAS, as RO is focused to help companies in the plastic sector to develop new products and increase their competitiveness through innovation, has more than 20 pilot plants representing the most relevant polymer/plastics/composites production technologies present in the industry nowadays and has state-of-the art test facilities for chemical, optical, morphological, mechanical and physical characterisation. These pilot lines and laboratories are used by many customer's every year allowing them to test new materials, optimize production processes and launch new products to the market, supported by AIMPLAS technical staff, resulting in more than 5000 assays, 170 technical assessments and 120 skills training actions to more than 1500 clients per year. AIMPLAS has state-of-the-art 8500 m2 facilities, including thermoplastics and thermoset pilot plants, analysis and testing laboratories (physical-mechanical, chemical, packaging, automotive and construction) and training areas.
Role in the Project: AIMPLAS will coordinate the project and participate in the different WPs, being its main involvement on WP3 in the Development of high barrier PGA hybrid coating and WP5 in the Preparation of packages for end-users validation in collaboration with MI-PLAST and AINIA. As Project Coordinator AIMPLAS will be key in WP8.

Scientific team:
Dr. Amador García (M), PhD in Organic Chemistry by the University of Valencia, working in AIMPLAS as researcher in the chemical laboratory since 2009. He has been involved in several regional and FP7 European projects focused in nanotechnology areas, materials from renewable resources and coatings with enhanced properties, etc. (SUPIN, TRAINER, NMP INNOREX, RECAN, KBBE BRIGIT, KBBE BISIGODOS, BIOREFINE-2G). Before joining to AIMPLAS, he worked as researcher assistant in CIPF (Centro de Investigación Principe Felipe) and in the University of Valencia. During 2007 he was in the chemical group of Professor Dennis P. Curran (University of Pittsburgh) as visiting researcher. In 2010 he received the Plastic & Innovation award.
Ms. Raquel Llorens-Chiralt (F), Master in Material Engineering by the Polytechnic University of Valencia, working in AIMPLAS as Senior Researcher since 2005 in the Compounding Department, focused in new material formulations. She has been involved as main researcher in the technical execution of several FP7 Projects focused on nano-materials, biopolymers, flame retardants, blends, mineral and natural fibres composites and PVC. Before joining AIMPLAS she worked in different private industries (paints, ceramics and metallurgical).
Dr. Ana Espert (F), PhD in Polymer Technology by the Royal Institute of Technology (KTH) in Stockholm

(Sweden). Since 2005, she is working as a Senior Researcher in AIMPLAS. In this position, she is in charge of the technical coordination of several of the Projects in which AIMPLAS participates (+75 Projects since 2000, 36 of them FP7 in 18 as Coordinator). Dr Espert has been the technical coordinator of several FP7 Projects focused in biotechnologies, materials from renewable resources and sustainability (Collective FLOWFREE, R4SMEs ECOBIONET, NMP BUGWORKERS) and is participating in others such as NMP ECOPLAST and KBBE BRIGIT. Before joining AIMPLAS, she worked at the Royal Institute of Technology, where she was also involved in EU projects Technical and Administrative Management of FP6 projects (ECOSITES, PRONACOM). She is the main author of more than 10 peer-reviewed articles in Scientific Journals and Conference proceeding in the field of Polymer Technology

Mr. Santiago Gálvez (M). Degree in Agricultural Engineering by the Polytechnic University of Valencia, and working for AIMPLAS as Project Manager in the International Projects Department since 2010. Mr Gálvez has participated in many EU projects focused in biotechnology, materials from renewable resources and sustainability; expertise in plastic materials and plastic processing, and in management of EU funded projects. Involved in the project coordination and/or management activities of some FP7 (R4SME THERMALCOND, R4SME REFRESH, DEMO XXL-REFRESH, NMP COALINE, R4SME BIOFIBROCAR, KBBE FIBRAGEN, JTI WAVECOM), CIP-Eco-Innovation (CAYLEY), LIFE+ (ECOFLEXOBAG, BREAD4PLA, PISA), Leonardo (ENERGYWISE PLASTICS, iPOLYCOND)and Interreg IVB (CARBONINSPIRED, CARBONINSPIRED 2.0) projects and in the preparation of some others.

Publications:

• Signori, F., Boggioni, A., Righetti, M. C., Rondán, C. E., Bronco, S. and Ciardelli, F. (2015), Evidences of Transesterification, Chain Branching and Cross-Linking in a Biopolyester Commercial Blend upon Reaction with Dicumyl Peroxide in the Melt. Macromol. Mater. Eng., 300: 153–160.

• Campanella, A., Wool, R. P., Bah, M., Fita, S. and Abuobaid, A. (2013), Composites from northern red oak (*Quercus robur*) leaves and plant oil-based resins. J. Appl. Polym. Sci., 127: 18–26. doi: 10.1002/app.36663.

• R.M. González, M.P. Villanueva19 – Starch-based polymers for food packaging, Multifunctional and Nanoreinforced Polymers for Food Packaging,2011, Pages 527–570

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• Martí-Ferrer, F., Vilaplana, F., Ribes-Greus, A., Benedito-Borrás, A. and Sanz-Box, C. (2006), Flour rice husk as filler in block copolymer polypropylene: Effect of different coupling agents. J. Appl. Polym. Sci., 99: 1823–1831

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Projects:

Project acronym/description. Scope End Year Website

BUGWORKERS: New tailor-made PHB-based nanocomposites for high performance applications produced from environmentally friendly production routes. FP7-NMP -2009-SME 2013 www.bugworkersproject.eu/.

BRIGIT: New tailor-made biopolymers produced from sugars obtained from lignocellulosic residues for flame retardancy highly demanding applications. FP7-KBBE-SME 2017 www.brigit-project.eu/

INNOREX: Realising the continuous, highly precise, metal-free polymerisation of PLA using alternative energies for an innovative reactive extrusion. FP7-NMP-2012 2016 www.innorex.eu/

BIOREFINE 2G : Development of 2nd Generation Biorefineries Production of Dicarboxylic Acids and Bio-based Polymers Derived Thereof FP7-KBBE 2013 2017 www.biorefine2g.eu/

BISIGODOS: High value-added chemicals and BIoreSIns from alGae biorefineries produced from CO2 provided by industrial emissions. FP7
KBBE 2013 2017 www.bisigodos.eu

Technical equipment:

- Equipment for mechanical, thermal, rheological and physical tests accredited to perform plastic characterization using the EN/ISO standards
- Rheometer, Optical and electronic Microscopy, DMA, DSC, Thermogravimetry. Raman Spectrometer.
- Twin-screw co-rotating extruders with gravimetric feeders (to feed solid and liquid additives) and other multiscrew planetary extruder.
- Thermoplastic injection machines.
- Multilayer blown film co-extrusion line and a pilot plant lamination unit.

PARTNER 2. **Full name**: BIOPOLIS, S.L. (BIOPOLIS). **Web page**: www.biopolis.es	**Country**: Spain **Type**: SME	BIOPOLIS Tailor-made Biotechnology

Brief description of the organization:
Biopolis, S.L. is a leading biotechnology company based in Valencia (Spain) offering contract research and manufacturing. Our core competences are the isolation, selection, design, characterization, validation and production of microorganisms and cellular metabolites with applications in agri-food, pharmaceutical and chemical industries. This results into the development of microbial strains and metabolites (most of them biopharmaceuticals) by strain and bioprocess design followed by fermentation and downstream process development and the production at an industrial scale. Prototype projects are (i) the selection, validation and production of starters and probiotics, (ii) the selection and production of enzymes, (iii) the identification, characterization and process development of functional ingredients, (iv) the overexpression of recombinant protein in microbial systems, (v) the overproduction of synthesis intermediates and active pharmaceutical ingredients by fermentation, (vi) the design of novel processes based on microbial and enzymatic catalysis.
Role in the Project: BIOPOLIS will participate in WP2, in the production of modified mcl-PHA; in WP6 doing the economic analysis of the outcome of the project; in WP7 in the Exploitation of results and dissemination and communication tasks: and in WP8.
Scientific team:
Dr Daniel Ramón Vidal (M) CEO and CSO of Biopolis S.L, with more than 160 publications and patents in microbial biotechnology (selection of microbial starters including probiotics, fermentation development and scale-up, strain improvement) and biotechnological production of microbial metabolites
Dr Marta Tortajada (F) Microbial Biotechnology Manager at Biopolis (Chemical Engineer, PhD in Automatics and System Engineering). She has 12 years' experience in protein overexpression in microbial systems and strain and bioprocess design for biochemical and bulk chemicals overproduction. From 2005 to the present she has been active in the preparation, management, coordination and / or development of several European R & D projects in the areas of bio-based industries, sustainability and revalorization of urban and agricultural wastes/residues (i.e. Integrated project FP6, KBBE-FP7, ERA-IB, EUROSTARS, NMP-FP7, CENIT, CIEN, etc.).
Dr Antonia Rojas (F) (Chemical Engineer, PhD in Biotechnology), head of Bioenergy Laboratory at Biopolis. She has large experience in microbial systems and strain selection, validation and bioprocess design for bulk chemicals overproduction.From 2009 to the present she has been active in the preparation, management, coordination and / or development of several European R & D projects in the areas of bio-based industries, sustainability and revalorization of urban and agricultural wastes/residues (i.e. Integrated project FP6, KBBE-FP7, ERA-IB, EUROSTARS, NMP-FP7, CENIT, CIEN, etc.).
Dr Silvia Segarra (F) (Organic Chemist, PhD in Biotechnology and Molecular Biology of Plants), head of Biocatalysis Laboratory at Biopolis.
Mr. Diego Rodriguez (M) Biochemist, MSc Biochemistry and Molecular Biology), responsible of bench-top fermentation and scale-up, both experienced in metabolic engineering and whole cell biocatalysis.

Five relevant publications, patents or products:
• Rojas A, Segarra S, Montesinos A, Tortajada M, Ramón D, Santos VE, Ladero M, García-Ochoa F, Ripoll V, Rodríguez A (2012) P201231119. Process for the production of 2,3-butanediol using improved strains of *Raoultella planticola*. BIOPOLIS SL – Universidad Complutense de Madrid. Priority date 17/07/2012.
• Boodhoo KVK, Cartwright CD, Vicevic M, Prieto MA, Tortajada M. Development of a Higee Bioreactor (HBR) for production of polyhydroxyalkanoate: hydrodynamics, gas-liquid mass transfer and fermentation studies. 2010. Chemical Engineering and Processing. 49, 748-758.
• Ripoll V, de Vicente G, Moran B, Rojas A, Segarra S, Tortajada M, Ramon D, Ladero M, Santos V. Novel catalysts for glycerol conversion into 2,3-butanediol. Process Biochemistry 51(6) 2016
• del Cerro C, Felpeto-Santero C, Rojas A, Tortajada M, Ramón D, García JL. Genome sequence of the butanol hyperproducer Clostridium saccharoperbutylacetonicum N1-4. Genome Announcements. 2013, e00070-13.
• Tortajada M, Ferreira da Silva L, Prieto MA. Second-generation functionalized medium chain-length polyhydroxyalkanoates: the gateway to high-value bioplastic applications. International Microbiology, 2013. 16, 1-15.

• del Cerro C, García JM, Rojas A, Tortajada M, Ramón D, Galán B, Prieto MA, García JL. Genome sequence of the methanotrophic poly-b-hydroxybutyrate producer *Methylocystis parvus* OBBP. Journal of Bacteriology. 2012, 194, 5709-5710.

Five Relevant Projects:
Project acronym/description. Scope End Year Website
SYNPOL. Biopolymers from Syngas Fermentation FP7 2016 http://www.synpol.org/
Optimized oxidoreductases for medium and large scale industrial biotransformations (INDOX) FP7 2017 http://www.indoxproject.eu/
Production of Organic Acids for Polyester Synthesis (POAP) 7FP. ERA-IB 2016 http://www.era-ib.net/poap
Production and Upgrading of 2.3-Butanediol from Biomass (PUBB) 6FP ERA-IB 2012 http://www.era-ib.net/pubb
3R2020+: From waste to resource by recycling Spain 2019 //3r2020.com/
Technical equipment:
9 specialized research laboratories (Biochemistry, Molecular Biology, Cellular Biology, Microbiology, Filamentous Fungi, Cell Culture, Chemistry, Analytics, and Bench-top fermentation) and an industrial plant with a total capacity of 9,000 L fermentation.

PARTNER 3 **Name:** Asociación de Investigación de la Industria Agroalimentaria (AINIA) **Name: Web Page**: www.ainia.es	**Type: RS** **Country: Spain**	ainia

Brief description of the organization: AINIA is a private non-profit industry association, formed mainly by companies from the Agri-Food Sector and related industries (packaging, cosmetic…). It was created in 1987 by the initiative of the industrial sector in conjunction with the Valencia Regional Institute of Small and Medium Industries (IMPIVA). With over 700 member companies and 1,200 clients, is one of Europe's technology centres with a largest business social base. It employs more than 190 people (most of them with a PhD and/or university degree). Our mission is essentially to "*add value to businesses by leading innovation and technological development in a responsible and committed way*". AINIA is accredited with Nº 3 in the Register of Centres of lnnovation and Technology of Spain; AINIA has a clear global vocation oriented to give an effective response to business needs from different perspectives aligning different disciplines such as food and packaging technologies and environment innovation with demonstration of industrial processes solutions and their later market replication. ainia belongs to the EEN (Enterprise Europe Network), to the Spanish Federation of Technology Centres (FEDIT), the Network of Technological lnstitutes of the Valencian Region (REDIT), the European Food lnstitutes Association (EFI Group), etc., as well as it has numerous collaboration agreements with entities and associations at national and international level. It work for the following sectors:
- Research & Development. Our main areas of research are the following: Biotechnology; Nanotechnology; Food technology; Chemical technologies; Electronics and Communications; Environmental and energetic technologies. The main applications of these technologies are: Food & Health; Food Quality & Safety; Industrial Design & Production; Sustainability
- Technological assistance. Product cost reduction, risk assessment and innovation management.
- Analytical Services. Chromatography (MSMS and Proteomic analysis), Chemical (nutritional and Isotopic Mass analysis) and Physical analysis.
- Specialised training for the food sector. E-learning, blended learning and in-classroom training.
- Food legislation activities. Public affairs. Advising companies in food legislation.
- Technology foresight. Innovation strategy plans development for companies.
- Consumolab. Consumer research unit (food sensory and market studies)

Ainia technology centre offers solutions covering the entire typology of innovation needs of small, medium and large companies:
With 13,000 m2 of facilities, **Ainia** has 10 pilot plants, 8 laboratories, 10 classrooms and training rooms, **Consumolab** (centre for analysis of consumer behavior through sensory analysis) and **Altex** (high-tech plant to provide a wide range of services based on supercritical CO2 extraction).

With a highly qualified team (over 70% of the staff are university graduates and PhD), multidisciplinarity, confidentiality and flexibility to develop customized solutions tailored to the needs of each client are different elements of **ainia** as part of its corporate culture.

Role in the project: Ainia will carry out the research work on developing active coatings for spoilage and pathogens microorganisms, combining antimicrobials with a broad spectrum for the spoilage ones and bacteriophages (endolysins) against pathogen one.

Scientific team:

Dr. José Angel Garde (M), PhD in Pharmacy and Master in Food Science and Technology by Institute of Agrochemistry and Food Technology. He has 18 years of experience in AINIA, in projects of scientific research and technological development with SMEs related to packaging in Interaction package-product area (migration and active packaging), food packaging and biodegradable material. He has been involved, as main researcher, in projects funded by Spanish Ministry: Biological degradation of polymer by means of selection and production of starter cultures, Development of functional polymeric structures with antimicrobial activity by polyethylene coating; and Active Packaging combining polyolefin with natural extracts.

Carlos Enguix (M), Head of Packaging Technologies Department, Industrial Engineer and master degree on Business Administration, on Marketing and on Quality of Plastic Industries. With more than 15 years of experience coordinating projects related to packaging in four main areas: Interaction package-product, Packaging Machinery, Packaging Waste Minimisation and Industrial Packaging Design (CAD/CAE/CAM).

Mr. Miguel Alborch (M), Engineer specialized in Agricultural and Food Industries. He has more than 10 years of experience in project management about characterization and improvement of food products and manufacturing processes aimed at developing higher-quality food from a sensory, nutritional and food security point of view. Expertise in EU regulations about organic products, hygiene and quality standards of foodstuffs in general.

Leonor Pascual (F), Chemical Engineer, specialized in Industrial Processes. Master's degree on Plastic Materials Transformation Processes She has experience in the Compounding Department of a Polymeric Research Centre, taking part in national and international research projects. Also in Polymeric Materials Migration Modelling at the reference research centre INRA.

Publications:

• Garde, J.A.; Catala, R.; Gavara, R. (2002) Migration kinteics. Antioxidan tmigration from polypropylene Migración de componentes y residuos de envases en contacto con alimentos, pp.219-232. Catala, R.; Gavara,R. (Ed.). Instituto de Agroquímica y Tecnología de Alimentos CSIC. Burjassot(Valencia,Spain).ISBN:84-920942-4-9.

• Imbernón, A.; Enguix, C.; Garde J.A.; Castrillón T.; Furman,S.; Ammala,A.; Casey,P.; White, P.(2006).In corporation of zincoxidenano particules in polilacticacid (PLA)for improving oxygen barrier. National Congress on Food Technology and Engineering (CESIA). CÓRDOBA (Spain).

• Enguix, C. Imbernon, A.,Ferrer, J.M.(2008):Environmentally compatible food packaging: Chapter 22 Legislation and certification of environmentally compatible packaging in the European Union.pp521 556.Wood head publishing in food science, technology and nutrition. England

• Active packaging for meat products: with the present getting near the future. (2010)Eurocarnenº186, pp 114-126.

• Active packaging and foods:the future is here] (2011) Reed Business Information. Alimentación, Equipos y Tecnología(nº263,pp52-57).Madrid.España

Projects:

Project acronym/description. Scope End Year Website

SUCCIPACK: Active, intelligent and sustainable food packaging using polybutylenesuccinate. FP7 KBBE, 2014, www.succipack.eu

PHBOTTLE: New sustainable, functionalized and competitive PHB material based in fruit by-products getting advanced solutions for packaging and non-packaging applications. FP7-NMP 2015 www.phbottle.eu/

WHEYPACK: Reduction of CO_2 emissions by the phb use obtained from whey: demonstration in dairy products packaging LIFE13 2016 www.wheypack.eu

APT-PACK: Advanced knowledge of polymer deformation for tomorrow's packaging. FP6 -S Project (STRP). 2007, Not available.

PACKTECH: Assimilation and standardisation of environmentally friendly packaging technologies within the food industry. G1RT-CT-2002-0, 2005 Not available.

Technical equipment:
- A semiautomatic machine RK Print-Coat Instruments, Ltd. Model KCC 202, will be utilized for coating application, with rods for different thickness and speed control.
- Laboratory Class 3, designed for microorganisms under this category such as E. coli O157, whose low infection doses require filtered air negative pressure and restricted access.

PARTNER 4 **Name**: Fraunhofer-Gesellschaft zur Förderung der angewandten Forschung e. V. **Web page**: www.fraunhofer.de	**Type: RTD** **Country: Germany**	Fraunhofer ICT

Brief description of the organization:
The Fraunhofer-Gesellschaft has more than 80 research units, including 66 Fraunhofer Institutes, at different locations in Germany. The majority of the 21000 staff are qualified scientists and engineers. The annual research budget is about € 1.6 billion. The Fraunhofer-Gesellschaft carries out internationally networked applied research which is both of direct use to industry and of benefit to society.
The Fraunhofer Institute for Chemical Technology (ICT) carries out contract research for industry and works together with commercial companies on research projects. Its grounds cover 200000 square meters, and subsequent to recent construction work around 12000 square meters of pilot plants, test facilities and workshops are available, as well as a further 12000 square meters of laboratories, infrastructure and offices.
Around 500 people are currently employed at the Fraunhofer ICT. The department of Polymer Engineering works in the whole process chain of thermoplastic and thermoset processing, developing new processes and products. A special focus is set on the processes being available in pilot scale for industrial applications.
The R&D expertise of the Fraunhofer ICT is primarily addressed towards products and processes in the fields of polymer technology. The facilities and expertise focus on the conception, computation and development of products and components, and on process technology, research into materials and constituent substances, characterisation, testing and quality assurance.
In the field of polymer development, the ICT has substantial experience in innovative compounding processes, reactive extrusion and process intensification by minimizing process steps. Furthermore, intensive research activities in the field of polymer chemistry, modification, assembling and degradation of polymer systems and interaction of different components in compounding processes have been undertaken by the ICT.

Role in the Project:
The Fraunhofer ICT will be responsible for the polymerisation of glycolic acid. This will be realized by thee researchers investigating different promising processing technologies:
- (Semi) continuous reactor based reaction polymerisation will be evaluated for maximum control of the resulting material properties and easy state of art further material utilisation
- Reactive extrusion polymerisation will be evaluated for highly efficient, solvent free continuous polymerisation
- Finally, the combination of both will be evaluated, realizing initial pre-polymerisation completed by reactive extrusion, enabling the opportunity for further material modification in continuous, solvent free processes.

Scientific team:
Dr. Jan Diemert (M), deputy head department Polymer Engineering. Dr. Diemert studied mechanical engineering at the Technical University of Aachen and specialized in polymer processing. He achieved his doctoral degree (Dr.-Ing.) at the University of Stuttgart with his work on "Fundamentals of plasticizing polyvinylchloride with microwaves". He has 10 years of experience in the extrusion and the compounding of thermoplastic materials.
Björn Bergmann (M), group leader Compounding & Extrusion. He studied chemistry at the Johannes Gutenberg-University Mainz and specialized in the field of macromolecular chemistry, in which he finished his studies with a Thesis at the Max-Planck Institute for Polymer research in Mainz. His expertise at the ICT is the molecular interactions of polymer matrices to additives in compounding and reactive extrusion. Nowadays he is leading the Compounding and Extrusion Group at the ICT and coordinating multiple national and international projects.

Andrei Holzer, (M) Senior Researcher Compounding. After having finished his apprenticeship as physics laboratory assistant in 1989, Andrei studied chemical engineering (specialisation in food process engineering) at the University of Karlsruhe (KIT). He has now 14 years of experience in compounding and profile extrusion of thermoplastic materials.
Irma Mikonsaari, (F) EU-Project management. Irma studied process engineering at the Technical University of Karlsruhe. She is specialised in particle technology. During her work at Fraunhofer ICT she has gained 17 years of experience on the field of particle and nanoparticle processing and characterisation as well as on interphase phenomena. She supports her colleagues in setting up new funded projects as well in their coordination.

Related publications, patents or products:
- Moser K, Holzer A, Bergmann B. Environmentally-Friendly, Heat-Resistant Profiles -Development of Materials Tailored to Process. Kunststoffe international, Ausgabe 2014/09, Carl Hanser Verlag, München.
- Moser K, Bergmann B, Holzer A, Diemert J, Elsner P. Chain alignment for improved properties - Optimization of PLA and PHBV by crystallization and reinforcement. 29th International Conference of the Polymer Processing Society (PPS-29), Nürnberg.
- Satya Dubey, Hrushikesh Abhyankar, Veronica Marchante, James Brighton, Kim Blackburn, Clive Temple, Björn Bergmann, Giang Trinh, Chantal David; Modelling and Validation of Synthesis of Poly Lactic Acid Using an Alternative Energy Source through a Continuous Reactive Extrusion Process; 10.3390/polym 8040164
- Björn Bergmann, Jan Diemert, Jean-Marie Raquez, Thomas Hielscher, Markus Reichmann, Chantal David, Satya P Dubey, and Carolina Losada Fernández, "InnoREX – European Project Reveals Impact of Microwave and Ultrasound Energy on Polymerization of PLA via Reactive Extrusion" Polymer Processing Society (PPS-32), 2016

Related Projects:
Project acronym/description. Scope End Year Website
InnoREX: Realising the continuous, highly precise, metal-free polymerisation of PLA using alternative energies for an innovative reactive extrusion. FP7-NMBP 2016 www.innorex.eu
StarTex: Production of thermoplastic starch by reactive extrusion and inline modification thereof for property tailoring German Project 2018 NA
BUGWORKERS New tailor-made PHB-based nanocomposites for high performance applications produced from environmentally friendly production routes FP7-NMBP 2013 www.bugworkersproject.eu/
PLA4FOOD Active Multilayer Packaging based on optimised PLA formulations. FP7-R4SME 2013 http://www.aimplas.es/proyectos/pla4food/

Technical equipment:
- 6 Co-rotational twin screw extruders (screw diameter: 18-32 mm)
- Gravimetric feeding units for bulk, liquid and gas
- In-line extrusion characterisation equipment in terms of NIR spectroscopy and viscosity
- Simulation software of twin screw extrusion process
- Reaction and separation lab holding numerous reactors (continuous and discontinuous), pumps, purification devices and peripheral equipment
- Chemical characterisation equipment (FT-IT, GPC, DSC…)
- Testing laboratory (tensile test, impact test, HDT, Vicat, SEM, light microscopy, durability, colour, gloss);

PARTNER 5 Name: Johann Heinrich von Thünen Institute (THUNEN) Web page: www.thuenen.de	Type: RO Country: Germany	THÜNEN

Brief description of the organization:
The Thünen Institute (THUNEN, formerly Federal Agricultural Research Centre) is attached to the German Federal Ministry for Food and Agriculture. Its mission is to provide the scientific basis for designing and implementing efficient agricultural policies, and to extend research findings to benefit society at large. The TI comprised of 14 specialty institutes and pursues interdisciplinary research in the following areas: economics, technology, material use of renewable natural resources, climate, biodiversity, organic farming.
The Thünen Institute of Agricultural Technology will be involved in the project: This institute is working with an interdisciplinary team of scientists and technical staff mainly in the fields of environmental protection, conversion of renewables and sustainability assessment of bio-based systems. Profound expertise exists in the biotechnological and chemo-catalytic conversion of renewables for the production of bio-based chemicals as well as in the production and characterisation of bio-based polymers and materials

Role in the Project: THUNEN will mainly participate in WP2 to develop an efficient process for the production of glycolic acid (GA) and will give contributions to the WPs 1, 6, 7 and 8.

Scientific team:
PD Dr. Ulf Prüße (M) (chemist, PhD and habilitation in technical chemistry) is the acting vice head of the institute and scientific director of the research topic renewables conversion technology. He received his diploma in chemistry in 1994 at the Technical University Braunschweig (TU BS) in Germany. In 1995 he joined the institute. Since his PhD in 1999 (TU BS) he has research experience in the fields of catalysis, biotechnology, immobilisation technologies, polymer chemistry in several national and EU projects (FP6, FP7, ERA-IB, ERA-Net WoodWisdom). Since 2003 he also lectures on catalysis, industrial chemistry and conversion of renewables at TU BS, where he also received his habilitation in technical chemistry in 2013. He has over 200 papers, patents and conference contributions.
Dr. Anja Kuenz (F) (biotechnologist, PhD in biotechnology) is the group leader biotechnology and down-stream processing within the research topic renewables conversion technology. She received her diploma from TU BS in 2004 followed by her PhD in 2008. She joined the institute in 2004 and has since that time research experience in the bioconversion of renewables as well as in the isolation and purification of chemicals from bio- and chemical conversion processes in national and ERA-IB projects. She has over 50 papers and conference contributions.
Ms. Evgenia Feldmann (F) (MSc in chemistry) works on the chemo-catalytic conversion of renewables. She received a grade as technician in chemistry in 2003 in Jarowslawl (Russia) and her MSc in chemistry at TU BS in 2013. She has two years research experience in polymer chemistry at universities and half a year in catalysis in industry. She joined the institute in early 2016 where she now works in catalysis research

Related publications, patents or products:

• Heidkamp K, Aytemir M, Vorlop KD, Prüße U. (2013) Ceria supported gold-platinum catalysts for the selective oxidation of alkyl ethoxylates, Catal. Sci. Technol., 3, 2984-2992.
• Heidkamp K, Decker N , Prüße U. (2013) Catalyst development for the selective oxidation of ethoxylates to their corresponding ether carboxylic acids, Catal. Commun., 40, 88-92.
• Bonrath W, Grünewald E, Karge R, Medlock J, Prüße U, Vorlop K. (2013) WO2013/010811, Process for the oxidation of sorbose, Priority date 15/07/2011.
• Prüße U, Herrmann M, Baatz C, Decker N. (2011) Gold-catalyzed selective oxidation of glucose in concentrated solutions at elevated oxygen partial pressures, Appl. Catal. A: Gen., 406, 89-93.
• Decker N, Franke O, Prüße U, Stankowiak A, Vorlop KD, (2009) WO2008/125241, Method for producing alkyl polyglycol carboxylic acids and polyglycol dicarboxylic acids, Priority date 12/04/2007.

Related Projects
Project acronym/description. Scope End Year Website
FAPA. Fumaric Acid for Polymer Applications (FAPA) ERA-IB 2019 www.era-ib.net/6thjointcall/FAPA
COSEPA. Controlled Separation and Conversion Processes for Wood Hemicelluloses. ERA-Net WoodWisdom 2017 www.woodwisdom.net/wp-content/uploads/2014/ 09/summary_cosepa.pdf
POAP. Production of Organic Acids for Polyester Synthesis. ERA-IB 2016 www.era-ib.net/poap
PUBB. Production and Upgrading of 2.3-Butanediol from Biomass. ERA-IB 2012 www.era-ib.net/pubb
HMF and FDCA from sugars for polyesters and plasticizers Germany 2017 www.thuenen.de/en/at/projects/products-from-renewables/hmf-from-sugars/

Technical equipment: More than 30 fully equipped research laboratories (analytics, biotech, catalysis, polymers), various batch and continuous-flow reactors (low - high temperatures and pressures), catalyst production up to kg scale, catalyst characterisation equipment

PARTNER 6 **Full Name:** Mi-Plast d.o.o., za proizvodnju, trgovinu i usluge (MIPLAST) **Web site:** www.mi-plast.eu	**Country:** **CROATIA** **Type: SME**	mi-plast bio-company

Organization description: Mi-PLAST is a company based in Rijeka, Croatia, founded in 1983. Company´s main activity is the processing of LDPE and HDPE materials, packaging and production of these materials, as well as regeneration and recycling of these materials. The products obtained by processing LDPE and HDPE are mainly used in the construction sector, agriculture and in retail sector. These products are: agriculture mulch, shopping bags, garbage bags, construction film foils, lamination foils etc. The company is placing its products mainly to the countries in the South Eastern Europe, Slovenia and Italy. MI-PLAST provides agriculture products to public institutions in the field of agriculture and to private companies. The company markets its products through its retail stores (field sales) and by direct contact with customers. MI-PLAST is a member of Croatian Economy Chamber and Croatian plastic and rubber products manufacturers association. The company is constantly trying to invest in the modernization of the production process and aims at increasing biobased products range.

Role in the Project: Production of PHA and Bio-PE films for package savoury products and cereals.

Scientific team:
Mr. Davorin Miketa (M): Holds a Master degree in economics, he is an Owner and Chief executive officer at MI-PLAST. Mr. Davorin Miketa has more than 40 years of experience in Industrial manufacturing and applied technology of plastic products. He was one of the first private entrepreneurs in former Yugoslavia and owned private small factory from the year 1980. He is responsible for company´s further industrial development and green process implementation.
Mr. Alan Drljačić (M): Master degree in chemical engineering, he has more than 20 years of experience in running conventional polymerisation plants and development of raw materials. Mr. Drljačić will be responisble for technical processes and scale-up actions.
Ms. Slavica Miketa (F): As a Chief administrative Officer and coordinator of EU projects, she will be responsible for management and reporting of the project. She is involved in implementation of FP7 BIO-QED project, and has extensive knowledge of project management.

Five publications/patents related to the project: N/A

Five Related Projects:
Description of experience/project Scope End Year Website (for information)
BIO-QED. Development of innovative and sustainable industrial processes for the production of bio chemicals from renewable sources. FP7-KBBE 2013 2017 http://bio-qed.eu/

HYPERBIOCOAT. High performance biomass extracted functional hybrid polymer coatings for food, cosmetic and medical device packaging BBI-2015 2019 NA
FUNGUSCHAIN. Demonstration of an integrated process with more than 40% valorisation of agro-residues from mushroom industrial cultivation residues valorised into high value bio-based products. BBI-2015 2020 NA
PULPACKTION. Optimised moulded pulp for renewable packaging solutions (Under Negotiation). BBI-2015 2020 NA

Technical equipment.
Twin-screw extruder; Blown extruders: Equipment for mechanical, thermal, rheological and physical tests

PARTNER 7 **Full Name:** Innovacio i Recerca Industrial i Sostenible (IRIS) **Web page**: http://www.iris.cat	**Country:** Spain **Type:** SME	

Organization description: IRIS is an advanced engineering company that specialises in process monitoring and control, and process optimisation. Our commercial division offers advanced monitoring and control solutions based on: NIR chemical monitoring (handheld device and in-line solutions); hyperspectral imaging solutions for foreign body detection and chemical composition analysis; data management solutions for data processing, decision support, automation and data analysis (big production data).
Our research division has 4 core research lines:
1. Optical and spectroscopy technologies (e.g. MEMs spectrophotometers, fluorescence and Raman) for process monitoring for on-line and PAT applications in food, pharmaceuticals and the process industries
2. Data Processing & Information Systems (using eg. Cloud technologies, Predictive modelling, Data mining, Deep Learning) for the development of intelligent data management and decision support systems for smart manufacturing, processing and agriculture.
3. Novel processing technologies (ultrasound, plasma, microwave…) for process intensification and non-thermal decontamination.
4. Sustainable processing technologies: waste valorisation strategies, biocompounds and biopackaging for the circular economy.

The company was established in 2007 and currently employs 48 staff with a turnover of €3.8 M. We are based over two sites (Barcelona and Dublin) that are equipped with engineering workshops (3), food testing and wet chemistry laboratory, optics labs (2), and an electronics/telecommunications lab.

IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i.
IRIS is an industry member if the following sectoral networks: SPIRE, SPIE and SECPHO.
IRIS has a matrix of multidisciplinary knowledge groups in the company. The main Groups that will be involved in this project are: food science and materials science (who have expertise in developing bio-based coatings and testing resulting packaging as well as active packaging).
IRIS' research team consists of biologists, food and material scientists, physicists, among other, who have the expertise to validate the technologies developed with support in terms of process from its broad cross-disciplinary engineering team, including electronics engineers, mechatronics, mechanical, industrial design engineers, system engineers, application programmers, web developers, etc.

Role in the Project: Building on its significant experience in projects related with the development of biobased barrier coatings and active packaging, IRIS will lead WP4 in terms of the development of active coatings solutions for PGA-derived plastics as well as their validation in contact with different food products as well as make contributions to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr. Ing. Elodie Bugnicourt (F). Masters in Material Science & Engineering (2002), PhD in Polymers and Composite Materials (2005), both from National Institute of Applied Science (INSA), Lyon, France. Leader of the Eco-Materials business unit in IRIS since 2009. Coordinated several European projects in the fields of bio-based packaging materials development, process monitoring, surface treatment. Previous industry experience: R&D in chemical industry especially related to reactive polymer formulations: i) Arkema (ex. Cray Valley, 2002, Spain) in the field of polyester powder coating; Wacker Chemie (2002-2005, Germany and France), Silicone division that funded her PhD thesis: development of nanocomposites based on an epoxy matrix and silica; iii) Ashland (2006-2009, located sequentially in US, Spain and France) in the Composite Polymers division, focussing on bio-sourced resins, flame retardant resins, SMC resins, PU and epoxy structural adhesives. Led and participated in many R&D projects aimed at developing new formulations, understanding mechanisms and structure-properties relationships, developing new analytical methods and modelling (for improved quality management); as well as collaborations with academic labs and other players in the supply and value chain. Author of number of papers in peer-reviewed and technical journals, and has presented her work in numerous international conferences.

Dr. Laura Rodríguez Turienzo (F). B.S in Food Science & Technology (2008) Masters in Innovation in Safety & Food Technologies (2010), PhD in Biological Sciences (2011), all from University of Santiago de Compostela (USC), Spain. She has participated in a number of R&D regionally and nationally funded projects aimed at developing and validating new bio-based active edible packaging, physico-chemical characterization of novel foods and valorisation of food industry wastes. She is author of several papers in peer-reviewed journals and has participated in different national and international conferences.

Dr. Edurne Gaston (F). PhD in Food Technology (Dublin Inst. of Tech.). Thesis: Visible and hyperspectral imaging systems for the detection and discrimination of mechanical and microbiological damage of mushrooms. Postgrad. Dipl. in Food Quality and Safety, BSc in Food Science and Technology (Uni. of the Basque Country). Leader of Food Tech Group in IRIS. Expertise in R&D projects, sensorial analysis. Industry expertise (Aguas Municipales de Vitoria) performing physical, chemical, and microbiological analysis of crude and drinking water, sensory analysis protocol for routine quality control. Author of numerous publications. Project management expertise.

Dr. Sara Peiró (F) PhD student performing her thesis at IRIS. Research topic: Natural antioxidants from plants: extraction, characterization and incorporation to food models and products. Co-directed by UPC and IRIS, Food Tech Group. She has the degree in Biology specialize in Vegetal Biology and Ecology by UAB, 2006.

Dr. Ana García Sanz (F) PhD degree in Biophysics from the Autónoma University de Madrid (PhD thesis: Electron and positron interactions with molecules of biological interest"). Masters in Biophysics from the Autónoma University in Madrid and Master-degree in Physics from the Complutense University de Madrid. Prior to joining IRIS, she has worked at the National Research Council Centre (CSIC) and has been invited as a research intern in Universitá La Sapieza di Roma, The Open University and Sherbrooke University. Her research has been focused on the study of the damage induced by radiation (electrons, positrons and X-rays) in biological systems at molecular level, by means of both theoretical (quantum chemistry calculations) and experimental (spectroscopy, HPLC, LC-MS/MS) techniques. Author of 17 scientific papers in peer reviewed Journals.

Five relevant publications, patents or products:

• "Whey protein coated films"; **world patent** IB2011/053271; Inventors: M. Schmid, K. Noller, F. Wild, **E. Bugnicourt**; positive PCT examination; Application date: July 22, 2011.

• 'Method to prepare a coating made of whey-proteins treated with ultrasounds. **L. Rodríguez Turienzo** et al. **Patent** application 201130582

• Effects of edible coatings based on ultrasound treated whey proteins in quality attributes of frozen Atlantic salmon (Salmo salar), **Rodríguez-Turienzo, L**., Cobos, A., & Díaz, O., (2012), Innovative Food Science & Emerging Technologies, 14, pp. 92-98.

• Processing and Validation of Whey-Protein-Coated Films and Laminates at Semi-Industrial Scale as Novel Recyclable Food Packaging Materials with Excellent Barrier Properties. **Bugnicourt, E,** Schmid, M., Mc. Nerney, O., Wildner, J., Smykala, L., Lazzeri, A., Cinelli, P. (2013) Adv in Materials Sci Eng, 10 p.

• State of the art in the development and properties of protein-based films and coatings and their applicability to cellulose based products: an extensive review; M.B. Coltelli*, F. Wild*, **E. Bugnicourt***, P. Cinelli, M. Lindner, M. Schmid, V. Weckel, K. Müller, P. Rodriguez, A. Staebler, L. Rodriguez, A. Lazzeri; Coatings 2016, 6, 1. (*equal contribution of corresponding authors)

Five Relevant Projects:
Project acronym/description. Scope End Year Website
BIOBOARD Development of sustainable protein-based paper and paperboard coating systems to increase the recyclability of food and beverage packaging materials FP7-SME 2015 www.bioboard.eu
WHEYLAYER2 Barrier biopolymers for sustainable packaging. FP7-DA 2014 www.wheylayer.eu
OPTINANOPRO Processing and control of novel nanomaterials in packaging, automotive and solar panel processing lines H2020-NMBP 2018 www.optinanopro.eu
OLI-PHA A novel and efficient method for the production of PHA polymer-based packaging from olive oil waste water FP7-NMP 2015 www.olipha.eu
AgriMax Agri & food waste valorisation crops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications (Grant Agreement no: 720719, in contract negotiations) BBI-V 2019 N/A

Technical equipment: Spectroscopic Laboratory equipped with several instruments, namely, a FTIR-ATR (Interspectrum), 2 UV-VIS (PerkinElmer and Ocean Optics), an Extended-SWIR (StellarNet) and a NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are properly trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes).
Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others.
IRIS' optical engineers work with ZEEMAX software for optimally designing of optical systems and to setup the designs by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware").
Food and wet chemistry lab equipped with a DSC, a Surface Energy analyser, an HPLC, a Cobb tester, microbiological equipments, microspcope and microtome for structural analysis, migration cells, colourimeter (for colour and opacity measurement), Ford cup viscometers, water vapour permeation cells, turbidimeter, climatic chambers, manual and automatic coaters, electrospray, different drying technologies, plasma surface activation, etc.

PARTNER 8. **Full name**: Agencia Estatal Consejo Superior de Investigaciones Científicas (CSIC) – Centro de Investigaciones Biologicas (CIB-CSIC) **Web page**: http://cib.csic.es/	**Country:** Spain **Type:** Research Organisation	CSIC CONSEJO SUPERIOR DE INVESTIGACIONES CIENTÍFICAS

Brief description of the organization:

The Spanish National Research Council (CSIC) is the largest public institution dedicated to research in Spain and the third largest in Europe. Belonging to the Spanish Ministry of Economy and Competitiveness through the Secretary of State for Research, Development and Innovation, its main objective is to develop and promote research that will help bring about scientific and technological progress. It has a staff of more than 13,000 employees. The CSIC has 71 institutes or centres distributed throughout Spain CSIC has considerable experience in both participating and managing R&D projects and training of research personnel. Under the 7th Framework Programme CSIC has signed 724 actions (including 62 coordinated by CSIC and 45 ERC projects). Within the first Work Programme of H2020 (2014-2015), CSIC has actively participated in different calls, having special success in some programmes, such as Research Infrastructures, ERC and Marie Curie Actions, as well as in the Societal Challenges of "Food Security, Sustainable Agriculture and Forestry". In June 2016, CSIC has obtained 203 projects in the first Work Programme with a total EU financial contribution of 88, 5 million euros. As E-CORDA points out CSIC is listed the 1st organisation in Spain and the 10th in Europe in H2020 funding. CSIC is also an active member in Knowledge and Innovation Communities (KIC), Raw Materials and Health, of the European Institute for Innovation and Technology (EIT).

The Biological Research Centre (CIB) is a public funded national research institute that belongs to the Spanish National Research Council (CSIC). At the CIB´s Department of Environmental Biology, we are devoted to advance the understanding on how living organisms, plants, animals and microorganisms, interact and respond to their environment. Organisms are studied on every scale, from molecular to whole-body level, and using multidisciplinary experimental approaches. The diversity of the permanent research staff, which operates in 11 groups, combined with modern equipped labs provide for variety and excellence in both, fundamental and applied research. The significance of the research developed in our Department is reflected in highly-cited publications in leading journals, as well as in collaborations with other scientists from top institutes all over the world. In this scenario, the Department provides a unique entourage for graduate students to perform their Ph.D. thesis.

Role in the Project: CSIC will participate in the production of PHA, and in the exploitation of results and dissemination and communication tasks.

Scientific team:

Dr. Auxiliadora Prieto (F). Head of the Polymer Biotechnology Lab. The significance of the research developed in the group is reflected in highly-cited publications in leading journals (more than 65; > 2500 citations; h-index 27,), as well as in collaborations with other scientists from top institutes all over the world. Dr Prieto has participated in, led and coordinate national (11) and international projects (5). She is main inventor of five international patents related to biopolymer production, three of them licensed to Biotech SMEs dedicated to the production of bioplastics.

Prof. Dr. José Luis García (M), Head of the Environmental Biotechnology Lab, 240 WoS papers, >3000 citations; h-index 39, 18 Ph.D. students; 32 patents; participated in more than 60 National, EU and Industrial projects, many of them as PI. Teaching experience. Science policy maker experience. Founder of two spin-off companies.

Dr. Oliver Drzyzga (M), Environmental Microbiologist, Senior Researcher, 39 WoS papers, >900 citations, h-index 17, 11 years of university teaching experience, 5 master and PhD students, 4 Erasmus students, participated in 2 EU projects (1 as post-doctoral researcher, 1 as project manager).

Five relevant publications, patents or products:

- Dinjaski, N, and MA Prieto. 2015. Smart polyhydroxyalkanoate nanobeads by protein based functionalization. Nanomedicine: Nanotechnology, Biology and Medicine 11: 885-889.

- Drzyzga O., Revelles O., Durante-Rodríguez G., Díaz E., García J.L. and Prieto MA. 2015. New challenges for syngas fermentation: towards production of biopolymers. Journal of Chemical Technology & Biotechnology, 90:1735-1751.
- Prieto, MA, Escapa IF, Martínez V, Dinjaski N, Herencias C,de la Peña F, Tarazona N and Revelles O. 2016. A holistic view of polyhydroxyalkanoate metabolism in Pseudomonas putida. Environmental Microbiology 18: 341-357.
- Revelles O, N Tarazona, JL García, MA Prieto. 2016. Carbon roadmap from syngas to polyhydroxyalkanoates in *Rhodospirillum rubrum.* Environmental Microbiology 18: 708-720.

Martínez V, Herencias C, Jurkevitch E, Prieto MA. Engineering a predatory bacterium as a proficient killer agent for intracellular bio-products recovery: The case of the polyhydroxyalkanoates. Sci Rep. 2016 Apr 18;6:24381.

Five Relevant Projects:

Project acronym/description. Scope End Year Website

- SYNPOL - Biopolymers from Syngas Fermentation; FP7 KBBE, agreement no. 311815; ended in 09-2016; http://www.synpol.org/

- PREDATICS - Integration of lytic agents of biological origin in the design of new downstream methods for the production of bioplastics; national program BIO2013-44878-R; ends in 2017;

https://www.cib.csic.es/project/integration-lytic-agents-biological-origin-design-new-downstream-methods-produc

- NANOBIOSOMA - Design, development and production of nanocarriers and nanovehicles; national program S2013/MIT2807; ends in 2018; http://www.nanobiosoma.org/index.php/en/
- P4SB - From plastic waste to plastic value using *Pseudomonas putida* synthetic biology; H2020-LEIT-BIO-2014-1, agreement no. 633962; ends in 2019; http://p4sb.eu/

CELBICON - Cost-effective CO_2 conversion into chemicals via combination of capture, electrochemical and biochemical conversion technologies; H2020-ISIB-2015-2, agreement no. 679050; ends in 2019; http://www.celbicon.org/

Technical equipment:

The Biological Research Centre (CIB) is a public funded national research institute that belongs to the National Spanish Research Council (CSIC). The CIB is one of the best equipped research centers for developing projects on Biotechnology and Molecular Microbiology in Spain. Available facilities at CIB as central services are: automatic DNA sequencing, quantitative RT-PCR, Transcriptomic analysis platform, oligonucleotide and peptide synthesis, protein chemistry (protein sequencing, proteomics, spectroscopy, NMR, analytical ultracentrifugation, crystallization), chromatographic analyses (HPLC, GC-MS), microscopy (electronic, confocal), fermentation, bioinformatics. Particularly, the groups of Prof. J.L. García, Dr. A. Prieto and E. Díaz are additionally equipped with all equipment required for Molecular Biology and Microbiology (Bioreactors, PCRs, spectrophotometers, gel-doc, electrophoresis, PFGE, shaker, incubators, ultra-freezers, HPLC, FPLC, microscopes, anaerobic cabinet, centrifuges, computers, etc.).

PARTNER 10 **Name**: Carton Bros (T/A Manor Farm) **Web page**: www.manorfarm.ie	**Type: IND** **Country: Ireland**	

Brief description of the organization:

Carton Brothers - an Irish Family Tradition since 1775

Carton Brothers is the name of the company that produces Manor Farm chicken. It is one of the oldest family companies in Ireland and dates back to 1775. Manor Farm is the largest chicken processor in the Republic of Ireland currently producing approximately 900,000 birds on a weekly basis. The company, trading under its own flagship "Manor Farm" brand, distributes to all major retail outlets in the Republic of Ireland and many of the catering institutions. The company also produces "Own Label" product to all the major retailers in Ireland. The company is highly active in the export market where its products may be seen in the U.K., France, Spain, Eastern Europe, South Africa and the Far East. The company is vertically integrated in that it has its own hatchery, feed mill and processing plant. The farms where the birds are reared are under contract to the company and must follow all guidelines and procedures as laid down by the company. This allows complete traceability for the products. Processing moved to Shercock in Co. Cavan in 1970 becoming the first plant in Britain and Ireland to be EEC approved. The company whilst already at the forefront of the industry in terms of size has spent over €14m in 2005 increasing the size and capacity of its processing plant at Shercock. Further investment is currently underway with an additional €25million planned over the next 5 years. Consistent within Manor Farm at all times is dedication to Quality and the production of unique and very tasty chicken products from the best chicken that we ourselves produce. Manor Farm believes innovation is critical to their success and support new product development to ensure that we are at the forefront of new & emerging technologies. Manor Farm was first to market in Ireland with cook in the bag packaging technology & continues to invest resources in new packaging developments.

Role in the Project: To validate the performance of the new material in an industrial, commercial environment from both migration & shelf life perspectives in comparison to existing PET & PE materials. We distribute through all fresh food channels throughout the island of Ireland & are therefore in a unique position to test new materials.

Scientific team:
Mr JUSTIN CARTON (M) (Fellow of the Association of Chartered Management Accountants, MSC MIS, BSC Marketing and Management)**.** Director of Information Technology at Carton Bros Ltd, Ireland's largest poultry industry business. Has worked in the poultry industry since 1984 on all aspects of integrated poultry production. He is responsible for ensuring that the business maintains its leading-edge position in the poultry industry, by directing its research and development programme, with particular emphasis on the application of "smart systems" within the industry.
Mr Declan Kennedy (M) (Manor Farm processing director) qualification of BSC in Food Business from University College Cork. Prior to joining Manor Farm, Declan has over 20 years' experience in food production in senior management positions with Dawn Farm Foods, Irish Country Meats and Green Isle.
Mr Declan Prunty (M) (Manor Farm Project Manager) leads all major projects for the Manor Farm site to including new & emerging technologies. Qualifications include:
2013 Lean Six Sigma Green Belt / 5S
2009 L5 in AutoCAD Design & Computer Aided Draughting
2008 C&G Certificate in Handling Refrigerants (Ammonia)
2006 Project Management
1992 C&G Advanced Certificate Refrigeration & Air Conditioning (Part 3)
1990 National Craft Certificate Refrigeration (Part 1&2)
Mr Liam Hilliard (M) (Manor Farm Food Safety & Quality Assurance Manager). Liam has held Quality Management positions since 2003 working with large organisations including Kerry Foods, Irish Country Meats and Dawn Farm Foods before joining Manor Farm. Qualifications include:
DIT Cathal Brugha Street, MSc. Food Safety Management 2011-2013
DIT Kevin Street, Kevin Street, Dublin 8, Graduate Diploma (FETAC Level 8) in Food Science and Technology – 1995-1996
Carlow RTC, Kilkenny Road, Carlow, Diploma in Food Quality Management – 1992-1995
Dr Zandra Reed (Montgomery) (F) (Manor Farm NPD Manager). NPD Manager at Manor Farm since 2007, previously working as senior food technologist in Manor Farm since Oct 2005. Previously, NPD Manager at Dawn Farm Foods, The Mauldins, Naas, Co. Kildare (2003-2005), NPD Project Manager, Dawn Farm Foods, Oct 2001 – Nov 2003. 1998-2001, completed full time PhD in University of Ulster, thesis title "Sensory Quality & Consumer Acceptance of Chilled Ready Meals", September 2001. Graduated from University of Ulster with 1st Class Hons Degree in Consumer Studies with DIS in 1998.

Related publications, patents or products:
Haughton, P.N., Lyng, J., Fanning, S., Whyte, P. (2013). Potential of a commercially available water acidification product for reducing *Campylobacter* in broilers prior to slaughter. *British Poultry Science, 54, 319-324.*
Haughton, P.N., Lyng, J., Cronin, C., Fanning, S., Whyte, P. (2012). Effect of crust freezing applied alone and in combination with ultraviolet light on the survival of *Campylobacter* on raw chicken. *Food Microbiology, 32, 147-151.*
Haughton, P.N., Gomez Grau E., Lyng, J., Cronin, D., Fanning, S., Whyte, P. (2012). Susceptibility of *Campylobacter* to high intensity near ultraviolet/visible 395±5nm light and its effectiveness for the decontamination of raw chicken and contact surfaces. *International Journal of Food Microbiology, 159, 267-273.*
Haughton, P.N., Lyng, J.G., Cronin, D.A., Morgan, D.J., Fanning, S., Whyte, P. (2012). Efficacy of pulsed electric fields for the inactivation of indicator microorganisms and foodborne pathogens in liquids and raw chicken. *Food Control, 25, 131-135.*
Haughton, P.N., Lyng, J.G., Morgan, D.J., Cronin, D.A., Noci, F., Fanning, S., Whyte, P. (2012). An evaluation of the potential of high intensity ultrasound for improving the microbial safety of poultry. *Food and Bioprocess Technology, 5, 992-998. (doi 10.1007/s11947-010-372-y).*

Related Projects:
Project acronym/description. Scope End Year Website
No related projects.

Technical equipment:
- Manor Farm Innovation Centre

Contracted Laboratories for Migration & Shelf life Testing

PARTNER 11 **Name:** MAICERÍAS ESPAÑOLAS, S.A. (DACSA) **Web page:** www.dacsa.com	**Type:** IND **Country:** Spain	DACSA Maicerías Españolas, S.A.

Brief description of the organization: DACSA is a leading food company focused on cereal products, mainly those derived from corn and rice. DACSA is present in various European countries, with installations in Spain, France and United Kingdom. Innovation is a core part of the company, focusing research on both the improvement of the food products quality and the development of new packaging strategies, including packaging and bioplastics.

Role in the Project: DACSA will be involved in selecting the food to be tested and validating the final packaging products given their experience in food packaging and technical capabilities. They are also a large end-user company, assuring that the project will have a direct effect on consumer products.

Scientific team:
Dr. Mª del Carmen Vidal (F), Corporative responsible for innovation. She has previously been quality manager in DACSA and professor at the university.
Maria José Valero Barrera (F), Rice quality manager (Chemist). She has previously worked as a laboratory technician and control quality manager.
Noelia Ruiz Martínez (F), Innovation technician (Chemist). Four years of experience as a technician in DACSA.
Mª Begoña Bisbal Ramón (F), Corn Laboratory Manager. 16 years of experience in DACSA, she has attended courses on BRC global standard for food safety, internal auditor and gluten.
Lamberto Tarín (M), Corn area manager (Industrial Engineer). Previously he has been the plant manager for 3 years and has 30 years of experience in the textile industry.

Related publications, patents or products:
No related publications, patents or products.

Related Projects:
Project acronym/description. Scope End Year Website
BIOPLÁSTICOS. Production of bioproducts from alimentary waste (RTC-2015-3825-5). 2017. http://www.dacsa.com/blog/proyecto-bioproductos/
RIZEOPACK. Development of packaging activated with zeolites. 2016. http://www.itene.com/proyectos-de-difusion-abierta/i/7728/56/rizeopack
PLAGUEFREE. Packaging for plague control in dried food. 2014. http://www.dacsa.com/eng/dacsa-plaguefree-project/

Technical equipment:
- RVA equipment to analyze flour viscosity
- Soxhlet equipment for fat determination
- NIR analyzer for humidity, fat and protein determination

PARTNER 12 **Name**: EUFIC - European Food Information Council **Web page**: www.eufic.org/	**Type**: non-profit organisation **Country**: Belgium	EUFIC European Food Information Council

Brief description of the organization:
EUFIC (http://www.eufic.org) is a non-profit organisation whose mission is to communicate science-based information on food, health and nutrition to help consumers to be better informed when choosing a well-balanced, safe and healthful diet.
With its offices located in Brussels (Belgium), EUFIC counts on and liaises with a large European and international network (spanning academia, industry, European food and health stakeholders and policy makers, health professionals, media, and consumers) to enhance the impact and outreach of its communication instruments and programs in different countries. EUFIC actively participates in European initiatives by the European Commission Directorate Generals for Research (DG Research) and for Health and Food Safety (DG Santé). Since 2004, EUFIC has been involved in 20 EC-funded projects, seven of which are ongoing. EUFIC's main role is to lead the dissemination of project results and organise stakeholder consultations and workshops.
EUFIC is an active member of the EU Platform for Action on Diet, Physical Activity and Health of DG Santé, as well as of the Stakeholder Advisory Board of the Joint Programming Initiative Healthy Diet for a Healthy Life (JPI HDHL).
EUFICs participation in the EU funded projects:
- **H2020**: PROTEIN2FOOD, RICHFIELDS, OLEUM, STRENGTH2FOOD, NanoPack
- **FP7**: CHANCE, RECAPT, CONNECT4ACTION, INPROFOOD, Food4Me, NUAGE, EURO-DISH, CLYMBOL, DIVERSIFY, MyNewGut
- **FP6**: SAFEFOODS, HELENA, IDEFICS, EURECA, FLABEL, AGRIFOODRESULTS, EATWELL, FoodRisc

EUFIC has extensive expertise and experience in summarising scientific results and publications for a wider audience of relevant stakeholders, and has the tools and networks to ensure broad and effective dissemination. EUFIC issues several publications, such as FOOD TODAY, a newsletter that highlights subjects of current interest to scientists, educators, media and other communication multipliers, EUFIC Reviews on topical issues, or leaflets on healthy eating for kids and adults. EUFIC also uses new technologies such as podcasts and webinars to disseminate key messages quickly and efficiently. As an indication of outreach, EUFIC's website has >1 million visits a month (>13.6 million visits/year), EUFIC's newsletter FOOD TODAY, which exists in 11 languages, is distributed to > 47,000 health professionals, educators, media, scientists, and other communication multipliers. EUFIC has more than 12,000 followers in its social media accounts, which include Facebook, Twitter, LinkedIn, Google+ and YouTube.

Role in the Project: - Communications and Dissemination

Scientific team:
Dr Laura Fernández Celemín (F), Director General, holds a degree in human nutrition and a PhD in Biomedical Sciences from the Catholic University of Louvain, Brussels, Belgium. Laura's roles include the supervision of EUFIC's involvement in EU-funded research projects. She was in charge of coordinating the FLABEL project on behalf of EUFIC.
Dr Nina McGrath (F), Food Safety and Risk Communications Manager, holds a Master's degree in Chemistry and a PhD in Inorganic Chemistry and Materials Science and has published several peer-reviewed papers. In her current position, Nina manages the Food Safety and Quality interest group supporting their production of science-based communications materials on food safety related topics. Nina previously worked as Science Counsellor for Euro Chlor, an industry association representing European chlorine producers, where she provided scientific support on occupational health and environmental issues.
Frédéric Goffard (M), Administrative and Financial Advisor for EU projects, has a Bachelor in Accounting from École Pratique des Hautes Études Commercial, Brussels. Frédéric previously worked as Financial, HR and IT Manager at Digital Europe, Fortum Oil and Telematics and Services.
Dr Aleksandar Sokolović (M) is Scientific Advisor at EUFIC, where he focuses on coordination of EUFIC's communication and research activities on EU funded projects. Aleksandar has a degree in Biology from the

University of Belgrade, and a PhD in Biomedical Sciences from the University of Amsterdam. During the last two years he has contributed to EUFIC's work as a Scientific Consultant, bringing a valuable multicultural experience from scientific and commercial settings.

Related publications, patents or products:

- Grunert KG, Hieke S, Wills JM. Sustainability labels on food products: Consumer motivation, understanding and use. Food Policy. Food Policy 2014;44:177-189.
- Friel M. Comparison of Reporting of Food Benefits and Risks in European Newspapers. Annals of Nutrition and Metabolism, 2013;63(1):173.
- Wills JM, Storcksdieck genannt Bonsmann S, Kolka M, Grunert KG. European consumers and health claims - attitudes, understanding, and purchasing behaviour. Proc Nutr Soc. 2012;71(2):229–36. doi:10.1017/S0029665112000043.
- Grunert KG, Wills J, Fernández Celemín L, Lähteenmäki L, Scholderer J, Storcksdieck genannt Bonsmann S. Socio-demographic and attitudinal determinants of nutrition knowledge of food shoppers in six European countries. Food Qual Pref. 2012;26:166–77. doi:10.1016/j.foodqual.2012.04.007.
- Hieke S, Wills JM. Nutrition labelling - is it effective in encouraging healthy eating? CAB Reviews. 2012;7(031):1–7. doi:10.1079/PAVSNNR20127031

Related Projects:

- **FoodRisC -** Food Risk Communication - Perceptions and communication of food risks/benefits across Europe: development of effective communication strategies**. FP7, 2013.** www.foodrisc.org
- **Connect4action -** Strategies for improving communication between social and consumer scientists, food technology developers and consumers**. FP7, 2014.** http://www.connect4action.eu/
- **RECAPT -** Retailer and Consumer Acceptance of Promising Novel Technologies and Collaborative Innovation Management**. FP7, 2014.** http://www.recapt.org
- **NanoPack -** demonstrate and validate the feasibility and the advantages of HNTs-based antimicrobial films in three food packaging industrial environments. **H2020, 2018,** N/A

Technical equipment:

EUFIC has a broad outreach. EUFIC's website has >1 million visits a month (>13.6 million visits/year), EUFIC's newsletter FOOD TODAY, which exists in 11 languages, is distributed to > 47,000 health professionals, educators, media, scientists, and other communication multipliers. EUFIC has more than 12,000 followers in its social media accounts, which include Facebook, Twitter, LinkedIn, Google+ and YouTube.

PARTNER. 13 **Full name**: **APEX. Aperitivos y extrusionados SAU** **Web page**: www.grupoapex.es	**Country: Spain Type:**	

Brief description of the organization:

APERITIVOS Y EXTRUSIONADOS, S.A.U (APEX) was funded as a corporation in 1983, its main purpose is to manufacture, distribute, store, represent, import, export, pack and transport snacks and preparations, both with own brands and for third parties

The process of concentration of the Spanish sector of snacks, allowed the company to encompass all the products and marketing channels, gaining competitiveness and cost effectiveness. Such changes made Apex the Spanish leader in sales and referent in its sector, both own brand and brands of the Distributor, with the most modern factory in the sector, and providing the competitive advantage of Purchase of the raw material - directly to the farmers.

Also, in recent years, Apex has been expanding internationally, with sales in more than 15 countries such us France, Germany, United Kingdom, Holland, United States, Russia, China, Morocco, Libya, South Korea ...), and with new Projects in other areas (mainly Latin America, Persian Gulf and Africa).

The philosophy of the company includes several points: children's snacks Fun, healthy and tasty; Attractive and novel youth product lines; Broad Offer of dried and dried fruits; Total quality in products and services; Intense activity.
The values of APEX are innovation, differentiation and marketing. In other words, we work with products totally differentiated as much by the origin as by the quality of the raw materials, differing also in textures, shapes, packaging ... and ultimately in innovation that ends up generating greater added value to our products.
In relation to the technological and industrial capacity of the company, APEX counts with three factories in Spain, which main numbers are stated below.

Role in the Project: Definition of packaging requirements and validation on final product packaging
Scientific team:
D. Mario Gonzalez Morala, responsible for Quality and Agronomic Department of Apex (Potato Chips factory).
D. Laura Añón, responsible for Quality Department of Apex (Extruded snacks factory).

Five relevant publications, patents or products:
N/A
Five Relevant Projects:
Project acronym/description. Scope End Year Website

Technical equipment:
Business group with 2 factories and 1 logistic warehouse.
- 19 packers
- Protective atmosphere packaging (Nitrogen)
- Metal detectors
- 6 extruders
- 4 ovens
- 2 Trains pellets
- 2 Trains french fries
- 8 frying pan fry
- Waxy maize fryer and peeler
- IFS certification in production plants
- 2 Process Control Laboratories
- Quality department> 8 workers
- Technical service and maintenance> 10 workers

4.2. THIRD PARTIES INVOLVED IN THE PROJECT (INCLUDING USE OF THIRD PARTY RESOURCES)
No third parties involved

5. ETHICS AND SECURITY

5.1 ETHICS
There are no ethical issues associated with this project. The work conforms to all relevant national or international regulations and EC legislation. There are no foreseen negative ethical issues associated with the implementation of the project results. There will be no experimentation on animals, with human cloning, genetic modification or human embryos or stems cells as a part of the project.

5.2 SECURITY
Please indicate if your project will involve:
• Activities or results raising security issues: NO.

'EU-classified information' as background or results: NO.